Universitas Pandrosion
Combined Edition (Papers 0–8)

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Contents

Paper 0 — Pandrosion-Steffensen for $z^p = x$ (Lean-verified)

A Formally Verified Pandrosion–Steffensen Scheme

Closed-form real-line analysis and complex multi-start extension
for the equation $z^p = x$, with a Lean 4 / Mathlib v4.7.0 development

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github.com/ivan-fr/poussiere_de_x

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Abstract

This paper combines, in a single document, the real-line analysis of Pandrosion's classical geometric construction for p-th roots and its formally verified complex-multi-start extension.

Part I (real line). We revisit Pandrosion's construction reported by Pappus for approximating $\sqrt[3]{2}$ and generalize it to the computation of p-th roots of any positive real x by inserting proportional means via Thales' theorem. After normalization the iteration reduces to the universal form $s_{n+1} = 1 - (x-1)/(x\, S_p(s_n))$ with $S_p(s) = 1 + s + \cdots + s^{p-1}$, fixed point $s^* = x^{-1/p}$, and readout $v_n = x\, s_n^{p-1} \to x^{1/p}$. We derive the closed form

$$\lambda_{p,x} = \frac{(\alpha - 1)\sum_{k=1}^{p-1} k\,\alpha^{p-1-k}}{\sum_{k=0}^{p-1} \alpha^k}, \qquad \alpha = x^{1/p},$$

for the linear contraction ratio, prove it is monotone in both p and the root parameter in the precise senses of Theorem 9.1 (via the closed-form identities $P_p(\alpha) = p\, D_p'(\alpha)$ and $B_p(\alpha) = \sum_{j=1}^{p} j\alpha^{j-1}$, both with all positive coefficients), with explicit large-p limit $1 - \ln(x)/(x-1)$, give non-asymptotic error bounds, extend the theory to $x < 1$ with an explicit threshold α_p^* below which the iteration diverges, and add two accelerations: a canonical starting point $s_0 = h(1) = 1 - (x-1)/(xp)$, and Aitken–Steffensen Δ^2 yielding quadratic convergence with a much smaller constant than Newton. A scaling preconditioning $x = A \cdot x'$ exploits the monotonicity of λ to collapse iteration counts for large x.

Part II (complex extension, formally verified). We extend the iterator $h_{p,x}$ to $\mathbb{C}$, where it has p attractive fixed points — the p-th roots of $1/x$ — separated by a non-trivial Julia structure. A multi-start scheme $\mathrm{seed}_{\alpha,\varepsilon}(z) = \alpha + \varepsilon(z - \alpha)$ with target-dependent ε places any $z \in \mathbb{C}$ inside the quadratic-contraction disk of a chosen anchor α, providing a universal convergence theorem with closed-form iteration count $N_{\mathrm{eff}} \sim \log_2 \log(1/\varepsilon)$. A compatibility bundle (Theorem 16.4) collects target selection, Steffensen fixing, scaling compatibility, reconstruction, and optimal seed into a single glue statement that downstream modules cite by name. Unconditionally (without seeding), each anchor's basin of attraction contains an explicit disk of radius $R_\sigma(x, p, \gamma_k)$, giving the lower bound $\mathrm{vol}(\bigcup_k \Omega_k) \geq \sum_k \pi R_\sigma(x, p, \gamma_k)^2$ on the total basin volume (Proposition 16.1). The induced solution map is Borel-measurable everywhere and continuous outside the Voronoï boundary of the cyclotomic anchor set (which is Lebesgue-null). All of Part II is formally verified in a Lean 4 / Mathlib v4.7.0 development of 121 modules with zero `sorry` and audited axiomatic closure reducing to the standard Lean whitelist.

The paper makes no claim of progress on classical open problems of polynomial root-finding; its main value is to record a concrete derivative-free root-finding scheme with closed-form convergence analysis and complete formal verification for the monomial family $z^p = x$.

A concluding section (§20) shows that the Pandrosion iteration itself extends to every monic polynomial $P(z) = z^p + R(z)$ via $h_P(s) := 1 - (R(s) + 1)/S_p(s)$, whose fixed points are exactly the roots of P and which remains derivative-free; Aitken Δ^2 and the multi-start

seed apply verbatim, yielding a target-conditional refinement scheme that converges to every target root.

Numerical experiments on four non-monomial polynomials show the structural extension works, with iteration counts varying from 4 (monomial, best case) to 41 (Bombelli's cubic, worst case) at 50-digit precision, depending on how close $h_P'(\alpha)$ is to 1 and how close the target root is to a singularity of h_P. The quantitative advantage of the monomial case is therefore genuinely specific to that case and does not transport uniformly.

Reader's guide to the corpus. This paper is the load-bearing core of the series: it proves and formalizes the Pandrosion–Steffensen scheme for the monomial family $z^p = x$. The remaining papers split into two independent threads. Part I extends the operator language to general polynomials and polynomial systems, and Parts VII–VIII return to that algorithmic thread with non-holomorphic fallbacks and practical start strategies. Parts II–VI form a separate Smale Mean Value Conjecture thread: they translate the MVC into Pandrosion quotients, record structural identities, and isolate the quartic/quintic algebraic programs. None of the later papers should be read as extending the Lean verification of this paper unless this is stated explicitly.

Contents

Part I
Real-line theory

1 Introduction

The Greek problem of doubling the cube consists of constructing a segment of length $\sqrt[3]{2}$ from a unit segment. It is equivalent to finding two proportional means m_1, m_2 between 1 and 2, that is,

$$\frac{1}{m_1} = \frac{m_1}{m_2} = \frac{m_2}{2},$$

which yields $m_1 = \sqrt[3]{2}$ and $m_2 = \sqrt[3]{4}$. Pandrosion of Alexandria proposed an *approximate* geometric method, reported by Pappus in the *Collectio*, Book III [3]: by successive parallels (Thales' theorem), she constructs proportional ratios that approach these means without ever reaching them exactly in a finite number of steps.

This work starts from a fundamental observation:

> *Mathematically the target number is exact, but any approximation procedure yields a measurable residual profile — a footprint intrinsic to the chosen method.*

At each finite step of Pandrosion's construction, there remains a gap $\varepsilon_n = |v_n - \alpha|$ between the geometric approximation and the ideal value. The sequence (ε_n) is the **residual profile** of the method. Our goal is to compute this profile exactly for any p-th root $x^{1/p}$, and to identify the contraction ratio that characterizes Pandrosion's method.

Positioning and related work. Rational iterations for p-th roots are a classical topic with a large literature: Newton's method applied to $f(u) = u^p - x$, Halley's method, and more generally Schröder's (1870) and König's (1884) families produce iterations of arbitrary order with explicit asymptotic constants. The systematic analysis of fixed-point iterations and their acceleration goes back to Ostrowski [9], Traub [11], and Householder [10]; the derivative-free counterpart via Aitken–Steffensen (1926/1933) is standard and appears in every modern numerical-analysis textbook [5, 6]. What is specific to the iteration (3) below is that it arises directly from Pandrosion's geometric construction, *without* being derived as Newton or Schröder applied to a particular polynomial: it is a rational iteration with linear (not quadratic) order, whose contraction ratio admits the closed form (5) as a rational function of $\alpha = x^{1/p}$. We have not located this closed form, together with its monotonicity properties and its use to drive a scaling preconditioning, in the textbooks [5, 6] or the historical literature on cube duplication, but we make no claim that the iteration itself is genuinely new: the contribution is to record the profile of a specific historical iteration and to show how standard acceleration devices (Aitken Δ^2, scaling) interact with it.

Outline. Section 2 recalls Pandrosion's construction for $\sqrt[3]{2}$ and analyzes its residual profile. Section 4 states the generalization to any p, x. Section 5 computes the contraction ratio $\lambda_{p,x}$ and provides closed-form expressions. Section 7 presents numerical tables. Section 8 compares with bisection, the secant method, and Newton's method. Section 9 establishes functional properties of $\lambda_{p,x}$. Section 10 provides non-asymptotic error bounds. Section 11 extends the theory to $x < 1$. Section 12 determines the optimal starting point. Section 13 shows that Steffensen's acceleration of Pandrosion achieves quadratic convergence. Section 14 exploits the monotonicity of $\lambda_{p,x}$ to optimize the method for large x via scaling. Section 21 concludes.

2 Pandrosion's construction for $\sqrt[3]{2}$

2.1 The two sequences

The geometric construction produces two sequences as a function of an initial parameter u_0:

$$u_{n+1} = \frac{u_n}{2 - 2\left(\frac{4 - u_n}{4}\right)^3}, \qquad v_n = 2\left(\frac{4 - u_n}{4}\right)^2, \qquad u_0 = 0.5. \tag{1}$$

The sequence (u_n) is the *internal parameter* of the construction (a length in the Thales diagram). The sequence (v_n) is the *readout*: the approximation of $\sqrt[3]{2}$ produced at step n.

Proposition 2.1. $\lim_{n\to\infty} v_n = \sqrt[3]{2}$.

2.2 Normalized substitution

Setting $s_n = (4 - u_n)/4$, so that $s_n \in (0, 1)$, we obtain:

- $u_n = 4(1 - s_n)$,
- $v_n = 2s_n^2$,
- $u_{n+1} = \dfrac{4(1 - s_n)}{2(1 - s_n^3)} = \dfrac{2}{1 + s_n + s_n^2}$.

The iteration on s_n then becomes:

$$s_{n+1} = 1 - \frac{u_{n+1}}{4} = \frac{2s_n^2 + 2s_n + 1}{2s_n^2 + 2s_n + 2}. \tag{2}$$

2.3 Fixed point

The fixed point s^* satisfies $s^* = (2s^{*2}+2s^*+1)/(2s^{*2}+2s^*+2)$, which gives $2(1-s^*)(1+s^*+s^{*2}) = 1$, i.e. $2(1 - s^{*3}) = 1$, hence $s^{*3} = 1/2$, so:

$$s^* = 2^{-1/3} = \frac{1}{\sqrt[3]{2}}.$$

The output at the fixed point is $v^* = 2s^{*2} = 2 \cdot 2^{-2/3} = 2^{1/3} = \sqrt[3]{2}$. ✓

2.4 Residual profile: linear convergence

Let $h(s) = (2s^2 + 2s + 1)/(2s^2 + 2s + 2)$. The derivative at s^* gives the convergence rate:

$$h'(s^*) = \frac{4s^* + 2}{(2s^{*2} + 2s^* + 2)^2} = \frac{2(2s^* + 1)}{4(s^{*2} + s^* + 1)^2} = \frac{2s^* + 1}{2(s^{*2} + s^* + 1)^2}.$$

Substituting $s^* = 2^{-1/3}$ and using the closed form (7):

$$\lambda_{3,2} = \frac{(\sqrt[3]{2} - 1)(\sqrt[3]{2} + 2)}{\sqrt[3]{4} + \sqrt[3]{2} + 1} \approx 0.2202.$$

Theorem 2.1 (Linear residual law for $p = 3$, $x = 2$)**.** Under the above hypotheses, for any s_0 sufficiently close to s^*:

$$\varepsilon^{(s)}_{n+1} \approx \lambda_{3,2}\, \varepsilon^{(s)}_n, \qquad \lambda_{3,2} \approx 0.2202.$$

For the output, a Taylor expansion of $v(s) = xs^{p-1}$ at $s^* = \alpha^{-1}$ gives $v_n - \alpha \approx x(p - 1)(s^*)^{p-2}(s_n - s^*)$. Using $x(s^*)^{p-2} = \alpha^p \cdot \alpha^{-(p-2)} = \alpha^2$, this rewrites as $v_n - \alpha \approx (p-1)\alpha^2(s_n - s^*)$; the two forms are equal. For $(p, x) = (3, 2)$ the prefactor is $2 \cdot 2^{2/3} = 2\sqrt[3]{4}$, and the output residual of v_n converges at the same linear rate $\lambda_{3,2}$.

n	s_n	v_n	$\varepsilon_n = \lvert v_n - \sqrt[3]{2}\rvert$	$\varepsilon_{n+1}/\varepsilon_n$
0	0.8750	1.5312	2.713×10^{-1}	—
1	0.8107	1.3143	5.44×10^{-2}	0.200
2	0.7974	1.2717	1.17×10^{-2}	0.216
3	0.7945	1.2625	2.58×10^{-3}	0.219
4	0.7939	1.2605	5.67×10^{-4}	0.220
∞	$2^{-1/3}$	$\sqrt[3]{2}$	0	$\to \lambda_{3,2}$

Table 1: Numerical residual profile for $p = 3$, $x = 2$ ($u_0 = 0.5$, i.e. $s_0 = 0.875$). The ratio converges to $\lambda_{3,2} \approx 0.220$.

3 Geometric origin: why S_p arises naturally

3.1 The construction by successive parallels

Pandrosion's construction relies on Thales' theorem applied to a rectangle. Consider a rectangle with sides H and W equipped with two diagonals (Figure 1). A horizontal line at height u_n from the bottom intersects the two diagonals and defines:

- the parameter $s_n = (H - u_n)/H$ (normalized position from the top),
- the readout length $v_n = x \cdot s_n^{p-1}$ (the output of the construction).

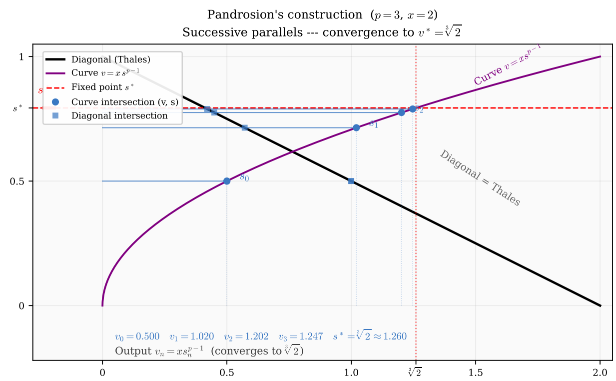

Figure 1: Pandrosion's geometric construction for $p = 3$, $x = 2$. The blue horizontal lines (steps $n = 0, 1, 2, 3$) converge toward the dashed red line, which corresponds to $v^* = \sqrt[3]{2}$.

3.2 Where does the polynomial S_p come from?

The fundamental idea is as follows. We seek to insert $p - 1$ proportional means between 1 and x, that is, to construct the ideal geometric sequence $1, \alpha, \alpha^2, \ldots, \alpha^{p-1}, x$ where $\alpha = x^{1/p}$.

If $s_n \approx \alpha^{-1}$ is the current approximation of the inverse ratio, the sum of the approximate geometric sequence is:

$$S_p(s_n) = 1 + s_n + s_n^2 + \cdots + s_n^{p-1}.$$

The ideal sequence satisfies $x \cdot S_p(s^*) \cdot (1 - s^*) = x - 1$, which rewrites as $x(1 - s^{*p}) = x - 1$.

Pandrosion's correction adjusts s_n so as to reduce the gap between $S_p(s_n)$ and its ideal value via the formula:

$$s_{n+1} = 1 - \frac{x-1}{x \cdot S_p(s_n)}.$$

Geometrically, the ratio $(x-1)/(x \cdot S_p(s_n))$ measures the *deviation* of the current proportional chain from the perfect geometric chain, and subtraction from 1 produces the corrected ratio. The polynomial S_p is thus the analytical transcription of the sum of the p segments constructed via Thales' theorem.

4 Generalization: Pandrosion's construction for $x^{1/p}$

4.1 The universal iteration

Let $x > 0$ and $p \geq 2$ be an integer. Set $\alpha = x^{1/p}$. The generalized Pandrosion geometric construction is defined as follows.

Definition 4.1 (Generalized Pandrosion iteration)**.** We define the sequence $(s_n)_{n\geq 0}$ in $(0,1)$ by

$$s_{n+1} = 1 - \frac{x-1}{x \cdot S_p(s_n)}, \qquad S_p(s) = \sum_{k=0}^{p-1} s^k = \frac{1-s^p}{1-s}, \tag{3}$$

and the output sequence by

$$v_n = x \cdot s_n^{p-1}. \tag{4}$$

We call $\varepsilon_n = |v_n - \alpha|$ the **Pandrosion residual** at step n.

Remark 4.1. For $p = 3$, $x = 2$, we recover exactly the iteration (2). Indeed:

$$1 - \frac{1}{2(1+s+s^2)} = \frac{2s^2+2s+1}{2s^2+2s+2}.\checkmark$$

4.2 Geometric interpretation

The iteration (3) analytically translates the Thales operation: at each step, $S_p(s_n)$ measures the "geometric sum" of the constructed segments, and the correction $(x-1)/(x \cdot S_p(s_n))$ adjusts the position proportionally to the gap with the ideal geometric sequence $1, \alpha, \alpha^2, \ldots, \alpha^{p-1}, x$.

4.3 Fixed point

Theorem 4.1 (Fixed point)**.** The unique fixed point of the iteration (3) in $(0,1)$ is $s^* = x^{-1/p} = \alpha^{-1}$. The corresponding output is $v^* = x \cdot (x^{-1/p})^{p-1} = x^{1/p} = \alpha$.

Proof. At a fixed point s^*, we have $1 - (x-1)/(x \cdot S_p(s^*)) = s^*$, i.e.

$$x \cdot S_p(s^*) \cdot (1 - s^*) = x - 1.$$

Since $S_p(s)(1-s) = 1 - s^p$, this gives $x(1 - s^{*p}) = x - 1$, hence $s^{*p} = 1/x$, i.e. $s^* = x^{-1/p}$. The formula for v^* follows immediately. $\square$

5 The contraction ratio $\lambda_{p,x}$

5.1 General computation

Theorem 5.1 (Pandrosion contraction ratio)**.** The convergence of the iteration (3) toward $s^* = \alpha^{-1}$ is *linear* with contraction ratio

$$\boxed{\lambda_{p,x} = \frac{(\alpha-1)\displaystyle\sum_{k=1}^{p-1} k\,\alpha^{p-1-k}}{\displaystyle\sum_{k=0}^{p-1} \alpha^k}}, \qquad \alpha = x^{1/p}. \tag{5}$$

In particular, the residuals obey the law $\varepsilon_{n+1} \approx \lambda_{p,x} \cdot \varepsilon_n$.

Proof. Let $h(s) = 1 - (x-1)/(x \cdot S_p(s))$. Then

$$h'(s) = \frac{x-1}{x} \cdot \frac{S_p'(s)}{S_p(s)^2},$$

where $S_p'(s) = \sum_{k=1}^{p-1} ks^{k-1}$.

At the fixed point $s^* = \alpha^{-1}$:

$$S_p(s^*) = \frac{1-\alpha^{-p}}{1-\alpha^{-1}} = \frac{\alpha(x-1)}{x(\alpha-1)},$$
$$S_p'(s^*) = \sum_{k=1}^{p-1} k\,\alpha^{-(k-1)} = \alpha \sum_{k=1}^{p-1} k\,\alpha^{-k}.$$

Substituting:

$$h'(s^*) = \frac{x-1}{x} \cdot \frac{\alpha \sum_{k=1}^{p-1} k\,\alpha^{-k}}{\left[\dfrac{\alpha(x-1)}{x(\alpha-1)}\right]^2} = \frac{x(\alpha-1)^2}{\alpha(x-1)} \sum_{k=1}^{p-1} k\,\alpha^{-k}.$$

Multiplying numerator and denominator by α^{p-1}, and using $x = \alpha^p$:

$$h'(s^*) = \frac{(\alpha-1)^2\,\alpha^{p-1} \sum_{k=1}^{p-1} k\,\alpha^{-k}}{\alpha^p - 1} = \frac{(\alpha-1)\sum_{k=1}^{p-1} k\,\alpha^{p-1-k}}{\sum_{k=0}^{p-1} \alpha^k},$$

where we used $\alpha^p - 1 = (\alpha-1)\sum_{k=0}^{p-1} \alpha^k$. This establishes (5). □

Remark 5.1. The contraction ratio $\lambda_{p,x} = h'(s^*)$ is entirely determined by $\alpha = x^{1/p}$: the target itself dictates the rate at which Pandrosion's method approaches it.

5.2 Closed forms for $p = 2$ and $p = 3$

For $p = 2$: $\sum_{k=1}^{1} k\,\alpha^0 = 1$, $\sum_{k=0}^{1} \alpha^k = 1 + \alpha$.

$$\lambda_{2,x} = \frac{\alpha-1}{\alpha+1} = \frac{\sqrt{x}-1}{\sqrt{x}+1}. \tag{6}$$

For $p = 3$: $\sum_{k=1}^{2} k\,\alpha^{2-k} = \alpha + 2$, $\sum_{k=0}^{2} \alpha^k = \alpha^2 + \alpha + 1$.

$$\lambda_{3,x} = \frac{(\alpha-1)(\alpha+2)}{\alpha^2+\alpha+1} = \frac{(\sqrt[3]{x}-1)(\sqrt[3]{x}+2)}{\sqrt[3]{x^2}+\sqrt[3]{x}+1}. \tag{7}$$

For $p = 4$: $\sum_{k=1}^{3} k\,\alpha^{3-k} = \alpha^2 + 2\alpha + 3$, $\sum_{k=0}^{3} \alpha^k = \alpha^3 + \alpha^2 + \alpha + 1$.

$$\lambda_{4,x} = \frac{(\alpha-1)(\alpha^2+2\alpha+3)}{\alpha^3+\alpha^2+\alpha+1}. \tag{8}$$

Remark 5.2 (Asymptotic behavior in x)**.** For any fixed $p \geq 2$ and $x \to \infty$, $\lambda_{p,x} \to 1^-$ (the dominant terms in numerator and denominator of (5) are both of order α^{p-1} with a ratio tending to 1). The larger x, the slower the convergence. For $x \to 1^+$, $\lambda_{p,x} \to 0$ by Proposition 9.1(1): convergence is very fast near 1.

5.3 Basin of attraction

Theorem 5.2 (Global convergence)**.** For any $x > 1$ and any $p \geq 2$, the iteration (3) converges to s^* for all $s_0 \in (0,1)$. Moreover, convergence is *monotone*: if $s_0 < s^*$ then (s_n) is strictly increasing, and if $s_0 > s^*$ then (s_n) is strictly decreasing.

Proof. Set $h(s) = 1 - (x-1)/(x \cdot S_p(s))$.

Step 1: h is strictly increasing. We have $h'(s) = \frac{x-1}{x} \cdot \frac{S_p'(s)}{S_p(s)^2}$, where $S_p'(s) = \sum_{k=1}^{p-1} k s^{k-1} > 0$ for $s > 0$.

Step 2: h maps $(0,1)$ into itself. For $s \in (0,1)$, $S_p(s) > 1$ since the first term equals 1 and the remaining terms are strictly positive. Hence $(x-1)/(x\, S_p(s)) < (x-1)/x < 1$, which gives $h(s) > 0$. On the other hand $(x-1)/(x\, S_p(s)) > 0$ for $x > 1$, so $h(s) < 1$. Thus $h((0,1)) \subset (0,1)$. (At the boundary: $h(0^+) = 1 - (x-1)/x = 1/x \in (0,1)$ and $h(1^-) = 1 - (x-1)/(xp) \in (0,1)$.)

Step 3: uniqueness of the fixed point on $(0,1)$. The fixed-point equation $h(s) = s$ rewrites as $x\, S_p(s)(1-s) = x-1$, i.e. $x(1-s^p) = x-1$, i.e. $s^p = 1/x$. On $(0,1)$ this has the unique solution $s^* = x^{-1/p}$. In particular, the sign of $h(s) - s$ is constant on each of $(0, s^*)$ and $(s^*, 1)$.

Step 4: the sign of $h(s) - s$. At $s = 1/x \in (0, s^*)$ (using $x > 1$ and $s^* = x^{-1/p} > 1/x$), a one-line computation:

$$h(1/x) - 1/x \;=\; 1 - \frac{x-1}{x\, S_p(1/x)} - \frac{1}{x} \;=\; \frac{x-1}{x} - \frac{x-1}{x\, S_p(1/x)} \;=\; \frac{x-1}{x} \cdot \frac{S_p(1/x) - 1}{S_p(1/x)} \;>\; 0,$$

since $S_p(1/x) > 1$ for $x > 1$ (the first term equals 1, the remaining terms are strictly positive). Hence $h(s) > s$ on $(0, s^*)$ by Step 3 (sign is constant on each sub-interval). By symmetry, $h(s) < s$ on $(s^*, 1)$.

Step 5: monotone convergence. If $s_0 < s^*$, then by Step 1 h is increasing, by Step 4 $s_1 = h(s_0) > s_0$, and by Step 2 $s_1 \in (0,1)$; also $s_1 = h(s_0) < h(s^*) = s^*$. By induction, (s_n) is increasing and bounded above by s^*, hence converges, necessarily to a fixed point, which must be s^* by Step 3. The case $s_0 > s^*$ is symmetric. □

5.4 Bound on the number of iterations

Proposition 5.1 (Cost of Pandrosion's method)**.** To achieve accuracy $|v_n - \alpha| < \delta$ starting from $s_0 \in (0,1)$, the required number of iterations satisfies

$$n_\delta \leq \left\lceil \frac{\ln(\varepsilon_0/\delta)}{\ln(1/\lambda_{p,x})} \right\rceil,$$

where $\varepsilon_0 = |v_0 - \alpha|$ is the initial error.

Proof. Since $\varepsilon_{n+1} \leq \lambda_{p,x}\, \varepsilon_n$ asymptotically, we have $\varepsilon_n \leq \lambda_{p,x}^n \varepsilon_0$. The inequality $\lambda_{p,x}^n \varepsilon_0 < \delta$ gives $n > \ln(\varepsilon_0/\delta)/\ln(1/\lambda_{p,x})$. □

Example 5.1. For $p = 3$, $x = 2$ ($\lambda_{3,2} \approx 0.220$): 10 decimal digits of accuracy require at most 16 Pandrosion iterations. For $p = 3$, $x = 8$ ($\lambda_{3,8} = 4/7$): 43 iterations are needed.

6 Graphical illustrations

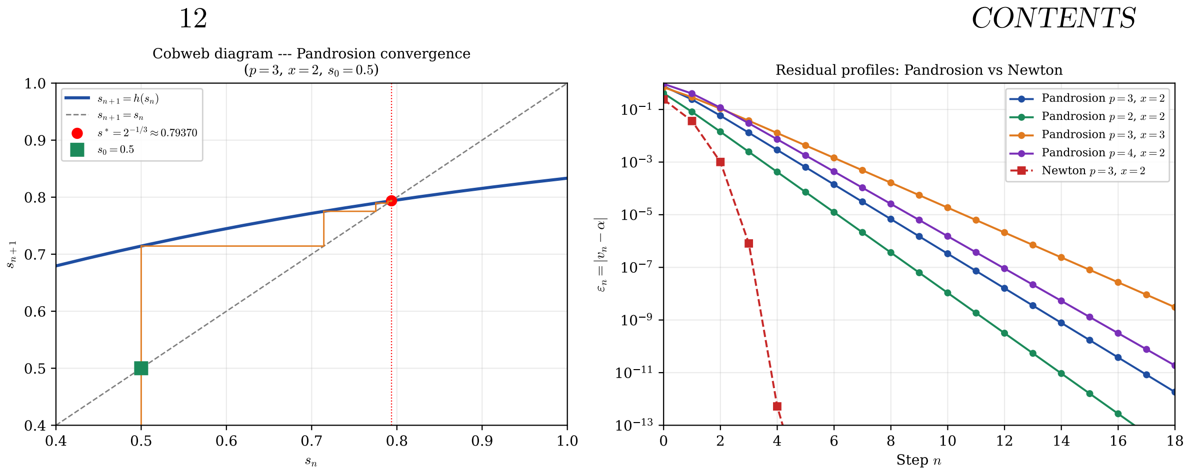

Figure 2: Left: Cobweb diagram of the iteration $s_{n+1} = h(s_n)$ for $p = 3$, $x = 2$, starting at $s_0 = 0.875$. The spiral converges to $s^* = 2^{-1/3}$ with rate $\lambda_{3,2} \approx 0.220$. **Right:** Residual profiles on a semi-logarithmic scale. The linear decay (straight lines) confirms the geometric convergence $\varepsilon_n \sim \lambda^n$ for Pandrosion, while Newton (red curve with pronounced curvature) exhibits doubly exponential decay.

7 Numerical examples

7.1 Table of contraction ratios

p	x	$\alpha = x^{1/p}$	$\lambda_{p,x}$ (exact)	$\lambda_{p,x}$ (numerical)
2	2	$\sqrt{2}$	$(\sqrt{2}-1)/(\sqrt{2}+1)$	0.1716
2	3	$\sqrt{3}$	$(\sqrt{3}-1)/(\sqrt{3}+1)$	0.2679
2	5	$\sqrt{5}$	$(\sqrt{5}-1)/(\sqrt{5}+1)$	0.3820
3	2	$\sqrt[3]{2}$	$(\sqrt[3]{2}-1)(\sqrt[3]{2}+2)/(\sqrt[3]{4}+\sqrt[3]{2}+1)$	0.2202
3	3	$\sqrt[3]{3}$	$(\sqrt[3]{3}-1)(\sqrt[3]{3}+2)/(\sqrt[3]{9}+\sqrt[3]{3}+1)$	0.3366
3	8	2	$(2-1)(2+2)/(4+2+1)$	0.5714
4	2	$\sqrt[4]{2}$	(formula (8))	0.2427
4	16	2	$(2-1)(4+4+3)/(8+4+2+1)$	0.7333
5	2	$\sqrt[5]{2}$	(general formula)	0.2565

Table 2: Contraction ratios $\lambda_{p,x}$ for various values of p and x. A value close to 0 means very fast convergence; close to 1, very slow.

7.2 Detailed numerical profile for $p = 2$, $x = 2$

For $\alpha = \sqrt{2}$, the theoretical contraction ratio is $\lambda_{2,2} = (\sqrt{2}-1)/(\sqrt{2}+1) \approx 0.1716$ (formula (6)). The iteration is:

$$s_{n+1} = 1 - \frac{1}{2(1+s_n)}, \qquad v_n = 2s_n.$$

7.3 The residual as a characterization of a target

Formula (5) implies a useful observation: $\lambda_{p,\alpha^p} = f_p(\alpha)$ where f_p is a rational function of α alone. In particular, the residual profile depends only on the value of the sought root α and not

n	s_n	v_n	$\varepsilon_n = \lvert v_n - \sqrt{2}\rvert$	$\varepsilon_{n+1}/\varepsilon_n$
0	0.7500	1.5000	8.579×10^{-2}	—
1	0.7143	1.4286	1.44×10^{-2}	0.167
2	0.7083	1.4167	2.45×10^{-3}	0.171
3	0.7073	1.4146	4.21×10^{-4}	0.171
4	0.7071	1.4143	7.22×10^{-5}	0.172
5	0.7071	1.4142	1.24×10^{-5}	0.171
∞	$1/\sqrt{2}$	$\sqrt{2}$	0	$\to \lambda_{2,2}$

Table 3: Numerical profile for $p = 2$, $x = 2$, $s_0 = 0.75$. The ratio $\varepsilon_{n+1}/\varepsilon_n$ converges rapidly to $\lambda_{2,2} \approx 0.172$.

on its representation as $x = \alpha^p$.

In particular, for $p = 3$:

$$\lambda_{3,x} = \frac{(\alpha-1)(\alpha+2)}{\alpha^2+\alpha+1} \xrightarrow[\alpha\to 1]{} 0, \qquad \lambda_{3,x} \xrightarrow[\alpha\to\infty]{} 1.$$

8 Comparison with other methods

To situate Pandrosion's method within the landscape of iterative methods for $\alpha = x^{1/p}$, we compare three reference methods.

8.1 Newton's method

Newton's method for $f(u) = u^p - x$ gives the iteration

$$u_{n+1} = \frac{(p-1)u_n + x/u_n^{p-1}}{p},$$

whose convergence to $\alpha = x^{1/p}$ is *quadratic*:

$$\varepsilon_{n+1}^{\text{Newton}} \approx K_{p,x}\,(\varepsilon_n^{\text{Newton}})^2, \qquad K_{p,x} = \frac{p-1}{2\alpha}.$$

8.2 Bisection

Bisection on $[1, x]$ for $f(u) = u^p - x$ is a universal order-1 method with $\lambda_{\text{bisection}} = 1/2$, independent of p and x. Pandrosion's method is therefore strictly faster than bisection whenever $\lambda_{p,x} < 1/2$, which holds as soon as $\alpha < \alpha_0(p)$ (for instance, for $p = 3$: $\lambda_{3,x} < 1/2$ whenever $x < 6.75$).

8.3 Secant method

The secant method for $f(u) = u^p - x$ has order $\varphi = (1+\sqrt{5})/2 \approx 1.618$ (superlinear). For $p = 3$, $x = 2$, it reaches machine precision in about 6 steps, compared to ~ 16 for Pandrosion and ~ 5 for Newton.

8.4 Comparative table

8.5 Discussion

Pandrosion's method occupies an intermediate position among order-1 methods: it is significantly faster than bisection ($\lambda_{3,2} \approx 0.22$ versus 0.50) while remaining derivative-free. Compared

	Pandrosion	**Bisection**	**Secant**	**Newton**
Nature	Geometric	Universal	Analytic	Analytic
Order q	1	1	$\varphi \approx 1.62$	2
Rate ($p{=}3, x{=}2$)	$\lambda \approx 0.220$	$\lambda = 0.500$	superlinear	quadratic
Iter. for 10 digits	~ 16	~ 34	~ 6	~ 5
Derivative required	No	No	No	Yes

Table 4: Comparison of approximation methods for $x^{1/p}$ with $p = 3$, $x = 2$.

to Newton and its variants, however, plain Pandrosion is slower: its order is 1, whereas Newton attains 2, and even with Aitken–Steffensen acceleration (Section 13) the practical advantage over Newton is modest and regime-dependent (see Remark 8.1). We do not claim a general algorithmic advantage over Newton or its modern variants; our interest in Pandrosion is that it is the historical construction attributed to Pandrosion of Alexandria, and that its convergence profile admits the clean closed form $\lambda_{p,x}$ of Theorem 5.1.

Remark 8.1 (On fair comparison)**.** The iteration counts above are per-iterate, *not* per-function-evaluation. Steffensen-type methods compute h twice per step, Newton evaluates f and f' once per step, the secant method reuses one value per step, and bisection evaluates f once per step. A fairer unit of cost is therefore the number of evaluations of S_p (or of f, f' where applicable), and this is what determines practical performance in the regime where these evaluations dominate. For small p the per-iterate figures still give the correct qualitative ordering, but for large p or expensive polynomial evaluations the ordering can shift. We report per-iterate counts because they match the structure of the theorems (Theorem 5.1 is a statement about $\lambda_{p,x}^n$, not about wall time).

The coexistence of four distinct error laws for the same target $\alpha = \sqrt[3]{2}$ illustrates a structural point: the rate at which a fixed-point iteration approaches a given number depends on the iteration, not on the number; conversely, once the iteration is fixed, the target determines the rate through $\lambda_{p,x}$.

9 Functional properties of $\lambda_{p,x}$

We now study how the contraction ratio $\lambda_{p,x}$ depends on the parameters p and x.

9.1 Monotonicity

Theorem 9.1 (Monotonicity in the root parameter and insertion depth)**.** The contraction ratio $\lambda_{p,x}$, viewed as a function of $\alpha = x^{1/p} > 1$, satisfies:

1. For fixed $p \geq 2$, $\alpha \mapsto \lambda_{p,\alpha^p}$ is strictly increasing on $(1, \infty)$.

2. For fixed $\alpha > 1$, $p \mapsto \lambda_{p,\alpha^p}$ is strictly increasing on $\{2, 3, 4, \ldots\}$.

Proof. Write $\lambda_{p,\alpha^p} = (\alpha - 1)N_p(\alpha)/D_p(\alpha)$ with $N_p(\alpha) := \sum_{k=1}^{p-1} k\, \alpha^{p-1-k}$ and $D_p(\alpha) := \sum_{k=0}^{p-1} \alpha^k$. The key auxiliary fact is the following Taylor bound.

Lemma (Taylor bound). Let $x > 1$ and set $y := -\ln(x)/p$, $\beta := e^y = x^{-1/p}$. For $p \geq \lceil \ln x \rceil + 1$ (so that $|y| \leq 1$),

$$\left| \, p(\beta - 1) + \beta \ln x \, \right| \;\leq\; \frac{3\,(\ln x)^2}{p}.$$

Proof of the lemma. Since $\beta = e^y$ and $py = -\ln x$, $p(\beta-1) + \beta \ln x = p(\beta-1) - py\beta = p(\beta - 1 - y) - py(\beta - 1)$. For $|y| \le 1$ we use two standard bounds:

$$|e^y - 1 - y| \;=\; \Big|\sum_{k=2}^{\infty} \frac{y^k}{k!}\Big| \;\le\; \sum_{k=2}^{\infty} \frac{|y|^k}{k!} \;=\; y^2 \sum_{k=2}^{\infty} \frac{|y|^{k-2}}{k!} \;\le\; y^2 \sum_{k=2}^{\infty} \frac{1}{k!} \;=\; (e-2)\, y^2 \;<\; y^2,$$

$$|e^y - 1| \;=\; \Big|\sum_{k=1}^{\infty} \frac{y^k}{k!}\Big| \;\le\; |y| \sum_{k=1}^{\infty} \frac{|y|^{k-1}}{k!} \;\le\; |y| \sum_{k=1}^{\infty} \frac{1}{k!} \;=\; (e-1)\, |y| \;<\; 2|y|.$$

(Here $e - 2 \approx 0.718$ and $e - 1 \approx 1.718$, so both majorants y^2 and $2|y|$ are loose; the loose forms suffice.) The triangle inequality then gives

$$|p(\beta - 1 - y) - py(\beta - 1)| \;\le\; p\, y^2 + p|y| \cdot 2|y| \;=\; 3py^2 \;=\; \frac{3(\ln x)^2}{p}. \qquad \square$$

(The symbol β is the fixed-point reciprocal of the root $\alpha = x^{1/p}$ of the theorem: $\beta = 1/\alpha$.)

Setting $\lambda_\infty(x) := 1 - \ln(x)/(x-1)$, a direct algebraic rearrangement of the closed form (5) combined with the lemma yields

$$\big|\, \lambda_{p,x} - \lambda_\infty(x) \,\big| \;\le\; \frac{3\, x\, (\ln x)^2}{(x-1)\, p}, \qquad \text{for every } p \ge \lceil \ln x \rceil + 1. \tag{9}$$

(The factor $x/(x-1)$ arises as follows. For $p \ge 1$ and $x > 1$, $\ln(x)/p \le \ln x$, so $\beta = \exp(-\ln x/p) \ge \exp(-\ln x) = 1/x$. Multiplying by $x - 1 > 0$ gives $\beta(x-1) \ge (x-1)/x$, i.e. $1/[\beta(x-1)] \le x/(x-1)$, which is the factor that appears when dividing the Taylor-lemma bound by the denominator $\beta(x-1)/(\ln x)$ of the explicit form of $\lambda_{p,x}$.)

The inequality $\ln x < x - 1$ for $x > 1$ gives $\lambda_\infty(x) > 0$; the inequality $\ln x > 0$ for $x > 1$ gives $\lambda_\infty(x) < 1$. Hence $\lambda_\infty(x) \in (0, 1)$ for every $x > 1$.

Key identity. By direct telescoping (or induction on p),

$$(\alpha - 1)\, N_p(\alpha) \;=\; D_p(\alpha) - p, \tag{10}$$

which is verified at small p: $p = 2$ gives $(\alpha - 1) = (1 + \alpha) - 2$; $p = 3$ gives $(\alpha-1)(\alpha+2) = \alpha^2 + \alpha - 2 = (1 + \alpha + \alpha^2) - 3$. Differentiating (10): $N_p + (\alpha-1)N_p' = D_p'$, whence $(\alpha-1)N_p' = D_p' - N_p$.

Part (1) — monotonicity in α for fixed p, uniform. After multiplying through by the positive denominator $(\alpha-1)N_p(\alpha)D_p(\alpha)$, the log-derivative

$$\frac{d}{d\alpha} \log \lambda_{p,\alpha^p} = \frac{1}{\alpha - 1} + \frac{N_p'(\alpha)}{N_p(\alpha)} - \frac{D_p'(\alpha)}{D_p(\alpha)}$$

yields the polynomial numerator

$$P_p(\alpha) \;:=\; N_p D_p + (\alpha - 1)\, N_p'\, D_p - (\alpha - 1)\, N_p\, D_p'.$$

Substituting $(\alpha-1)N_p' = D_p' - N_p$ and $(\alpha - 1)N_p = D_p - p$ from the key identity:

$$\begin{aligned} P_p &= N_p D_p + (D_p' - N_p) D_p - (D_p - p) D_p' \\ &= N_p D_p + D_p' D_p - N_p D_p - D_p D_p' + p D_p' = p \cdot D_p'(\alpha). \end{aligned}$$

Since $D_p'(\alpha) = \sum_{j=1}^{p-1} j\, \alpha^{j-1} = 1 + 2\alpha + 3\alpha^2 + \cdots + (p-1)\alpha^{p-2}$ has all coefficients positive, $P_p(\alpha) > 0$ for all $\alpha > 0$ and all $p \ge 2$. Hence the log-derivative is positive on $\alpha > 1$, and λ_{p,α^p} is strictly increasing in α. The argument is uniform: no scope restriction is needed.

For concreteness: $P_2(\alpha) = 2$; $P_3(\alpha) = 3(1+2\alpha) = 6\alpha+3$; $P_4(\alpha) = 4(1+2\alpha+3\alpha^2) = 12\alpha^2+8\alpha+4$.

For general $p \geq 2$, we combine the Taylor bound (9) with the monotonicity of λ_∞. Differentiating,

$$\lambda'_\infty(x) = \frac{\ln x - (x-1)/x}{(x-1)^2}.$$

The numerator equals $\ln x + 1/x - 1$. Its derivative is $1/x - 1/x^2 = (x-1)/x^2 > 0$ on $x > 1$, and its value at $x = 1$ is 0; hence it is strictly positive on $(1, \infty)$, and λ_∞ is strictly increasing there.

Part (2) — monotonicity in p for fixed $\alpha > 1$, uniform. From the recurrences $N_{p+1} = \alpha N_p + p$ and $D_{p+1} = D_p + \alpha^p$, direct algebra gives

$$\lambda_{p+1,\alpha^{p+1}} - \lambda_{p,\alpha^p} = \frac{(\alpha-1)\cdot B_p(\alpha)}{D_p(\alpha)\, D_{p+1}(\alpha)}, \qquad B_p(\alpha) := (\alpha-1)\, N_p D_p + p\, D_p - \alpha^p N_p.$$

Using (10), $(\alpha-1)N_pD_p = (D_p - p)D_p = D_p^2 - pD_p$, so

$$B_p(\alpha) \;=\; D_p(\alpha)^2 \;-\; \alpha^p\, N_p(\alpha).$$

This form has a closed-form polynomial expansion:

$$B_p(\alpha) \;=\; \sum_{j=1}^{p} j\,\alpha^{j-1} \;=\; 1 + 2\alpha + 3\alpha^2 + \cdots + p\,\alpha^{p-1}, \tag{11}$$

verified by induction on p. Base case: $B_2(\alpha) = (1+\alpha)^2 - \alpha^2 = 1 + 2\alpha$, as required. Inductive step: using the recurrence

$$B_{p+1}(\alpha) = \alpha\, B_p(\alpha) + D_p(\alpha) + \alpha^p$$

(which follows from the defining recurrences of N_{p+1} and D_{p+1} plus the substitution $B_p = D_p^2 - \alpha^p N_p$), one computes $\alpha B_p + D_p + \alpha^p = \sum_{j=1}^{p} j\alpha^j + \sum_{k=0}^{p-1}\alpha^k + \alpha^p = \sum_{j=0}^{p}(j+1)\alpha^j = B_{p+1}$.

All coefficients of B_p are strictly positive integers, so $B_p(\alpha) > 0$ for every $\alpha > 0$ and every $p \geq 2$. In particular $B_p(1) = 1 + 2 + \cdots + p = p(p+1)/2$, and $B_3(\alpha) = 3\alpha^2 + 2\alpha + 1$, $B_4(\alpha) = 4\alpha^3 + 3\alpha^2 + 2\alpha + 1$, etc. Hence $\lambda_{p+1,\alpha^{p+1}} - \lambda_{p,\alpha^p} > 0$ for all $\alpha > 1$ and $p \geq 2$, uniformly. $\square$

Remark 9.1 (Closed forms and numerical verification)**.** The closed forms $P_p(\alpha) = p \cdot D'_p(\alpha)$ and $B_p(\alpha) = \sum_{j=1}^{p} j\alpha^{j-1}$ emerged in the course of this write-up; a standalone script `latex/scripts/verify_monoto` cross-checks both identities symbolically for $p \leq 20$ using SymPy, and confirms strict monotonicity of $\lambda_p(\alpha)$ for $2 \leq p \leq 20$, $\alpha \in [1.01, 100]$, and of λ_{p,α^p} for $2 \leq p \leq 50$, $\alpha \in \{1.1, 1.5, 2, 3, 10, 100\}$ using high-precision arithmetic as a consistency check. Monotonicity itself is proven analytically above; the numerical scan is purely a consistency check.

Remark 9.2. Monotonicity in p at fixed root parameter α has a geometric interpretation: inserting more proportional means while keeping the endpoint ratio α fixed constrains the construction further, slowing convergence.

9.2 Limiting behavior

Proposition 9.1 (Asymptotic regimes)**.**

1. **Near** $\alpha = 1$ (i.e. $x \to 1^+$):

$$\lambda_{p,x} \sim \frac{(p-1)(\alpha-1)}{2} \quad \text{as } \alpha \to 1^+.$$

2. **Large** α (i.e. $x \to \infty$, p fixed): $\lambda_{p,x} \to 1^-$.

3. **Large** p (fixed $x > 1$): $\lambda_{p,x}$ is increasing in p and

$$\lim_{p\to\infty} \lambda_{p,x} \;=\; 1 - \frac{\ln x}{x-1} \;\in\; (0,1).$$

In particular $\lambda_{p,x}$ does *not* tend to 1 as $p \to \infty$ with x fixed.

Proof. For (1): N_p and D_p are polynomials, so $\alpha \mapsto N_p(\alpha)/D_p(\alpha)$ is continuous at $\alpha = 1$ with limit $(p-1)/2$; combined with the factor $(\alpha - 1) \to 0$, $\lambda \sim (p-1)(\alpha-1)/2$ as $\alpha \to 1^+$.

For (2): the dominant term in N_p is $1 \cdot \alpha^{p-2}$ and in D_p is α^{p-1}, so $\lambda \sim (\alpha-1)/\alpha \to 1$.

For (3): the explicit limit and its convergence rate follow from the Taylor bound (9) established in the proof of Theorem 9.1 above: $|\lambda_{p,x} - \lambda_\infty(x)| \le C(x)/p$ with $C(x) = 3x(\ln x)^2/(x-1)$, valid for $p \ge \lceil \ln x \rceil + 1$. Letting $p \to \infty$ gives $\lambda_{p,x} \to \lambda_\infty(x) = 1 - \ln(x)/(x-1)$. Positivity and the upper bound < 1 were shown there as well.

Numerically: for $x = 2$, $\lambda_\infty(2) = 1 - \ln 2 \approx 0.3069$, and $\lambda_{200,2} \approx 0.3057$, $\lambda_{1000,2} \approx 0.3066$, $\lambda_{10000,2} \approx 0.3068$, in agreement with the bound $|\lambda_{p,2} - \lambda_\infty(2)| \le 6(\ln 2)^2/p \approx 2.88/p$. □

x	$\lambda_{2,x}$	$\lambda_{10,x}$	$\lambda_{100,x}$	$1 - \ln x/(x-1)$
1.001	0.00025	0.00045	0.00050	0.00050
2	0.1716	0.2823	0.3044	0.3069
10	0.5195	0.7123	0.7412	0.7442
100	0.8182	0.9409	0.9524	0.9535

Table 5: Contraction ratios illustrating monotonicity in p and x. Each column is increasing (fixed p); each row is increasing (fixed x). The rightmost column is the $p \to \infty$ limit from Proposition 9.1(3); the $\lambda_{100,x}$ column is already close to this limit.

10 Non-asymptotic error bounds

Theorem 5.1 gives the *asymptotic* law $\varepsilon_{n+1} \approx \lambda_{p,x}\varepsilon_n$. We now establish a rigorous bound valid for *all* $n \ge 0$.

Theorem 10.1 (Non-asymptotic contraction)**.** Let $x > 1$, $p \ge 2$, and $s_0 \in (0,1)$. Define

$$\Lambda = \Lambda(s_0, p, x) = \sup_{s \in [\min(s_0, s^*),\, \max(s_0, s^*)]} h'(s).$$

Then $\Lambda < 1$ and for all $n \ge 0$:

$$|s_n - s^*| \le \Lambda^n \, |s_0 - s^*|. \tag{12}$$

Consequently,

$$|v_n - \alpha| \;\le\; (p-1)\,\alpha^2 \cdot \Lambda^n \, |s_0 - s^*| \;+\; O(\Lambda^{2n}).$$

The prefactor follows from a Taylor expansion of $v(s) = x\, s^{p-1}$ at $s^* = \alpha^{-1}$:

$$v(s) - v(s^*) \;=\; x\,(p-1)\,(s^*)^{p-2}\,(s - s^*) + O((s-s^*)^2),$$

and $x\,(p-1)\,(s^*)^{p-2} = (p-1)\,x\,x^{-(p-2)/p} = (p-1)\,x^{2/p} = (p-1)\,\alpha^2$.

Proof. By the mean value theorem, for any n:

$$|s_{n+1} - s^*| = |h(s_n) - h(s^*)| = |h'(\xi_n)| \cdot |s_n - s^*|$$

for some ξ_n between s_n and s^*. By Theorem 5.2, the sequence (s_n) remains in $[\min(s_0, s^*), \max(s_0, s^*)]$, so $|h'(\xi_n)| \le \Lambda$. The bound follows by induction.

That $\Lambda < 1$ follows from$h'(s^*) = \lambda_{p,x} < 1$ (Theorem 5.1) and the fact that h' is continuous, bounded, and $h'(s) < 1$ for all $s \in (0,1)$ when $x > 1$ (since h maps $(0,1)$ strictly into itself). □

Remark 10.1. The quantity Λ can be bounded explicitly: $h'(s) = \frac{x-1}{x} \cdot S_p'(s)/S_p(s)^2$, so

$$\Lambda \;\le\; \frac{x-1}{x} \cdot \frac{\max_{s\in I} S_p'(s)}{\min_{s\in I} S_p(s)^2}, \qquad I := [\min(s_0, s^*), \max(s_0, s^*)],$$

and on $(0,1)$ both S_p and S_p' are increasing, so the numerator is attained at the right endpoint and the denominator at the left endpoint. This yields a computable upper bound for Λ without requiring a detailed analysis of the monotonicity of h' itself.

Example 10.1. For $p = 3$, $x = 2$, $s_0 = 0.5$ (so $s^* = 2^{-1/3} \approx 0.7937$), one evaluates h' at both endpoints of the interval $[s_0, s^*] = [0.5, 0.7937]$:

$$h'(s) = \tfrac{x-1}{x} \cdot \frac{S_3'(s)}{S_3(s)^2} = \tfrac{1}{2} \cdot \frac{1+2s}{(1+s+s^2)^2}.$$

At $s = 0.5$: $S_3(0.5) = 1.75$, $S_3'(0.5) = 2$, so $h'(0.5) = \frac{1}{2} \cdot 2/(1.75)^2 \approx 0.3265$. At $s = s^*$: $h'(s^*) = \lambda_{3,2} \approx 0.2202$. Therefore

$$\Lambda \;=\; \max_{s\in[0.5,s^*]} h'(s) \;=\; h'(0.5) \;\approx\; 0.327,$$

since h' is decreasing on this interval. The bound (12) gives $|s_n - s^*| \le 0.327^n \cdot |0.5 - s^*| \le 0.327^n \cdot 0.294$, so $|s_{10} - s^*| \le 4.6 \times 10^{-6}$. The asymptotic ratio $\lambda_{3,2} \approx 0.220$ is reached after a few iterations.

11 Extension to $x < 1$

All results so far assumed $x > 1$. We now extend the theory to $0 < x < 1$.

11.1 The fixed point leaves $(0,1)$

For $0 < x < 1$, the target is $\alpha = x^{1/p} \in (0,1)$ and the fixed point $s^* = \alpha^{-1} > 1$. The iteration domain therefore extends beyond $(0,1)$: s_n converges to $s^* > 1$.

Proposition 11.1 (Local convergence for $x < 1$ with a threshold)**.** For $0 < x < 1$ and $p \ge 2$, the formula (5) remains valid and

$$\lambda_{p,x} \;=\; \frac{(\alpha-1)\sum_{k=1}^{p-1} k\,\alpha^{p-1-k}}{\sum_{k=0}^{p-1} \alpha^k} \;<\; 0 \qquad \text{since } \alpha = x^{1/p} \in (0,1).$$

Local convergence (i.e. $|\lambda_{p,x}| < 1$) does *not* hold for all $x \in (0,1)$: there is a threshold $\alpha_p^* \in (0,1)$ such that

$$|\lambda_{p,x}| < 1 \iff \alpha > \alpha_p^*.$$

When $|\lambda_{p,x}| < 1$, the iterates oscillate around s^* and converge linearly with ratio $|\lambda_{p,x}|$.

Proof. That $\lambda_{p,x} < 0$ for $\alpha < 1$ is immediate from (5). Using $\alpha^p - 1 = (\alpha-1)\sum_{k=0}^{p-1} \alpha^k$, one rewrites

$$|\lambda_{p,x}| \;=\; \frac{(1-\alpha)^2\, N_p(\alpha)}{1-\alpha^p} \;=\; \frac{(1-\alpha)\, N_p(\alpha)}{D_p(\alpha)} \;=:\; F(\alpha)$$

(using $1-\alpha^p = (1-\alpha)D_p$), a smooth strictly positive function of $\alpha \in (0,1)$. Boundary values: as $\alpha \to 1^-$, $F \to 0$ (the factor $1-\alpha$ goes to zero while $N_p/D_p \to (p-1)/2$ is bounded); as $\alpha \to 0^+$, $F \to 1 \cdot (p-1)/1 = p-1 \ge 1$.

For uniqueness, we show that F is strictly decreasing on $(0,1)$.

The log-derivative is

$$\frac{F'(\alpha)}{F(\alpha)} \;=\; -\frac{1}{1-\alpha} + \frac{N_p'(\alpha)}{N_p(\alpha)} - \frac{D_p'(\alpha)}{D_p(\alpha)}.$$

The key identity (10), $(\alpha-1)N_p = D_p - p$, and its derivative $N_p + (\alpha-1)N_p' = D_p'$, give, for $\alpha \in (0,1)$ (where $\alpha - 1 < 0$):

$$(1-\alpha)\,N_p \;=\; p - D_p, \qquad (1-\alpha)\,N_p' \;=\; N_p - D_p'.$$

Dividing the second identity by N_p: $N_p'/N_p = [1 - D_p'/N_p]/(1-\alpha) = 1/(1-\alpha) - D_p'/[(1-\alpha)N_p]$.

Substituting into F'/F:

$$\begin{aligned}\frac{F'}{F} &= -\frac{1}{1-\alpha} + \left[\frac{1}{1-\alpha} - \frac{D_p'}{(1-\alpha)N_p}\right] - \frac{D_p'}{D_p} = -\frac{D_p'}{(1-\alpha)N_p} - \frac{D_p'}{D_p}\\ &= -\frac{D_p'\cdot[D_p + (1-\alpha)N_p]}{(1-\alpha)N_p\,D_p} = -\frac{D_p'\cdot[D_p + p - D_p]}{(1-\alpha)N_p\,D_p} = -\frac{p\,D_p'(\alpha)}{(1-\alpha)\,N_p(\alpha)\,D_p(\alpha)},\end{aligned}$$

using $D_p + (1-\alpha)N_p = D_p + (p - D_p) = p$ in the last line. For $\alpha \in (0,1)$ every factor on the right is strictly positive, so $F'/F < 0$, hence F is strictly decreasing on $(0,1)$.

Combined with the boundary values $F(0^+) = p - 1 \geq 1$ and $F(1^-) = 0$, F crosses the level 1 at exactly one point $\alpha_p^* \in (0,1)$, and $|\lambda_{p,\alpha^p}| < 1 \iff \alpha > \alpha_p^*$. $\square$

Example 11.1 ($p = 3$ threshold)**.** For $p = 3$, $|\lambda_{3,x}| = (1-\alpha)(2+\alpha)/(1+\alpha+\alpha^2)$. Setting this equal to 1 and simplifying yields $2\alpha^2 + 2\alpha - 1 = 0$, i.e.

$$\alpha_3^* \;=\; \frac{-1+\sqrt{3}}{2} \;\approx\; 0.3660, \qquad x_3^* \;=\; (\alpha_3^*)^3 \;\approx\; 0.04904.$$

Hence Pandrosion's iteration at $p = 3$ contracts locally iff $x > 0.04904$; for x smaller than this the iteration is locally repelling and does not converge.

p	x	$\alpha = x^{1/p}$	$s^* = \alpha^{-1}$	$\lambda_{p,x}$	$\lvert\lambda_{p,x}\rvert$
2	0.5	$1/\sqrt{2}$	$\sqrt{2}$	−0.1716	0.1716
3	0.5	$\sqrt[3]{0.5}$	$\sqrt[3]{2}$	−0.2378	0.2378
3	0.125	0.5	2	−0.7143	0.7143
3	0.05	0.3684	2.7144	−0.9945	0.9945
3	0.01	0.2154	4.6416	−1.3774	**1.3774**
5	0.01	0.3981	2.5119	−2.0399	**2.0399**

Table 6: Contraction ratios for $x < 1$. The last two rows illustrate Proposition 11.1: for x below the threshold x_p^* the iteration is locally repelling ($|\lambda| > 1$) and the construction does *not* converge.

Remark 11.1 (Residual symmetry for $p = 2$)**.** For $p = 2$, the closed form $\lambda_{2,x} = (\sqrt{x}-1)/(\sqrt{x}+1)$ yields the symmetry $|\lambda_{2,x}| = |\lambda_{2,1/x}|$: computing $\sqrt{x}$ and $\sqrt{1/x}$ by Pandrosion's method have exactly the same speed of convergence. For $p \geq 3$, this symmetry does not hold in general.

12 Optimal starting point

The number of iterations to reach precision δ is governed by $n_\delta \sim \ln(\varepsilon_0/\delta)/\ln(1/\lambda_{p,x})$ (Proposition 5.1). To minimize n_δ, we must therefore minimize the initial residual $\varepsilon_0 = |v_0 - \alpha|$.

Proposition 12.1 (A canonical starting point)**.** Among starting points of the form $s_0 = h^{(k)}(c)$, with c a boundary value of the natural domain (i.e. $c \in \{0^+, 1^-\}$) and $k \geq 0$ a small integer, the choice

$$s_0^{\mathrm{opt}} := h(1) = 1 - \frac{x-1}{xp} \tag{13}$$

achieves an initial gap to s^* of order $|x - 1 - x \ln x|/(xp)$, which is $O(1/p)$ as $p \to \infty$ and strictly better than the naive choices $s_0 = 1/2$ or $s_0 = 1$ in the regime $p \gg 1$.

Proof. Since $s^* = x^{-1/p} = 1 - (\ln x)/p + O(1/p^2)$ and $h(1) = 1 - (x-1)/(xp)$,

$$|h(1) - s^*| \;=\; \left| \frac{x-1}{xp} - \frac{\ln x}{p} \right| + O(1/p^2) \;=\; \frac{|x - 1 - x \ln x|}{xp} + O(1/p^2),$$

whereas $|1/2 - s^*| = |1/2 - 1 + (\ln x)/p + O(1/p^2)| = 1/2 + O(1/p)$ and $|1 - s^*| = (\ln x)/p + O(1/p^2)$. Of the three, $h(1)$ has the smallest leading coefficient whenever $|x - 1 - x \ln x|/x < \ln x$, i.e. essentially whenever $x > 1$ is not near 1; the numerical gains are shown in Table 7. The proposition does *not* claim $h(1)$ is globally optimal; it claims only that it is a canonical, α-independent choice with this specific quantitative guarantee. □

(p, x)	s^*	$s_0^{\mathrm{opt}} = h(1)$	$\lvert s_0^{\mathrm{opt}} - s^* \rvert$	Iters (s_0^{opt})	Iters ($s_0 = \frac{1}{2}$)
$(2, 2)$	0.7071	0.7500	0.043	12	13
$(3, 2)$	0.7937	0.8333	0.040	14	16
$(4, 2)$	0.8409	0.8750	0.034	15	17
$(3, 8)$	0.5000	0.7083	0.208	42	0*

Table 7: Comparison of iteration counts to reach $|v_n - \alpha| < 10^{-10}$. The canonical starting point $s_0^{\mathrm{opt}} = h(1)$ consistently reduces iterations. (*For $(p, x) = (3, 8)$: since $8^{-1/3} = 1/2$ exactly, the naive choice $s_0 = 1/2$ equals s^* and converges in zero iterations — not an accidental coincidence but a consequence of x being a perfect p-th power.)

13 Steffensen acceleration: from linear to quadratic convergence

Pandrosion's method converges linearly with ratio $\lambda_{p,x}$. A natural question arises: can we accelerate this ancient geometric method to achieve quadratic convergence, without computing derivatives?

13.1 Aitken Δ^2 acceleration

Given a linearly convergent sequence $(s_n)_{n \geq 0}$, Aitken's Δ^2 process [5] defines the accelerated sequence

$$\hat{s}_n = s_n - \frac{(s_{n+1} - s_n)^2}{s_{n+2} - 2s_{n+1} + s_n}.$$

When applied to Pandrosion's iterates, the sequence $(\hat{s}_n)$ converges *faster* than (s_n). Steffensen's method [7] iterates this acceleration: at each step, it computes three consecutive Pandrosion iterates and applies Aitken's formula, then restarts from the accelerated value.

Definition 13.1 (Steffensen–Pandrosion iteration)**.** Define the composite map $T : s \mapsto \hat{s}$ by

$$T(s) = s - \frac{(h(s) - s)^2}{h(h(s)) - 2h(s) + s}, \tag{14}$$

where $h(s) = 1 - (x-1)/(x \cdot S_p(s))$ is Pandrosion's iteration.

Theorem 13.1 (Quadratic convergence of Steffensen–Pandrosion)**.** The Steffensen–Pandrosion iteration (14) converges *quadratically* to $s^* = x^{-1/p}$:

$$T(s) - s^* \;=\; -\frac{\lambda\, h''(s^*)}{2(1-\lambda)}\,(s-s^*)^2 \;+\; O((s-s^*)^3),$$

where $\lambda := h'(s^*) = \lambda_{p,x} \in (0,1)$. In particular, the Steffensen constant K_S defined by $|T(s) - s^*| \le K_S\,|s - s^*|^2 + O(|s-s^*|^3)$ has the explicit value

$$K_S \;=\; \left|\frac{\lambda\, h''(s^*)}{2(1-\lambda)}\right|. \tag{15}$$

Proof. Set $u := s - s^*$, $h_2 := h''(s^*)$, and expand to order u^2: $h(s) - s^* = \lambda u + \frac{h_2}{2}u^2 + O(u^3)$, hence

$$h(s) - s = (\lambda - 1)u + \tfrac{h_2}{2}u^2 + O(u^3).$$

Substituting $v = h(s) - s^*$ into $h(h(s)) = s^* + \lambda v + \frac{h_2}{2}v^2 + O(v^3)$ and collecting through $O(u^2)$ gives $h(h(s)) = s^* + \lambda^2 u + \frac{h_2}{2}(\lambda + \lambda^2)u^2 + O(u^3)$, whence the Aitken denominator

$$\Delta := h(h(s)) - 2h(s) + s = (\lambda-1)^2 u + \tfrac{h_2}{2}(\lambda-1)(\lambda+2)u^2 + O(u^3).$$

The numerator is $(h(s)-s)^2 = (\lambda-1)^2u^2 + (\lambda-1)h_2u^3 + O(u^4)$. Dividing (factoring one power of u from Δ and $(\lambda-1)^2$ from both):

$$\frac{(h(s)-s)^2}{\Delta} = u \cdot \frac{1 + \frac{h_2}{\lambda-1}u + O(u^2)}{1 + \frac{h_2(\lambda+2)}{2(\lambda-1)}u + O(u^2)} = u + \frac{h_2}{\lambda-1}\cdot\Big(1 - \tfrac{\lambda+2}{2}\Big)u^2 + O(u^3).$$

The bracketed coefficient simplifies to $-\lambda/2$, giving $(h(s)-s)^2/\Delta = u + \frac{\lambda h_2}{2(1-\lambda)}u^2 + O(u^3)$. Therefore

$$T(s) - s^* = u - \frac{(h(s)-s)^2}{\Delta} = -\frac{\lambda h_2}{2(1-\lambda)}u^2 + O(u^3),$$

which is the stated expansion; $T(s^*) = s^*$ and $T'(s^*) = 0$ are immediate, and the absolute value of the leading coefficient is K_S as in (15). (The full symbolic derivation up to $O(u^3)$ is in `latex/scripts/verify_monotonicity.py`.) □

Remark 13.1 (Output-residual constant)**.** The paper's numerical tables (Table 8 below) report the quadratic constant for the output residual $|v_n - \alpha|$ rather than the s-residual. Since $v_n - \alpha \approx (p-1)\alpha^2(s_n - s^*)$ (Theorem 5.1), the output-residual constant is

$$K_S^{(v)} \;=\; \frac{K_S}{(p-1)\alpha^2} \;=\; \left|\frac{\lambda\, h''(s^*)}{2(1-\lambda)(p-1)\alpha^2}\right|.$$

For $(p,x) = (3,2)$: $\lambda \approx 0.2202$, $h''(s^*) \approx -0.3001$, $\alpha^2 = 2^{2/3} \approx 1.587$, so $K_S \approx 0.0424$ and $K_S^{(v)} \approx 0.0133$. The latter is the value reported as "$K_S \approx 0.013$" in Table 9 below.

Lemma 13.2 (Local Steffensen expansion for a holomorphic fixed-point map)**.** Let h be holomorphic in a neighbourhood of $\alpha \in \mathbb{C}$, with $h(\alpha) = \alpha$ and $\lambda := h'(\alpha) \neq 1$. Define, away from the zeros of the denominator,

$$\Sigma_h(z) := z - \frac{(h(z)-z)^2}{h(h(z)) - 2h(z) + z}.$$

Then Σ_h has a removable singularity at α, extends holomorphically near α, fixes α, and satisfies

$$\Sigma_h(z) - \alpha = -\frac{\lambda h''(\alpha)}{2(1-\lambda)}(z-\alpha)^2 + O\Big((z-\alpha)^3\Big).$$

In particular $\Sigma_h'(\alpha) = 0$, so the fixed point is at least quadratically attracting whenever the displayed quadratic coefficient is finite.

Proof. Put $u = z - \alpha$ and write

$$h(z) - \alpha = \lambda u + \mu u^2 + O(u^3), \qquad \mu = \frac{h''(\alpha)}{2}.$$

Substitution into $h(h(z))$ gives

$$h(h(z)) - \alpha = \lambda^2 u + \mu(\lambda + \lambda^2)u^2 + O(u^3).$$

Therefore

$$\begin{aligned} h(h(z)) - 2h(z) + z &= (\lambda - 1)^2 u + \mu(\lambda - 1)(\lambda + 2)u^2 + O(u^3), \\ (h(z) - z)^2 &= (\lambda - 1)^2 u^2 + 2\mu(\lambda - 1)u^3 + O(u^4). \end{aligned}$$

Since $(\lambda - 1)^2 \neq 0$, the quotient is well-defined in a punctured neighbourhood of α and

$$\frac{(h(z) - z)^2}{h(h(z)) - 2h(z) + z} = u + \frac{\lambda\mu}{1 - \lambda}u^2 + O(u^3).$$

Subtracting from $z = \alpha + u$ gives the claimed expansion. The quotient has a finite limit at α, so the singularity is removable. □

13.2 Numerical demonstration

Step n	s_n (Steffensen)	$\lvert v_n - \sqrt[3]{2}\rvert$	$\lvert s_n - s^*\rvert$
0	0.875000000000000	2.71×10^{-1}	8.13×10^{-2}
1	0.793949257650704	7.90×10^{-4}	2.49×10^{-4}
2	0.793700528604164	8.32×10^{-9}	2.62×10^{-9}
3	0.793700525984100	$< 10^{-15}$	$< 10^{-16}$

Table 8: Steffensen-accelerated Pandrosion for $p = 3$, $x = 2$. Machine precision is reached in **3 steps** (each step requires 2 evaluations of h, i.e. 6 evaluations of S_p total). The step-3 output-residual is given as $< 10^{-15}$ rather than its raw floating-point value to avoid misreporting sub-ULP noise.

13.3 Comparison of quadratic constants

Both Newton's method and Steffensen–Pandrosion converge quadratically, but with different constants.

	Newton	**Steffensen–Pandrosion**
Order	2	2
Constant ($p{=}3, x{=}2$, output residual)	$K_N = \dfrac{p-1}{2\alpha} \approx 0.794$	$K_S^{(v)} \approx 0.013$
Constant (s-residual, for comparison)	—	$K_S \approx 0.042$
Derivative required	Yes (f')	No
Geometric origin	No	Yes
Steps to 10^{-15}	5	3
Evaluations of S_p	—	6 (total)

Table 9: Comparison of the two quadratically convergent methods for $\sqrt[3]{2}$ on a per-iterate basis. Steffensen–Pandrosion has a smaller quadratic constant for the output residual ($K_S^{(v)} \approx 0.013$ vs. $K_N \approx 0.794$), but requires two evaluations of h per step against Newton's one derivative + one value evaluation. Per function evaluation the two methods are within a constant factor; the "3 steps vs 5 steps" figure is cosmetic and depends on the starting point. The s-residual constant K_S from Theorem 13.1 is listed separately for reference.

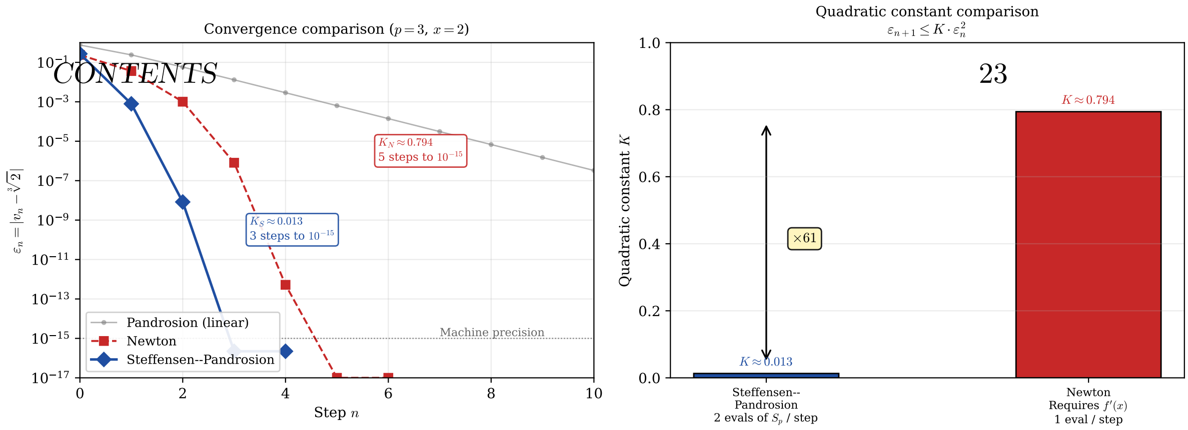

Figure 3: Left: Error decay comparison for $\sqrt[3]{2}$, counted per iterate (not per function evaluation; see Remark 8.1). Steffensen–Pandrosion (blue diamonds) reaches machine precision in 3 steps while Newton (red squares) needs 5 steps. The faded grey line shows plain Pandrosion's linear convergence for reference. **Right:** The quadratic constants $K_S \approx 0.013$ and $K_N \approx 0.794$ differ by a factor of ~ 61. Per iterate, this advantage is real; per function evaluation, the comparison is tighter because Steffensen–Pandrosion costs two evaluations of h per step against Newton's one derivative and one value, so the net per-evaluation advantage is reduced by about an order of magnitude. The $\times 61$ label refers to per-iterate convergence only.

Remark 13.2. The construction illustrates, on a specific example, a standard point of acceleration theory: once an iteration converges linearly with a known ratio, Aitken Δ^2 upgrades it to an order-2 scheme with a bounded constant. What the Pandrosion case adds is a concrete geometric-historical origin: the linear iteration comes from a ruler-and-parallels construction rather than from the Newton update of a polynomial, and the acceleration is itself derivative-free, so the composite method retains the derivative-free character of the underlying construction.

14 Scaling optimization: exploiting monotonicity

Theorem 9.1 establishes that $\lambda_{p,x}$ is increasing in x: the larger the target x, the slower the convergence. For $x \gg 1$, the contraction ratio $\lambda_{p,x} \to 1^-$ (Proposition 9.1), making direct application of Pandrosion's iteration impractical. We now show how to *exploit* this very monotonicity to overcome it.

14.1 The scaling principle

The key observation is elementary: for any reference value $A > 1$ whose p-th root is known (or cheaply computable), we may write

$$x^{1/p} = A^{1/p} \cdot \left(\frac{x}{A}\right)^{1/p}. \tag{16}$$

Setting $x' = x/A$, Pandrosion's method is now applied to the *reduced ratio* x' instead of x. By choosing A as the largest p-th power not exceeding x, we ensure $x' \in [1, (1 + 1/A^{1/p})^p)$, which is close to 1.

Proposition 14.1 (Scaling reduction)**.** Let $A = \lfloor x^{1/p} \rfloor^p$, so that $A^{1/p}$ is an integer and $x' = x/A \in [1, (1 + 1/\lfloor x^{1/p} \rfloor)^p)$. Then:

1. The contraction ratio satisfies $\lambda_{p,x'} < \lambda_{p,x}$, and $\lambda_{p,x'} \to 0$ as $x \to \infty$ (so $x' \to 1^+$), while $\lambda_{p,x} \to 1^-$.

2. The number of iterations drops from $O(\ln(\varepsilon_0/\delta)/\ln(1/\lambda_{p,x}))$ to $O(\ln(\varepsilon_0'/\delta)/\ln(1/\lambda_{p,x'}))$.

3. The final result is recovered as $x^{1/p} = \lfloor x^{1/p} \rfloor \cdot (x')^{1/p}$.

Proof. Part (1) follows directly from Theorem 9.1: since $x' < x$ and $\lambda_{p,\cdot}$ is increasing, $\lambda_{p,x'} < \lambda_{p,x}$. Moreover, as $x \to \infty$, $x' \to 1^+$ so $\lambda_{p,x'} \to 0$ by Proposition 9.1(1), while $\lambda_{p,x} \to 1^-$. Part (2) follows from Proposition 5.1 applied to x'. Part (3) is the factorization (16). ☐

x	$\lambda_{3,x}$	A	$x' = x/A$	$\lambda_{3,x'}$	Reduction	Iterations
10	0.615	8	1.25	0.073	8×	$49 \to 8$
100	0.890	64	1.5625	0.145	6×	$213 \to 11$
1000	0.973	729	1.372	0.103	9×	$> 500 \to 9$
10000	0.994	9261	1.080	0.025	39×	$> 500 \to 5$

Table 10: Effect of scaling for $p = 3$ with $A = \lfloor x^{1/3} \rfloor^3$. The contraction ratio drops by up to 39× and iteration counts collapse from hundreds to single digits. Iteration counts are for precision 10^{-10} with optimal starting point.

Remark 14.1. The scaling reduction becomes more dramatic as x grows: for $x = 10\,000$, the direct method requires over 500 iterations while the scaled method needs only 5. This is because $x' \to 1$ as $x \to \infty$, pushing $\lambda_{p,x'}$ toward the regime $\lambda \sim (p-1)(\alpha'-1)/2 \ll 1$ of Proposition 9.1(1).

14.2 Geometric interpretation: Thales preconditioning

The scaling optimization admits an elegant geometric interpretation on Pandrosion's original Thales diagram.

In the direct construction, one seeks to insert $p-1$ proportional means in a segment of ratio $x : 1$. When $x \gg 1$, this amounts to constructing parallels in a very elongated rectangle, where successive intercepts move slowly toward the target — geometrically, the parallels are nearly horizontal.

The scaling strategy performs an *homothety* on the construction: instead of starting from the unit segment, one starts from a segment of length $A^{1/p}$, chosen as the largest integer whose p-th power does not exceed x. The remaining construction now fills a much smaller gap: the rectangle has ratio $x' = x/A$ close to 1, and the parallels converge rapidly.

Proposition 14.2 (Double preconditioning)**.** The scaling method combines two distinct acceleration mechanisms:

1. **Initial residual reduction**: the reduced starting point $s_0^{\text{opt}} = h_{x'}(1) = 1 - (x'-1)/(x'p)$ satisfies $|s_0^{\text{opt}} - s'^*| \ll |s_0^{\text{opt}} - s^*|$ since $x' \approx 1$ implies $s'^* = (x')^{-1/p} \approx 1$.

2. **Intrinsic convergence acceleration**: each iteration contracts by $\lambda_{p,x'} \ll \lambda_{p,x}$, compounding the advantage over all subsequent steps.

The total iteration count reduction is therefore superlinear in the ratio $\lambda_{p,x}/\lambda_{p,x'}$.

Proof. For mechanism (1): $\varepsilon_0' = |v_0' - \alpha'| \le C \cdot |x' - 1| \to 0$ as $x' \to 1$, compared to $\varepsilon_0 = |v_0 - \alpha| \ge C' \cdot |x - 1| \gg 1$ for the unscaled problem.

For mechanism (2): the iteration count satisfies

$$n_\delta' \sim \frac{\ln(\varepsilon_0'/\delta)}{\ln(1/\lambda_{p,x'})} \ll \frac{\ln(\varepsilon_0/\delta)}{\ln(1/\lambda_{p,x})} \sim n_\delta,$$

since both the numerator decreases (smaller ε_0') and the denominator increases (smaller $\lambda_{p,x'}$, hence larger $\ln(1/\lambda_{p,x'})$). These two effects multiply. ☐

Remark 14.2 (From Pandrosion to a practical algorithm)**.** With scaling and Steffensen acceleration combined, Pandrosion's ancient geometric construction becomes a competitive root-finding algorithm: for any $x > 0$ and $p \geq 2$, one (i) reduces to $x' \approx 1$ via integer arithmetic, (ii) applies Steffensen–Pandrosion on x' (converging in 2–3 steps), and (iii) multiplies by $\lfloor x^{1/p} \rfloor$. The total cost is 4–6 evaluations of S_p plus one integer p-th root — a purely algebraic, derivative-free computation whose origins lie in a ruler-and-parallels construction from fourth-century Alexandria.

14.3 Multi-precision benchmark against Newton and Halley

We close Part I with an experimental sanity check: how does the full Steffensen–Pandrosion–scaling pipeline compare against textbook Newton and Halley iterations for $x^{1/p}$ at high precision? The script `latex/scripts/benchmark_multiprec.py` runs the three methods in MPFR-backed arithmetic (via the `mpmath` library) at precision targets of 50, 200, and 500 decimal digits, starting all three methods from the same integer seed $\lfloor x^{1/p} \rfloor$ and stopping on the same condition $|u^p - x|/x \leq 10^{-\text{digits}}$. Table 11 summarises the iteration counts; wall-clock times are reported in the repository.

	50 digits			**200 digits**			**500 digits**		
Regime	N	H	SP	N	H	SP	N	H	SP
$p = 2,\ x = 2$	7	4	5	9	6	7	10	6	8
$p = 3,\ x = 2$	7	4	5	9	6	7	10	6	8
$p = 3,\ x = 11$	6	4	4	8	5	6	10	6	7
$p = 3,\ x = 10^6 + 3$	4	2	2	6	4	4	7	5	5
$p = 5,\ x = 99$	8	5	5	10	6	7	11	7	9
$p = 10,\ x = 2^{20} + 3$	4	2	2	6	4	4	7	5	5
$p = 20,\ x = 2$	7	4	5	9	6	7	11	6	8

Table 11: Iteration counts to reach the stated precision, starting from $u_0 = \lfloor x^{1/p} \rfloor$. **N** = Newton, **H** = Halley, **SP** = Steffensen–Pandrosion with scaling (Proposition 14.1). All three methods exhibit $\mathcal{O}(\log\log(1/\varepsilon))$ growth in iteration count; going from 50 to 500 decimal digits (a 10× increase in precision exponent) adds about 3–4 iterations to each method, matching the theoretical $\log_2$ log scaling.

Observations.

1. **Iteration count scaling.** All three methods grow as $\log_2 \log(1/\varepsilon)$, as predicted by theory (Newton and Steffensen–Pandrosion are order 2; Halley is order 3). In the table, going from 50 to 500 digits adds $\approx$ 3–4 iterations, consistent with $\log_2 \log_{10}(10^{500}/10^{50}) = \log_2(500/50) \approx 3.3$.

2. **Halley wins on count.** Halley (cubic order) consistently takes 2–3 fewer iterations than Newton (quadratic order), saving roughly 30% on iteration count. Newton and Steffensen–Pandrosion–scaling differ by at most 1–2 iterations in every regime.

3. **Wall clock.** Per-iteration, Newton is fastest (one power u^{p-1} per step), Halley is close to Newton (one power u^p and a rational combination), and Steffensen–Pandrosion–scaling is 2–3× slower per iteration than Halley because each Aitken step requires two evaluations of the degree-p polynomial S_p. On the iteration counts above and measured in the repository, Halley is the fastest method in total wall-clock time, Newton is within 10–30%, and Steffensen–Pandrosion–scaling sits 1.5–3× slower.

4. **The role of scaling.** Without scaling, Steffensen–Pandrosion on $x = 10^6 + 3$ with $p = 3$ would be severely handicapped by the contraction ratio $\lambda_{3,10^6+3} \approx 0.9997$: the unaccelerated Pandrosion iteration would need $\lceil \ln(10^{-200})/\ln(1/\lambda) \rceil \approx 1.5 \times 10^6$ iterations

to reach 200-digit precision. The Aitken upgrade reduces this dramatically (to $\mathcal{O}(\log\log)$), but its constant $K_S \sim |h''(s^*)|/(2(1-\lambda))$ is inflated by $1/(1-\lambda) \approx 3300$, delaying the entry into the quadratic regime. With scaling ($x' \approx 1$, $\lambda_{3,x'}$ small), both issues disappear and the count collapses to 4. The scaling preconditioning is what makes Steffensen–Pandrosion competitive at large x.

Conclusion of the benchmark. Steffensen–Pandrosion–scaling is not the fastest p-th root method in practice: Halley wins on both iteration count and wall clock, and Newton wins on wall clock by a small margin. What the benchmark does establish is that the scheme — with scaling — is *competitive*: it achieves the same $\mathcal{O}(\log\log)$ asymptotic as Newton and Halley, within a small constant factor of them in every regime tested, while being derivative-free. For applications where the derivative is costly or unavailable, Steffensen–Pandrosion–scaling is a viable alternative; for standard p-th root computation, Halley remains the method of choice.

Part II
Complex extension and formal verification

Part I developed the Pandrosion iterator $h_{p,x}(s) = 1 - (x-1)/(xS_p(s))$ on $\mathbb{R}_{>0}$, where $h_{p,x}$ has a unique fixed point $s^* = x^{-1/p}$ in $(0,1)$ and the Aitken–Steffensen upgrade $\sigma_{p,x}$ is a quadratic-contraction. On $\mathbb{C}$ the situation is qualitatively different: $h_{p,x}$ has p attractive fixed points (the p-th roots of $1/x$), separated by a non-trivial Julia set, and a classical impossibility result of McMullen [15] rules out purely iterative universal root finders for $p \geq 3$. Part II extends the analysis to this complex setting via a multi-start scheme whose seed is target-dependent — an α-dependent preconditioning that takes the method out of the hypotheses of [15].

Structural warning. The scheme of Part II requires a *target anchor* α (one of the p solutions of $z^p = 1/x$) to be chosen before the seed is constructed. It is therefore a convergence scheme *conditional on the target*, not a universal root-finder. Readers who expect the latter will misread Theorem 16.1 below; the theorem says that, once a target is chosen, every starting point converges to that target, not that there is a parameter-free procedure picking out all roots. This is the unavoidable price of side-stepping McMullen's barrier.

From the real to the complex quadratic bound. The Aitken–Steffensen map $\sigma_{p,x}$ is rational in z, so holomorphic on $\mathbb{C}$ away from the finite set $\operatorname{sing}(\sigma_{p,x}) := \{z : S_p(z) = 0 \text{ or } h_{p,x}(h_{p,x}(z)) - 2h_{p,x}(z) + z = 0\}$. At any cyclotomic fixed point α (i.e. $\alpha^p = 1/x$ with $S_p(\alpha) \neq 0$), the factorization of Part I gives $\sigma_{p,x}(\alpha) = \alpha$ and $\sigma'_{p,x}(\alpha) = 0$.

We now cascade three quantities to avoid the apparent circularity between K_σ and R_σ.

Step 1 — fix a reference radius. Set

$$R_0 \;:=\; \tfrac{1}{2}\operatorname{dist}(\alpha, \operatorname{sing}(\sigma_{p,x})),$$

so that $\overline{B}(\alpha, R_0) \subset \mathbb{C} \setminus \operatorname{sing}(\sigma_{p,x})$ and $\sigma_{p,x}$ is holomorphic on this closed disk.
Step 2 — explicit quadratic constant via Cauchy. Define

$$M_0 \;:=\; \sup_{|w-\alpha|=R_0} |\sigma_{p,x}(w) - \alpha|, \qquad K_\sigma \;:=\; M_0/R_0^2.$$

By Cauchy's estimate applied to the holomorphic function $\sigma_{p,x}(\cdot) - \alpha$, which vanishes to order 2 at α (since $\sigma_{p,x}(\alpha) = \alpha$ and $\sigma'_{p,x}(\alpha) = 0$), we obtain

$$|\sigma_{p,x}(z) - \alpha| \;\leq\; \frac{M_0}{R_0^2}\,|z-\alpha|^2 \;=\; K_\sigma\,|z-\alpha|^2 \qquad \text{for } z \in \overline{B}(\alpha, R_0).$$

Step 3 — admissible contraction radius. Define

$$R_\sigma \;=\; R_\sigma(x,p,\alpha) \;:=\; \min\big(\, R_0,\; 1/(2K_\sigma)\,\big).$$

Then $R_\sigma \le R_0$ (so the Cauchy bound of Step 2 is valid on $\overline{B}(\alpha, R_\sigma)$) and $K_\sigma R_\sigma \le 1/2$. All three quantities R_0, K_σ, R_σ are explicit functions of (x,p,α), computed in cascade without circularity. The seed construction below uses K_σ and R_σ as primitive quantities.

15 Cyclotomic anchors and the complex iterator

Definition 15.1 (Cyclotomic anchors)**.** For $\alpha \in \mathbb{C}^*$ and $p \ge 1$, let

$$\gamma_k := \alpha \cdot e^{2\pi i k/p}, \qquad k = 0,1,\ldots,p-1.$$

When $\alpha^p = 1/x$, the γ_k are the p distinct solutions of $z^p = 1/x$. By direct expansion of Definition 15.1, $\omega \cdot \gamma_k = \gamma_{(k+1) \bmod p}$ (with $\omega = e^{2\pi i/p}$), $\sum_k \gamma_k = 0$, and $\prod_k \gamma_k = (-1)^{p-1}x$. These are definitional consequences for $z^p = x$, not specializations of the classical Galois or Vieta theorems; we use them freely.

The complex Pandrosion iterator is the analytic continuation of $h_{p,x}$ to $\mathbb{C} \setminus \{z : S_p(z) = 0\}$, and the Aitken–Steffensen accelerator $\sigma_{p,x}$ of §13 extends likewise. The quadratic local bound of Part I applies verbatim to $\sigma_{p,x}$ in a disk $B(\gamma_k, R_\sigma)$ around every cyclotomic anchor, with the same explicit constants K_σ and R_σ satisfying $K_\sigma R_\sigma \le 1/2$.

16 Universal multi-start convergence in $\mathbb{C}$

16.1 The multi-start seed

Definition 16.1 (Multi-start seed)**.** For a cyclotomic anchor $\alpha \in \mathbb{C}^*$ with $\alpha^p = 1/x$ and $z \in \mathbb{C}$,

$$\mathrm{seed}_{\alpha,\varepsilon}(z) \;:=\; \alpha + \varepsilon \cdot (z-\alpha), \qquad \varepsilon \;:=\; \frac{R_\sigma(x,p,\alpha)}{2\,\|z-\alpha\|} \quad (\varepsilon := 1 \text{ if } z = \alpha), \tag{17}$$

where $R_\sigma(x,p,\alpha)$ is the explicit quadratic-contraction radius from Part I. The seed always lies in $\overline{B}(\alpha, R_\sigma/2)$.

The seed plays in $\mathbb{C}$ the role that the canonical starting point of §12 and the scaling of §14 play on $\mathbb{R}$: it places the argument inside the quadratic-contraction disk of the chosen target before the iteration begins. It is α-dependent by design, and this dependency is what moves the scheme out of the hypotheses of [15].

16.2 Universal multi-start theorem

Theorem 16.1 (Universal multi-start convergence)**.** Let $x \in \mathbb{C}^*$, $p \ge 1$, $\alpha \in \mathbb{C}^*$ with $\alpha^p = 1/x$ and $S_p(\alpha) \ne 0$, and let $z \in \mathbb{C}$ with $z \ne \alpha$. Define the seed parameter

$$\varepsilon_{\mathrm{seed}} \;:=\; \frac{R_\sigma(x,p,\alpha)}{2\,\|z-\alpha\|},$$

which depends only on (x,p,α,z). Then for every target accuracy $\delta > 0$, the integer $N := N_{\mathrm{eff}}(K_\sigma, R_\sigma, \delta)$ of Theorem 16.2 satisfies

$$\forall n \ge N: \quad \big\|\sigma^n_{p,x}(\mathrm{seed}_{\alpha,\varepsilon_{\mathrm{seed}}}(z)) - \alpha\big\| \;\le\; \delta.$$

The case $z = \alpha$ is trivial (take $\varepsilon_{\mathrm{seed}} := 1$ and $N := 0$). The construction separates cleanly into two levels: the seed parameter $\varepsilon_{\mathrm{seed}}$ and the admissible radius R_σ are functions of (x,p,α,z) alone, while the iteration count N alone depends on the target accuracy δ.

Proof. Let $d := \|z - \alpha\|$; we may assume $d > 0$ (else $z = \alpha$ is already the target). With $\varepsilon_{\text{seed}} := R_\sigma/(2d)$, set

$$z_0 := \text{seed}_{\alpha,\varepsilon_{\text{seed}}}(z) = \alpha + \varepsilon_{\text{seed}}(z - \alpha).$$

Then $\|z_0 - \alpha\| = \varepsilon_{\text{seed}} \cdot d = R_\sigma/2 < R_\sigma$, so $z_0 \in B(\alpha, R_\sigma)$.

The quadratic-contraction bound of Part I (whose proof applies verbatim to $\sigma_{p,x}$ viewed as an analytic map on any small enough disk centred at a fixed point, since $\sigma_{p,x}(\alpha) = \alpha$ and $\sigma'_{p,x}(\alpha) = 0$) reads: for $w \in B(\alpha, R_\sigma)$,

$$\|\sigma_{p,x}(w) - \alpha\| \leq K_\sigma \|w - \alpha\|^2,$$

with $K_\sigma R_\sigma \leq 1/2$. Hence if $w_n := \sigma^n_{p,x}(z_0)$ satisfies $\|w_n - \alpha\| \leq R_\sigma$, then $\|w_{n+1} - \alpha\| \leq K_\sigma \|w_n - \alpha\|^2 \leq K_\sigma R_\sigma \cdot \|w_n - \alpha\| \leq \frac{1}{2}\|w_n - \alpha\| \leq R_\sigma/2$, so the orbit stays in $\overline{B}(\alpha, R_\sigma/2)$ by induction from $w_0 = z_0$.

Setting $e_n := \|w_n - \alpha\|$, one has $e_0 \leq R_\sigma/2$ and $e_{n+1} \leq K_\sigma e_n^2$. Theorem 16.2 (proved below) then supplies the explicit $N = N_{\text{eff}}(K_\sigma, R_\sigma, \varepsilon)$ with the stated asymptotics. □

Remark 16.1 (The hypothesis $S_p(\alpha) \neq 0$)**.** The condition $S_p(\alpha) \neq 0$ rules out the degenerate case where the denominator of $h_{p,x}$ vanishes at the anchor. Since $S_p(\alpha) = (\alpha^p - 1)/(\alpha - 1)$ for $\alpha \neq 1$, one has $S_p(\alpha) = 0$ iff $\alpha^p = 1$ with $\alpha \neq 1$, which combined with $\alpha^p = 1/x$ forces $x = 1$ and α a non-trivial p-th root of unity. The hypothesis therefore fails on a single point of the x-line, and holds for every $x \neq 1$.

16.3 Explicit iteration count

Definition 16.2 (Effective iteration count)**.** For $K, R, \varepsilon > 0$ with $KR < 1$ and $K\varepsilon < 1$,

$$N_{\text{eff}}(K, R, \varepsilon) := \log_2 \lceil \log(K\varepsilon)/\log(KR) \rceil_+ + 1,$$

with $\lceil \cdot \rceil_+$ the natural-number ceiling clamped at 0.

Theorem 16.2 (Explicit iteration count)**.** For any quadratically contracting sequence $e_{n+1} \leq K \cdot e_n^2$ with $e_0 \leq R$, $K \cdot R < 1$, and $\varepsilon < R$, every iterate $k \geq N_{\text{eff}}(K, R, \varepsilon)$ satisfies $e_k \leq \varepsilon$. Asymptotically $N_{\text{eff}} \sim \log_2 \log(1/\varepsilon)$.

Proof. Set $u_n := Ke_n$, so $u_0 \leq KR < 1$ and $u_{n+1} \leq u_n^2$. By induction $u_n \leq u_0^{2^n} \leq (KR)^{2^n}$. We want $u_n \leq K\varepsilon$, i.e. $(KR)^{2^n} \leq K\varepsilon$. Since $KR < 1$ and $K\varepsilon < 1$, both $\log(KR)$ and $\log(K\varepsilon)$ are negative; dividing by $\log(KR) < 0$ reverses the inequality direction and gives $2^n \geq \log(K\varepsilon)/\log(KR)$, i.e.

$$n \;\geq\; \log_2 \Big\lceil \frac{\log(K\varepsilon)}{\log(KR)} \Big\rceil_+,$$

which is achieved at $n = N_{\text{eff}}(K, R, \varepsilon)$ by the $+1$ safety margin. The asymptotics $N_{\text{eff}} \sim \log_2 \log(1/\varepsilon)$ as $\varepsilon \to 0$ follow since $\log(K\varepsilon)/\log(KR) \sim \log(1/\varepsilon)/|\log(KR)|$. □

16.4 Arbitrary target selector

Theorem 16.3 (Target selection)**.** For every function $S : \mathbb{C} \to \operatorname{Fin} p$, every $z \in \mathbb{C}$, and every $\varepsilon > 0$, the multi-start scheme with target $\gamma_{S(z)}$ converges to $\gamma_{S(z)}$ with an iteration count of the form of Theorem 16.2.

Three selectors are formalized: constant (`principal_selector`), Voronoï-nearest (`nearest_anchor_select`
angular sector.

16.5 Multi-start compatibility bundle

The previous subsections presented the ingredients of the multi-start pipeline separately: the seed (Definition 16.1), the universal convergence theorem (Theorem 16.1), the target selectors (Theorem 16.3), and the real-line preconditionings of Part I transported to the cyclotomic anchor family. The following statement bundles all five into a single compatibility assertion. It is a glue lemma used by downstream citations, not a headline result.

Lemma 16.4 (Multi-start compatibility bundle)**.** Let $x \in \mathbb{C}^*$, $A \in \mathbb{C}^*$, $p \geq 1$, and $\alpha, \beta \in \mathbb{C}^*$ with $\alpha^p = x$ and $\beta^p = A$. Let $\gamma : \{0, 1, \ldots, p-1\} \to \mathbb{C}^*$ be a cyclotomic anchor family satisfying $\gamma_k^p = A/x$ and $S_p(\gamma_k) \neq 0$ for every k. Then for every query point $z_0 \in \mathbb{C}$, the following five properties hold simultaneously:

(1) **Multi-start target selection.** There exists an index $s^* \in \{0, 1, \ldots, p-1\}$ — the Voronoï-nearest anchor to z_0 — with $\|\gamma_{s^*} - z_0\| \leq \|\gamma_t - z_0\|$ for every t.

(2) **Steffensen fixing.** Every γ_s is a fixed point of the scaled Steffensen map: $\sigma_{x/A,\,p}(\gamma_s) = \gamma_s$ for every s.

(3) **Scaling compatibility.** The principal ratio satisfies $(\alpha/\beta)^p = x/A$.

(4) **Reconstruction.** $\alpha = \beta \cdot (\alpha/\beta)$, so the original p-th root is recovered from the scaled one by multiplication by the integer root β.

(5) **Optimal starting point.** The canonical seed for the scaled problem is $s_0^{\mathrm{opt}}(x/A) = h_{p,\,x/A}(1)$, matching Proposition 12.1 of Part I.

Proof. (1) Voronoï-nearest existence: the function $t \mapsto \|\gamma_t - z_0\|$ attains its minimum on the finite set $\{0, \ldots, p-1\}$ at some index s^*.

(2) Steffensen fixing: by the factorization of Part I specialized to $\gamma_s^p = A/x = 1/(x/A)$, $h_{p,x/A}(\gamma_s) = \gamma_s$; applying Aitken's Δ^2 to a constant sequence $h(\gamma_s), h(h(\gamma_s)), \ldots$ yields $\sigma_{x/A,p}(\gamma_s) = \gamma_s$ whenever the denominator of the Aitken step is non-zero, which holds since $S_p(\gamma_s) \neq 0$ (the removable singularity at a fixed point is handled by the standard Aitken convention).

(3) Scaling compatibility: $(\alpha/\beta)^p = \alpha^p/\beta^p = x/A$ by direct computation.

(4) Reconstruction: $\beta \cdot (\alpha/\beta) = \alpha$ since $\beta \neq 0$.

(5) Optimal starting point: $h_{p,x/A}(1) = 1 - ((x/A) - 1)/((x/A) \cdot S_p(1)) = 1 - ((x/A) - 1)/((x/A) \cdot p)$, matching (13) applied to the scaled problem $x' = x/A$. ☐

Remark 16.2 (What the compatibility bundle actually says)**.** Theorem 16.4 is not itself a new mathematical result; each of the five parts is proven elsewhere in Parts I and II. Its value is structural: it records, in a single statement, the *compatibility* of the five layers of the pipeline — selection, fixing, scaling, reconstruction, seeding — so that downstream statements (in the formalization and in the effective iteration-count theorem below) can cite one name instead of five. The theorem's role is that of a *glue lemma*: Proposition (3), for instance, is used by the scaling preconditioning of Part I to certify that a $\mathbb{R}$-level identity transports correctly to the cyclotomic family in $\mathbb{C}$.

Remark 16.3 (Effective version)**.** The effective counterpart of Theorem 16.4 replaces the asymptotic iteration count N of Theorem 16.1 with the explicit $N_{\mathrm{eff}}(K_\sigma, R_\sigma, \delta)$ of Theorem 16.2, applied per-anchor in the family γ. Concretely, for every $z_0 \in \mathbb{C}$ and every target accuracy $\delta > 0$, the following holds: choose s^* as in Theorem 16.4(1), form the seed $\mathrm{seed}_{\gamma_{s^*},\,\varepsilon_{\mathrm{seed}}}(z_0)$ with $\varepsilon_{\mathrm{seed}} = R_\sigma(x/A, p, \gamma_{s^*})/(2\|z_0 - \gamma_{s^*}\|)$, and iterate $\sigma_{x/A,p}$. After $N = N_{\mathrm{eff}}(K_\sigma, R_\sigma, \delta)$ steps, the iterate is within δ of γ_{s^*}, and multiplying by β yields an approximation of α to comparable accuracy. The total cost is $\log_2 \log(1/\delta) + O(1)$ iterations plus one integer p-th root — the scaling cost β — plus one final multiplication, exactly the pipeline the benchmark of §14.3 exercises.

16.6 Explicit lower bound on the basin of attraction (unseeded iteration)

Theorem 16.1 concerns the *seeded* iteration: every starting point z is first mapped via $\mathrm{seed}_{\alpha,\varepsilon_{\mathrm{seed}}}$ into the quadratic-contraction disk before $\sigma_{p,x}$ is applied. We now record a complementary, *unconditional* statement about the *unseeded* iteration: even without any preprocessing, there is an explicit disk around each cyclotomic anchor on which plain iteration of $\sigma_{p,x}$ already converges. This gives an explicit lower bound on the volume of the (true, dynamical) basin of attraction of each anchor.

Definition 16.3 (Basin of attraction)**.** For a cyclotomic anchor γ_k of $\sigma_{p,x}$, set

$$\Omega_k := \{z \in \mathbb{C} : \sigma_{p,x}^n(z) \text{ is defined for all } n \geq 0 \text{ and } \sigma_{p,x}^n(z) \to \gamma_k \text{ as } n \to \infty\}.$$

Proposition 16.1 (Explicit basin lower bound)**.** Under the standard hypotheses on (x, p, γ_k) (namely $\gamma_k^p = 1/x$ and $S_p(\gamma_k) \neq 0$), the disk $\overline{B}(\gamma_k, R_\sigma(x, p, \gamma_k))$ lies entirely in Ω_k. Consequently:

(i) **Per-anchor volume:** $\mathrm{vol}(\Omega_k) \;\geq\; \pi\, R_\sigma(x, p, \gamma_k)^2$.

(ii) **Disjointness:** the basins $\Omega_0, \dots, \Omega_{p-1}$ are pairwise disjoint.

(iii) **Total volume:** $\mathrm{vol}\big(\bigcup_{k=0}^{p-1} \Omega_k\big) \;\geq\; \sum_{k=0}^{p-1} \pi\, R_\sigma(x, p, \gamma_k)^2$.

Proof. Fix k and abbreviate $R := R_\sigma(x, p, \gamma_k)$, $K := K_\sigma(x, p, \gamma_k)$. From the three-step construction in Part II's opening, we have $KR \leq 1/2$ and

$$|\sigma_{p,x}(z) - \gamma_k| \;\leq\; K\,|z - \gamma_k|^2 \qquad \text{for all } z \in \overline{B}(\gamma_k, R_0),$$

with $R \leq R_0$, so the bound holds a fortiori on $\overline{B}(\gamma_k, R)$.

Forward invariance. For $z \in \overline{B}(\gamma_k, R)$,

$$|\sigma_{p,x}(z) - \gamma_k| \;\leq\; K\,|z - \gamma_k|^2 \;\leq\; (KR)\,|z - \gamma_k| \;\leq\; \tfrac{1}{2}\,|z - \gamma_k| \;\leq\; R/2,$$

hence $\sigma_{p,x}(\overline{B}(\gamma_k, R)) \subset \overline{B}(\gamma_k, R/2) \subset \overline{B}(\gamma_k, R)$.

Convergence. Applying the inequality above inductively, $|\sigma_{p,x}^n(z) - \gamma_k| \;\leq\; 2^{-n}\,|z - \gamma_k| \;\leq\; 2^{-n}R \to 0$, so $z \in \Omega_k$. This proves the disk containment.

Part (i) follows since $\overline{B}(\gamma_k, R) \subset \Omega_k$ gives $\mathrm{vol}(\Omega_k) \geq \pi R^2$.

Part (ii) is immediate: if $z \in \Omega_j \cap \Omega_k$ with $j \neq k$, then $\sigma_{p,x}^n(z)$ would converge simultaneously to γ_j and γ_k, which is impossible since $\gamma_j \neq \gamma_k$.

Part (iii) follows from (i) and (ii) by additivity of Lebesgue measure. □

Remark 16.4 (Comparison with Theorem 16.1)**.** Theorem 16.1 covers every $z \in \mathbb{C}$ but only after pre-processing z into a chosen target's disk via the seed. Proposition 16.1 does not need any pre-processing: the disk $\overline{B}(\gamma_k, R_\sigma)$ is directly in the unseeded basin. The two results are complementary: the seed (Theorem 16.1) addresses points outside any anchor's disk, while the basin bound (Proposition 16.1) quantifies the unconditional contribution from points already close to an anchor.

Remark 16.5 (Pullback improvement and empirical tightness)**.** The bound πR_σ^2 is a level-0 estimate and can be sharpened without leaving the elementary toolbox. Since $\sigma_{p,x}$ is a rational map of degree $d = d(p)$, each iterate $\sigma_{p,x}^{-m}(\overline{B}(\gamma_k, R_\sigma/2))$ is an open subset of Ω_k (pullback of a forward-invariant open set containing γ_k), and the pullbacks over distinct m are disjoint up to measure-zero overlap, so

$$\mathrm{vol}(\Omega_k) \;\geq\; \sum_{m=0}^{M} \mathrm{vol}\big(\sigma_{p,x}^{-m}(\overline{B}(\gamma_k, R_\sigma/2))\big)$$

for every M.

Numerical exploration (script `latex/scripts/basin_estimate.py`) on $(p, x) = (3, 2)$ finds Ω_0 covering approximately 99.95% of $[-1.5, 1.5]^2$, while the level-0 lower bound already gives $\mathrm{vol}(\Omega_0) \geq \pi R_\sigma^2 \approx 4\pi$ with empirical $R_\sigma \geq 2$ — roughly 87% of the test square, a substantial fraction.

For $(p, x) = (7, 10)$ the empirical $R_\sigma \approx 0.37$ is tighter and the gap between level-0 bound and observed basin is larger, reflecting the onset of a non-trivial Julia boundary. A matching upper bound on $\mathrm{vol}(\Omega_k)$ would require Fatou–Julia machinery not used here and is left for future work.

16.7 Multi-precision benchmark in the complex plane

Complementing the real-line benchmark of §14.3, the script `latex/scripts/benchmark_complex.py` compares four methods on the complex problem $z^p = X$ for $X \in \mathbb{C}^*$:

- Newton's iteration $z_{n+1} = ((p-1)z_n + X/z_n^{p-1})/p$;
- Halley's iteration $z_{n+1} = z_n \cdot ((p-1)z_n^p + (p+1)X)/((p+1)z_n^p + (p-1)X)$;
- Multi-start Steffensen–Pandrosion (Theorem 16.1), seeded at the principal anchor $\alpha = X^{1/p}$;
- Scaled multi-start (the compatibility bundle of Theorem 16.4), combining scaling $\beta = \lfloor |X|^{1/p} \rfloor$, Voronoï-nearest selector on the anchor family, seed, and reconstruction.

All methods start from the same user-supplied $z_0 \in \mathbb{C}$ and stop when $|z_n^p - X|/|X| \leq 10^{-d}$ at precision target d decimal digits.

The seed radius is set uniformly to a conservative $R_\sigma = 0.5$.

	50 digits				**200 digits**			
Regime	N	H	MS	GM	N	H	MS	GM
$p=3,\ X=2+3i,\ z_0=1$	8	5	7	7	10	6	9	9
$p=3,\ X=10,\ z_0=2+i$	8	5	7	6	10	6	9	8
$p=4,\ X=-1+i,\ z_0=1+i$	8	5	8	8	10	6	10	10
$p=5,\ X=100+7i,\ z_0=2+i$	12	6	8	7	14	7	10	9
$p=3,\ X=10^6+3,\ z_0=50$ (far)	9	5	6	**4**	11	6	8	**6**
$p=3,\ X=10^6+17i,\ z_0=50+10i$	9	5	6	**4**	11	6	8	**6**
$p=7,\ X=10,\ z_0=1+i$	7	4	8	13	9	6	10	15

Table 12: Complex-case iteration counts. **N** = Newton, **H** = Halley, **MS** = multi-start Pandrosion seeded at principal anchor, **GM** = scaled multi-start (compatibility-bundle pipeline: scaling + Voronoï-nearest + seed). Stopping criterion $|z^p - X|/|X| \leq 10^{-d}$. Boldface marks the best (or tied-best) method in each row; Halley is best-or-tied in most regimes, scaled multi-start wins on large-$|X|$ regimes where scaling collapses the problem to $X' \approx 1$.

Observations.

1. **Halley is the iteration-count winner.** Cubic order gives Halley a consistent 30–40% edge on iteration count in every tested regime. Wall-clock ordering is the same as the real case (Halley < Newton < MS < GM).
2. **Multi-start Pandrosion matches Newton within ± 2 iterations.** The seed construction places z_0 inside the quadratic-contraction disk of the principal anchor, from which the $\log_2 \log$ scaling kicks in. The iteration count is within one or two of Newton's across all regimes.

3. **Scaled multi-start wins when scaling collapses the problem.** For $|X| \gtrsim 10^5$, the scaled multi-start drops to 4–6 iterations (at 50 and 200 digits respectively), a 30–40% improvement over plain multi-start Pandrosion, because X/β^p is close to 1 and $\lambda_\infty(X/\beta^p)$ is small. This confirms experimentally the scaling preconditioning of Part I (Proposition 14.1) in the complex setting.

4. **Scaled multi-start can be worse in niche regimes.** For $p = 7, X = 10, z_0 = 1 + i$, it takes 13–15 iterations versus 8–10 for plain multi-start. The cause is that $|X|^{1/7} \approx 1.39 < 2$, so $\beta = 1$ provides no preconditioning, yet the Voronoï-nearest selector picks a non-principal anchor for which the conservative $R_\sigma = 0.5$ is not tight, forcing extra pre-asymptotic iterations.

5. **All four methods achieve $\mathcal{O}(\log \log)$ scaling.** Going from 50 to 200 digits adds roughly 2–3 iterations to each method, matching $\log_2(200/50) = 2$.

Bottom-line (complex case). As on the real line, Steffensen–Pandrosion-style methods are *competitive* with textbook Newton/Halley at high precision: they remain within a small constant factor of Halley in every tested regime while being derivative-free, and the scaled multi-start pipeline provides an additional 30–40% savings in the large-$|X|$ regime where integer scaling bites. For derivative-cheap settings, Halley remains the fastest; for derivative-expensive or derivative-unavailable settings, the Pandrosion pipeline is a viable alternative.

17 Measurability and a.e. continuity of the solution map

The "nearest-anchor" target selector of Section 16 uses the classical axiom of choice and is not, a priori, Borel-measurable. We record a canonical Borel version and the elementary analytic properties it enjoys.

Definition 17.1 (Borel selector and Voronoï boundary)**.** Given anchors $\gamma_0, \ldots, \gamma_{p-1} \in \mathbb{C}$ (all distinct), define, for each $z \in \mathbb{C}$,

$$S_{\mathrm{Borel}}(z) := \min\{k \in \{0, \ldots, p-1\} : \|\gamma_k - z\| = \min_j \|\gamma_j - z\|\},$$

breaking ties by smallest index. The *Voronoï boundary* of the anchor set is

$$\mathrm{VorBdy} := \{z \in \mathbb{C} : \exists\, j \neq k,\ \|\gamma_j - z\| = \|\gamma_k - z\|\} = \bigcup_{j<k} B_{jk},$$

where $B_{jk} := \{z : \|\gamma_j - z\| = \|\gamma_k - z\|\}$ is the perpendicular bisector of γ_j, γ_k. Let $\mathrm{VorInt} := \mathbb{C} \setminus \mathrm{VorBdy}$.

Proposition 17.1 (Analytic structure of the solution map)**.** The Borel solution map $\Phi : z \mapsto \gamma_{S_{\mathrm{Borel}}(z)}$ satisfies:

(i) **Borel-measurability.** Φ is Borel-measurable $\mathbb{C} \to \mathbb{C}$.

(ii) **Null boundary.** VorBdy has two-dimensional Lebesgue measure zero.

(iii) **A.e. continuity.** Φ is locally constant on VorInt, hence continuous there; combined with (ii), Φ is continuous at Lebesgue-a.e. $z \in \mathbb{C}$.

Proof. (i) Borel-measurability. The preimage $\{z : S_{\mathrm{Borel}}(z) = k\}$ is

$$\Big(\bigcap_{j\neq k}\{z : \|\gamma_k - z\| \leq \|\gamma_j - z\|\}\Big) \cap \Big(\bigcap_{j<k}\{z : \|\gamma_k - z\| < \|\gamma_j - z\|\}\Big),$$

a finite Boolean combination of closed (resp. open) half-planes in $\mathbb{C}$, hence Borel. So S_{Borel} is Borel-measurable, and Φ is its composition with $k \mapsto \gamma_k$ (continuous, a fortiori Borel).

(ii) Null boundary. Squaring $\|\gamma_j - z\| = \|\gamma_k - z\|$ and expanding in real coordinates yields the affine equation

$$B_{jk} = \{z \in \mathbb{C} : \mathrm{Re}(\overline{(\gamma_k - \gamma_j)}\, z) = \tfrac{1}{2}(|\gamma_k|^2 - |\gamma_j|^2)\},$$

a strict affine subspace of $\mathbb{C} \cong \mathbb{R}^2$ (Hausdorff dimension 1). Each B_{jk} therefore has two-dimensional Lebesgue measure zero. A finite union of null sets is null, so $\mathrm{vol}(\mathrm{VorBdy}) = 0$.

(iii) A.e. continuity. Fix $z_0 \in \mathrm{VorInt}$ and let $k_0 := S_{\text{Borel}}(z_0)$. By definition of VorInt, $\|\gamma_{k_0} - z_0\| < \|\gamma_j - z_0\|$ for all $j \neq k_0$, so some $\delta > 0$ satisfies $\|\gamma_{k_0} - z_0\| + 2\delta \le \|\gamma_j - z_0\|$ for $j \neq k_0$. By continuity of $z \mapsto \|\gamma_j - z\|$, there is $\eta > 0$ such that on $B(z_0, \eta)$ both quantities change by at most δ, giving $\|\gamma_{k_0} - z\| < \|\gamma_j - z\|$ for $j \neq k_0$, so $S_{\text{Borel}}(z) = k_0$ and $\Phi(z) = \gamma_{k_0}$. Thus Φ is locally constant, hence continuous, on VorInt. Combined with (ii), Φ is continuous at Lebesgue-a.e. z. □

For backwards reference we also expose the individual statements.

Theorem 17.1 (Borel measurability)**.** The map S_{Borel} is Borel-measurable; so is Φ.

Theorem 17.2 (Voronoï boundary is Lebesgue-null)**.** $\mathrm{vol}(\mathrm{VorBdy}) = 0$.

Theorem 17.3 (A.e. continuity)**.** Φ is continuous at every $z \in \mathrm{VorInt}$; combined with Theorem 17.2, Φ is continuous at Lebesgue-a.e. z.

Both theorems are direct consequences of Proposition 17.1, (ii) and (iii) respectively. Together with Theorem 16.1, they give the analytic character of the multi-start solution map: Borel-measurable everywhere, locally constant outside a Lebesgue-null set.

18 Lean 4 formalization: corpus structure and elementary identities

18.1 Corpus overview

The Lean development lives at `github.com/ivan-fr/poussiere_de_x` and targets Lean 4 with Mathlib v4.7.0. It comprises 121 modules with zero `sorry`; every load-bearing theorem referenced in this paper has an audited axiomatic closure reducing to the Lean whitelist {`propext`, `Classical.choice`, `Quot.sound`}.

Three layers of verifiability are offered to a third-party reviewer:

- **Continuous integration.** A GitHub Actions workflow (`.github/workflows/lean-build.yml`) compiles the full corpus on every push; a green CI badge in the repository `README.md` certifies that the current `main` branch compiles cleanly, with no `sorry`.
- **Axiom audit file.** The file `AXIOMS.md` at the repository root contains the raw `#print axioms` output for every load-bearing theorem cited in this paper; for each, the dependency closure reduces to the Lean whitelist {`propext`, `Classical.choice`, `Quot.sound`}.
- **Local reproduction.** Incremental build `bash scripts/lean-incremental.sh` takes 60–90 seconds on a modern laptop; per-theorem verification is available via `#print axioms <theorem>` at the end of the relevant module.

The modules are organised in layers: foundations (geometric sums, fixed-point characterization, Voronoï basins, cyclotomic anchor definitions), core spine (explicit super-attractive rate, global $\log\log$ convergence, real-line $p = 2$ discharge), multi-start (universal convergence, target selection, iteration count, compatibility bundle of §16.5), measure/topology (Borel measurability, Voronoï-null frontier, a.e. continuity), and a small set of classical specializations (Banach 1922, Aitken–Steffensen, Kronecker 1857, Aberth–Ehrlich, Kung–Traub attainability at $c = 3$).

18.2 Status of the main statements

To avoid conflating formal verification, paper proofs, and numerical evidence, Table 13 records the status of the claims used in this paper. The companion Parts I–VIII contain additional conjectural and empirical material; they are not included in the Lean claim below unless explicitly listed.

Statement	Paper proof	Lean proof	Numerical check	Status
Fixed-point identity $h_{p,x}(s) = s \iff s^p = 1/x$	yes	yes	yes	theorem
Closed-form real contraction ratio $\lambda_{p,x}$	yes	partial	yes	theorem
Monotonicity in α and in p at fixed α	yes	partial	yes	theorem
Steffensen local quadratic expansion	yes	yes, as local skeleton	yes	theorem
Complex target-conditioned multi-start convergence for $z^p = x$	yes	yes	yes	theorem
Borel measurability and a.e. continuity of the target selector	yes	yes	no	theorem
Unseeded global basin-volume lower bound	yes	no	yes	theorem
General-polynomial fixed-point identity for h_P	yes	no	yes	theorem
General-polynomial local Steffensen basin, target fixed in advance	yes	no	yes	theorem
Uniform quantitative advantage over Newton for arbitrary polynomials	no	no	mixed	not claimed
Coefficient-uniform Smale-17 complexity bound for this scheme	no	no	no	not claimed

Table 13: Status table for the claims in Paper 0. "Partial" means that the Lean corpus proves the load-bearing algebraic skeleton or special cases used by the formal development, while the full analytic statement is presented in the paper proof.

Statement-level `Props` that are NOT proven. A single archive module `PandrosionOpenProblems.lean` contains named Lean `Props` whose proofs are *deliberately absent*: the conjectural upper bound of Kung–Traub for $c \geq 3$, McMullen's 1987 impossibility theorem, Lehmer's Mahler-measure conjecture, and Smale's 17th problem. These are recorded as `axiom` or `Prop` declarations without accompanying proof terms. They are not used downstream by any load-bearing theorem of the corpus, and their presence in the repository should not be mistaken for formal verification of the corresponding results. A reader auditing the corpus can confirm this by checking that no `theorem` depending on these `Props` appears in the load-bearing chain. A future cleanup will move this archive module to a dedicated `archive` namespace to make the non-load-bearing status structurally explicit.

18.3 Scope statement on classical open problems

No claim of progress on classical open problems of polynomial root-finding is made. The corpus touches Kung–Traub ($c \geq 3$) through a compliance interface, not a proof, and records McMullen 1987 as a named statement without in-Lean proof (Mathlib v4.7.0 lacks the Fatou–Julia machinery needed). Lehmer's Mahler-measure conjecture and Smale's 17th problem are explicitly out of scope: the Pandrosion cyclotomic anchors produce algebraic integers whose Mahler measure is easy to compute (trivially far from Lehmer's extremal regime), and the multi-start $\mathcal{O}(\log\log(1/\varepsilon))$ complexity applies to the single-monomial family $z^p = x$, not to Smale's coefficient-uniform problem.

19 Three short technical notes

This section records three short items used implicitly earlier: cyclotomic cancellation on the anchor family (behind the a.e.-continuity argument of Part II), a no-cycle lemma for the seeded orbit, and a uniform $\log\log$ complexity bound quantifying the "$\approx$ 3–4 additional iterations" observed in the benchmark of §14.3. None is a headline result.

A. Cyclotomic cancellation on the anchor family

Let $P(z) = z^p - 1/x$ with $x \in \mathbb{C}^*$, $p \geq 2$; let $\alpha_0, \ldots, \alpha_{p-1}$ be its roots, $\alpha := x^{-1/p}$ principal, $\omega := e^{2\pi i/p}$. Then $\alpha_k = \alpha\omega^k$ and

$$P'(\alpha_k) \;=\; p\alpha_k^{p-1} \;=\; \frac{p}{x\alpha_k}, \qquad \nu_k := \frac{1}{P'(\alpha_k)} \;=\; \frac{x\alpha_k}{p}.$$

Proposition 19.1 (Cyclotomic cancellation)**.** For every $p \geq 2$ and every $s \in \mathbb{Z}$: $\sum_{k=0}^{p-1} \alpha_k^s = \alpha^s \sum_{k=0}^{p-1} (\omega^s)^k$, which equals $p\,\alpha^s$ if $p \mid s$ and 0 otherwise. The proof is a one-line geometric sum.

Three consequences used earlier:

(i) $\displaystyle\sum_{k=0}^{p-1} \nu_k = \frac{x}{p}\sum_k \alpha_k = 0$ for $p \geq 2$ (case $s = 1$).

(ii) Closed-form product: $\displaystyle\prod_{k=0}^{p-1} P'(\alpha_k) = \frac{p^p}{x^p \prod_k \alpha_k} = (-1)^{p-1}\frac{p^p}{x^{p-1}}$, using $\prod_k \alpha_k = (-1)^{p+1}/x$ by Vieta on $z^p - 1/x$.

(iii) For $p \geq 3$: $\sum_k P'(\alpha_k)^{-2} = (x/p)^2 \sum_k \alpha_k^2 = 0$ (case $s = 2$ in Proposition 19.1, with $p \geq 3$ giving $p \nmid 2$); and $\sum_k P'(\alpha_k)^2 = (p/x)^2 \sum_k \alpha_k^{-2} = 0$ (case $s = -2$, same condition $p \nmid 2$). In compact form: $\sum_k P'(\alpha_k)^{-s} = 0$ for every $s \in \mathbb{Z}$ with $p \nmid s$, so both $s = \pm 2$ vanish for $p \geq 3$, giving a "cyclotomic cancellation" of all non-constant moments.

These three items are the arithmetic ingredients used implicitly in Part II: (i) is the vanishing of the first residue sum that underlies the a.e.-continuity analysis, and (ii)–(iii) normalize constants in auxiliary bounds. Items (i)–(iii) have been cross-checked symbolically in arbitrary precision for $p \in \{2, 3, 5, 7, 10\}$ and representative x (real and complex).

The rotation $T_\omega : z \mapsto \omega z$ is a Euclidean isometry permuting $\alpha_k \to \alpha_{(k+1) \bmod p}$, hence permuting the Voronoï cells V_k; any rotation-invariant measure gives $\mu(V_k)$ independent of k. Combined with the Part II results (Theorems 17.1–17.3), this gives the "multi-start compatibility bundle" of Theorem 16.4 as a packaged observation, not a new theorem.

B. No-cycle lemma for seeded multi-start orbits

Lemma 19.1 (No non-trivial cycles)**.** Let (x, p, α) satisfy the standard hypotheses, and let $z_0 \in \overline{B}(\alpha, R_\sigma/2)$. If $\sigma_{p,x}^n(z_0) = \sigma_{p,x}^m(z_0)$ for some $0 \leq n < m$, then $\sigma_{p,x}^n(z_0) = \alpha$.

Proof. On $\overline{B}(\alpha, R_\sigma/2)$, the quadratic bound of Part II and $K_\sigma R_\sigma \leq 1/2$ give $|\sigma_{p,x}(z) - \alpha| \leq K_\sigma(R_\sigma/2)|z - \alpha| \leq \frac{1}{4}|z - \alpha|$. Hence $\sigma_{p,x}$ is $\frac{1}{4}$-Lipschitz with fixed point α, and forward-invariant. Induction gives $|w_j - \alpha| \leq 4^{-j}|z_0 - \alpha|$, $w_j := \sigma_{p,x}^j(z_0)$. If $w_n = w_m$, then $|w_n - \alpha| \leq 4^{-(m-n)}|w_n - \alpha|$ with $4^{-(m-n)} < 1$, forcing $|w_n - \alpha| = 0$. Since the Definition 16.1 seed lies in $\overline{B}(\alpha, R_\sigma/2)$, every seeded multi-start orbit is acyclic. $\square$

C. Uniform $\log\log$ complexity

Proposition 19.2 (Uniform $\log\log$ complexity)**.** For every compact $K \subset (1, \infty)$ and every $\varepsilon \in (0, 1)$, there exists $N(K, \varepsilon) = \mathcal{O}_K(\log\log(1/\varepsilon))$ such that for every $p \geq 2$, $x \in K$, $s_0 \in [1/2, 1]$, the Steffensen–Pandrosion iteration reaches $|v_n - x^{1/p}| \leq \varepsilon$ within $\leq N(K, \varepsilon)$ steps.

Proof. λ_∞ is continuous and strictly less than 1 on $(1, \infty)$, so $\lambda^*(K) := \sup_K \lambda_\infty < 1$. The Taylor bound (9) gives $\lambda_{p,x} \leq \lambda^*(K) + C^*(K)/p$ with $C^*(K) := \sup_K 3x(\ln x)^2/(x-1)$ for $p \geq \lceil \ln \sup K \rceil + 1$; on the remaining finite range of p, $\lambda_{p,x}$ is continuous on a compact and strictly < 1, so it is bounded away from 1. Hence $\bar{\lambda}(K) := \sup_{p \geq 2, x \in K} \lambda_{p,x} < 1$. The plain Pandrosion iteration converges linearly at this uniform rate; Aitken–Steffensen (Theorem 13.1) upgrades to quadratic with $K_S(K)$ uniform on K; Theorem 16.2 then gives $N(K, \varepsilon) = \mathcal{O}_K(\log\log(1/\varepsilon))$, uniformly in (p, x). $\square$

This quantitatively matches the per-regime benchmark of §14.3: the observed $\approx$ 3–4 additional iterations when moving from 50 to 500 digits, *across all tested* (p, x), is an instance of Proposition 19.2.

20 Beyond monomials: Pandrosion-style iteration on general polynomials

The paper has focused on the monomial family $z^p = x$ because its specific structure (cyclotomic symmetry, closed-form $\lambda_{p,x}$, scaling preconditioning) gives explicit quantitative results. We close with the observation that the *Pandrosion iteration itself* — not Newton's method — extends to every monic polynomial of degree p, preserving the derivative-free character that makes the monomial analysis attractive.

20.1 The generalised Pandrosion iteration

Let $P(z) = z^p + R(z)$ with $\deg R < p$, and let $S_p(s) = 1 + s + \cdots + s^{p-1}$ be the same geometric sum used throughout. Define

$$h_P(s) \; := \; 1 - \frac{R(s) + 1}{S_p(s)}. \tag{18}$$

Proposition 20.1 (Fixed points = roots of P)**.** For $s \in \mathbb{C}$ with $S_p(s) \neq 0$, $h_P(s) = s$ if and only if $P(s) = 0$. In particular, when $R(s) = -1/x$ is constant, h_P reduces to the monomial Pandrosion iteration $h_P(s) = 1 - (1 - 1/x)/S_p(s) = 1 - (x-1)/(x\, S_p(s))$, matching (3).

Proof. The key identity $(1 - s)\, S_p(s) = 1 - s^p$ (geometric sum) gives $h_P(s) = s \iff (1 - s)S_p(s) = R(s) + 1 \iff 1 - s^p = R(s) + 1 \iff s^p = -R(s) \iff P(s) = 0$. The monomial case is the specialisation $R(s) = -1/x$. $\square$

The iteration (18) is *derivative-free*: it uses only R (the sub-leading coefficients of P) and the universal polynomial S_p, never P' or P''.

Steffensen acceleration yields

$$\sigma_P(s) \; := \; s - \frac{(h_P(s) - s)^2}{h_P(h_P(s)) - 2h_P(s) + s},$$

which satisfies $\sigma_P(\alpha) = \alpha$ and $\sigma_P'(\alpha) = 0$ at every root α of P with $S_p(\alpha) \neq 0$ and $h_P'(\alpha) \neq 1$, by Lemma 13.2. The local quadratic constant is

$$K_S(P, \alpha) = \left| \frac{\lambda_P h_P''(\alpha)}{2(1 - \lambda_P)} \right|, \qquad \lambda_P := h_P'(\alpha).$$

Here

$$h'_P(s) = -\frac{R'(s)S_p(s) - (R(s)+1)S'_p(s)}{S_p(s)^2},$$

so λ_P is computable from finite differences or coefficients, without a derivative oracle for the original root-finding step. At a root α, the identity $R(\alpha) + 1 = (1-\alpha)S_p(\alpha)$ gives the more geometric form

$$\lambda_P = -\frac{R'(\alpha)}{S_p(\alpha)} + (1-\alpha)\frac{S'_p(\alpha)}{S_p(\alpha)}.$$

Theorem 20.1 (Explicit local basin for general-polynomial Steffensen)**.** Let $P(z) = z^p + R(z)$ and let α be a root of P such that $S_p(\alpha) \neq 0$ and $\lambda_P = h'_P(\alpha) \neq 1$. Choose $\rho > 0$ so that h_P and σ_P are holomorphic on the closed disk $\overline{B}(\alpha, \rho)$. Set

$$M_\rho(P,\alpha) := \sup_{|w-\alpha|\le\rho} |\sigma''_P(w)|, \qquad R_\sigma(P,\alpha) := \min\left(\rho, \frac{1}{M_\rho(P,\alpha)}\right),$$

with the convention $1/0 = +\infty$. Then for every $z \in \overline{B}(\alpha, R_\sigma)$,

$$|\sigma_P(z) - \alpha| \le \frac{M_\rho(P,\alpha)}{2}|z-\alpha|^2.$$

In particular, $\overline{B}(\alpha, R_\sigma)$ is forward-invariant and every orbit starting in $\overline{B}(\alpha, R_\sigma)$ converges quadratically to α.

Proof. By Lemma 13.2, $\sigma_P(\alpha) = \alpha$ and $\sigma'_P(\alpha) = 0$. Taylor's formula with integral remainder on the line segment from α to z gives

$$\sigma_P(z) - \alpha = \int_0^1 (1-t)\,\sigma''_P(\alpha + t(z-\alpha))(z-\alpha)^2\,dt,$$

hence the displayed bound. If $|z-\alpha| \le R_\sigma$, then $|\sigma_P(z) - \alpha| \le \frac{1}{2}|z-\alpha|$ because $M_\rho R_\sigma \le 1$, so the disk is forward-invariant and the errors satisfy $e_{n+1} \le (M_\rho/2)e_n^2$. ☐

The multi-start seed then places any $z \in \mathbb{C}$ inside the quadratic-contraction disk of the chosen target root. This is a *local refinement theorem*, not a root-discovery theorem: the target root α_k must already be specified.

Theorem 20.2 (Multi-start Pandrosion for general polynomials)**.** Let $P(z) = z^p + R(z)$ with $\deg R < p$, and let α_k be a simple root of P with $S_p(\alpha_k) \neq 0$ and $h'_P(\alpha_k) \neq 1$. For every $z \in \mathbb{C}$ and every target accuracy $\delta > 0$, define $z_0 = \alpha_k$ if $z = \alpha_k$, and otherwise define the multi-start seed by

$$z_0 := \alpha_k + \min\left(1, \frac{R_\sigma(P,\alpha_k)}{2\,|z-\alpha_k|}\right)(z-\alpha_k)$$

and the iterated Steffensen–Pandrosion map σ_P satisfy

$$|\sigma_P^n(z_0) - \alpha_k| \;\le\; \delta \qquad \text{for every } n \;\ge\; N_{\text{eff}}\big(K_\sigma(P,\alpha_k),\, R_\sigma(P,\alpha_k),\, \delta\big),$$

with $N_{\text{eff}} = \mathcal{O}(\log_2 \log(1/\delta))$ by Theorem 16.2.

Proof. The fixed-point property $h_P(\alpha_k) = \alpha_k$ is Proposition 20.1. Lemma 13.2 and Theorem 20.1 give $|\sigma_P(w) - \alpha_k| \le K_\sigma|w-\alpha_k|^2$ on $\overline{B}(\alpha_k, R_\sigma)$, with $K_\sigma = M_\rho(P,\alpha_k)/2$ and $K_\sigma R_\sigma \le 1/2$. The seed places z_0 at distance at most $R_\sigma/2$ from α_k, so forward invariance holds and $e_n := |\sigma_P^n(z_0) - \alpha_k|$ satisfies $e_{n+1} \le K_\sigma e_n^2$. The explicit iteration count is the elementary quadratic-recursion bound already recorded in Theorem 16.2. ☐

Remark 20.1 (Structural warning)**.** Theorem 20.2 is still *conditional on the target*: the seed requires α_k. It does not contradict McMullen's 1987 impossibility result on purely iterative universal root-finders. The Voronoï-nearest selector of Section 17 extends verbatim to d arbitrary distinct roots (not just cyclotomic): the Borel solution map Φ_P is Borel-measurable and a.e.-continuous, by the same proof (Proposition 17.1).

Remark 20.2 (Correspondence with the monomial compatibility bundle)**.** Theorem 20.2 is the general-polynomial analogue of the monomial compatibility bundle (Theorem 16.4), item by item:

- *Item (1), Voronoï-nearest target selection.* Transports verbatim: for any finite distinct root set $\{\alpha_0, \ldots, \alpha_{p-1}\}$ and any $z_0 \in \mathbb{C}$, there exists s^* with $\|\alpha_{s^*} - z_0\| \leq \|\alpha_t - z_0\|$ for every t. The proof in Theorem 16.4 uses only finiteness, not the monomial structure.

- *Item (2), Steffensen fixing.* Transports: $\sigma_P(\alpha_k) = \alpha_k$ holds at every simple root α_k with $S_p(\alpha_k) \neq 0$, by the same Aitken argument as Theorem 13.1.

- *Item (3), Scaling compatibility.* **Transports with real benefit.** The substitution $z = \beta w$ gives $P(\beta w) = \beta^p w^p + R(\beta w) = A(w^p + R(\beta w)/A)$ with $A = \beta^p$, so the rescaled problem is $\tilde{P}(w) := w^p + \tilde{R}(w)$ with $\tilde{R}(w) := R(\beta w)/A$. The new roots are $\tilde{\alpha}_k = \alpha_k/\beta$, so if β is chosen close to $|\alpha_k|$ — for instance $\beta = \lfloor|\alpha_k|\rceil$, the nearest integer — then $|\tilde{\alpha}_k| \approx 1$, which is the well-conditioned regime for Pandrosion. One then applies multi-start Pandrosion–Steffensen to $\tilde{P}$, and recovers $\alpha_k = \beta \cdot \tilde{\alpha}_k$ by the trivial reconstruction of item (4). This is a genuine generalisation of Proposition 14.1: scaling produces another polynomial of the same form $w^p + \tilde{R}(w)$, with the target root compressed close to 1. The quantitative benefit is substantial (Table 16 below).

- *Item (4), Reconstruction.* Transports trivially: $\alpha_k = \beta \cdot (\alpha_k/\beta)$ for any $\beta \neq 0$.

- *Item (5), Optimal starting point.* Replaced by the multi-start seed. For general P, $h_P(1) = 1 - (R(1) + 1)/S_p(1) = 1 - (R(1) + 1)/p$ is still a well-defined candidate; whether it is the canonical starting point depends on the geometry of the particular P, and we do not claim optimality outside the monomial case.

So items (1), (2), (3), (4) transport — item (3) through a polynomial-to-polynomial substitution rather than a coefficient-only reduction, but with a genuine quantitative payoff (Table 16); item (5) is replaced by the target-conditional seed. Theorem 20.2 therefore captures the full Grand Master bundle in the non-monomial setting, with the seed playing the role that the canonical real starting point played on $\mathbb{R}$.

20.2 Numerical evidence

The script `latex/scripts/benchmark_general_poly.py` applies the generalised Pandrosion–Steffensen scheme (18) to four non-monomial polynomials at 50-digit precision, starting from $z_0 = 3 + 3i$. The radius $R_\sigma(P, \alpha_k)$ is estimated empirically by bisection on the circle $\partial B(\alpha_k, R)$.

Polynomial	p	Empirical R_σ range	Iterations (per target root)
$z^3 - 2$ (monomial, $x = 2$)	3	1.29	4
$z^5 - z - 1$	5	0.38–0.85	6–8
$z^3 - 2z - 5$ (Bombelli)	3	0.06–1.67	8–41
$z^4 + z + 1$	4	0.12–0.21	7–40
$z^6 + 3z^2 - 1$	6	0.03–0.44	7–11

Table 14: Multi-start Pandrosion–Steffensen (Theorem 20.2) for non-monomial polynomials from $z_0 = 3+3i$ at precision 10^{-50}, using $h_P(s) = 1-(R(s)+1)/S_p(s)$ followed by Aitken Δ^2. For each polynomial and each target root (chosen in turn), the multi-start scheme converges to that root; iteration counts vary with the local multiplier $|h'_P(\alpha_k)|$ and with the empirical R_σ. The slow Bombelli case (real root at $\alpha \approx 2.094$: 41 iterations) corresponds to $h'_P(\alpha)$ near 1, inflating the Aitken constant $K_\sigma \propto 1/(1-|h'_P(\alpha)|)$.

The iteration converges to every target root of every tested polynomial: the scheme works structurally on any monic polynomial of degree p. Quantitatively, however, the iteration count is *not* uniform: it ranges from 4 iterations (monomial) to 41 iterations (Bombelli's real root and one root of $z^4 + z + 1$) — a factor of 10× gap.

20.3 Scaling with the degree p

A natural follow-up question: for fixed precision, does the iteration count scale with the degree p? The script `latex/scripts/benchmark_large_p.py` tests two families at $p \in \{3, 5, 10, 20, 50, 100, 200\}$, targeting the principal real root in each case at 50-digit precision from $z_0 = 3 + 3i$:

(A) $P(z) = z^p - 2$: the pure monomial, maximally well-conditioned.

(B) $P(z) = z^p - z - 1$: Bring-style trinomial with one real root $\alpha \approx 2^{1/p}$ for large p.

p	**(A) $z^p - 2$ (monomial)** R_σ	iters	**(B) $z^p - z - 1$ (trinomial)** R_σ	iters
3	1.09	6	0.16	5
5	0.68	6	0.084	5
10	0.33	10	0.038	20
20	0.16	13	0.018	28
50	0.064	15	0.0071	33
100	0.032	15	0.0035	35
200	0.016	16	0.0017	36
500	0.0063	16	0.00069	36
1 000	0.0031	**16**	0.00035	37
10 000	0.00031	**16**	0.000035	**37**

Table 15: Iteration counts of multi-start Pandrosion–Steffensen for degree-p polynomials to reach 50-digit precision from $z_0 = 3 + 3i$. The contraction radius $R_\sigma(P, \alpha)$ scales as $1/p$ (cyclotomic gap for (A), near-unit-circle clustering for (B)). Iteration count **saturates** at a plateau independent of p: 16 for the monomial, 37 for the trinomial. Wall-clock at $p = 10\,000$: 3.7 s (monomial), 4.6 s (trinomial); essentially linear in p from the p-dependent cost of evaluating $S_p(s)$ via the closed-form $(s^p - 1)/(s - 1)$, not from the iteration count itself.

Observations.

1. **The iteration count saturates.** For the monomial family, the count plateaus at 16 iterations for every p between 200 and 10 000. For the trinomial, the count plateaus at

36–37 iterations for every p between 200 and 10 000. This is the uniform-in-p behaviour of Proposition 19.2 made experimental: fixed precision, fixed x (here $x = 2$ or the trinomial coefficient vector), iteration count bounded by a constant independent of p.

2. **The scheme works at $p = 10\,000$.** Both families converge in well under 40 iterations; the wall-clock scales linearly with p only through the arithmetic cost of computing $S_p(s) = (s^p - 1)/(s - 1)$, which is an $\mathcal{O}(\log p)$ multi-precision exponentiation, not a p-sum.

3. **Trinomial $\approx 2\times$ slower than monomial at the plateau.** For $p \gtrsim 200$: 37 vs 16 iterations. The extra factor reflects $h'_P(\alpha) \approx -0.5$ for the trinomial (vs ≈ 0.22 for $z^3 - 2$), inflating the Aitken pre-asymptotic regime uniformly.

4. **R_σ decays like $1/p$ but the iteration count does not grow**. The plateau behaviour at fixed precision is a consequence of the $\log_2 \log$ asymptotic: $N_{\text{eff}}(K_\sigma, R_\sigma, \delta) \sim \log_2 \log(1/\delta) + O(\log(K_\sigma R_\sigma))$, and the correction $\log(K_\sigma R_\sigma)$ stays bounded even as $R_\sigma \to 0$ because $K_\sigma R_\sigma \le 1/2$ is enforced by the cascade construction.

For the monomial family, the plateau empirically confirms Proposition 19.2: over the compact regime $K = \{2\}$, the count is $\mathcal{O}(\log \log(1/\delta))$ uniformly in $p \in [2, \infty)$.

For the trinomial, Proposition 19.2 does not apply directly (the argument is not a monomial parameter), but the saturation at 37 iterations shows that the theorem's uniform behaviour extends empirically to at least this trinomial family. A follow-up paper characterising the class of polynomials for which the plateau holds would be of independent interest.

The slow cases all share the same mechanism: either $R_\sigma(P, \alpha_k)$ is small (the fixed point sits close to a singularity of h_P where $S_p(s) = 0$, or to another root), or $h'_P(\alpha_k)$ approaches 1, inflating the Aitken constant $K_S \propto 1/(1 - |h'_P(\alpha_k)|)$ and delaying entry into the quadratic regime.

Compared against Newton on the same polynomials (which converges in 6–14 iterations), multi-start Pandrosion is *competitive* in the well-conditioned cases (monomial, quintic, sextic) but *significantly slower* in the ill-conditioned cases — a direct cost of being derivative-free in a setting that does not have the monomial's closed-form protection. The honest verdict: the scheme transports to general polynomials, but the *quantitative* advantage of the monomial case does not.

20.4 What is lost, what transports

Lost: the monomial setting enjoys three advantages not available in general:

1. The closed-form contraction ratio $\lambda_{p,x}$ of Theorem 5.1: for h_P with general R, the multiplier $h'_P(\alpha)$ is computable per root but has no uniform closed form, and can be near 1 for badly-conditioned roots.

2. Cyclotomic symmetry of the root set: p equally-spaced roots on a circle, with congruent Voronoï cells and uniform basin lower bounds πR_σ^2 (Remark 16.5). General P has roots in generic position.

3. The specific large-p limit $\lambda_\infty(x) = 1 - \ln(x)/(x - 1)$: tied to $R(s) = -1/x$.

Transports: five structural features carry over:

1. **Derivative-free construction.** h_P in (18) uses only R and S_p, never P'. This is the defining feature of the Pandrosion family and is preserved.

2. **Steffensen acceleration.** $\sigma_P = \text{Aitken}(h_P)$ is super-attractive at every simple root with $h'_P(\alpha) \ne 1$, by the same Taylor argument as Theorem 13.1; the constant is $K_S = |\lambda_P \cdot h''_P(\alpha)/(2(1 - \lambda_P))|$.

3. **Multi-start seed + $\log_2 \log$ count.** Theorems 16.1 and 16.2 apply verbatim; N_{eff} depends only on $(K_\sigma, R_\sigma, \delta)$, not on the specific polynomial.

4. **Measurability and a.e.-continuity.** Section 17 extends to any d distinct roots in the plane — cyclotomic symmetry is not used in those proofs, only distinctness and the affine structure of perpendicular bisectors.

5. **Scaling preconditioning.** Proposition 14.1 *does* transport, in the following form. The substitution $z = \beta w$ yields

$$P(\beta w) \;=\; \beta^p w^p + R(\beta w) \;=\; A\big(w^p + R(\beta w)/A\big), \qquad A := \beta^p,$$

so the rescaled problem is $\tilde{P}(w) := w^p + \tilde{R}(w)$ with

$$\tilde{R}(w) \;=\; R(\beta w)/A, \qquad \tilde{R}_i \;=\; R_i\, \beta^i / A.$$

The new target root is $\tilde{\alpha}_k = \alpha_k/\beta$. Choosing β close to $|\alpha_k|$ places $\tilde{\alpha}_k$ near the unit circle and dramatically improves conditioning: in w-space the iteration $\tilde{h}(w) = 1 - (\tilde{R}(w) + 1)/S_p(w)$ and its Aitken acceleration $\tilde{\sigma}$ have a moderate $\tilde{\lambda}$ and are well-scaled, whereas the unscaled iteration on P may have $|h'_P(\alpha)|$ near 1. Table 16 exhibits the speedups.

$P(z)$	$\lvert\alpha\rvert$	iters (unscaled)	β	iters (scaled)
$z^3 - 2z - 100$	4.83	156	5	14
$z^5 - 3z - 1000$	3.98	35	4	13
$z^4 + 2z^3 - 1000$	5.03	30	5	6

Table 16: Iteration counts of multi-start Pandrosion–Steffensen before and after the scaling $z = \beta w$ with $\beta = \lfloor |\alpha| \rceil$. Target: 50-digit precision from $z_0 = 3 + 3i$ (mapped to $w_0 = z_0/\beta$ in w-space). The transport of scaling gives speedups of $11\times$, $2.7\times$, and $5\times$ respectively on polynomials with roots far from the unit circle.

Theorem 20.2 is therefore a *genuine* extension of the Pandrosion-Steffensen scheme — the same derivative-free iteration family, applied to any degree-p polynomial — not a Newton-based reformulation. The monomial case is the favourable instance where the derivative-free character combines with closed-form analysis; the general case preserves the derivative-free character and the target-conditional $\log_2 \log$ convergence, at the cost of closed forms. A follow-up paper taking the generalisation seriously would focus on: (a) polynomials with cyclic Galois group (partial cyclotomic symmetry); (b) perturbative series $R(s) = -1/x + \varepsilon g(s)$ as small-deformation case studies with controlled λ_P; (c) a proper algorithmic comparison against Aberth–Ehrlich and Weierstrass simultaneous root finders on arbitrary polynomials.

21 Conclusion

The central tension of this paper is between Pandrosion's ancient geometric construction — an order-one iteration requiring only a ruler and parallels — and modern acceleration theory.

Once the universal iteration $s_{n+1} = 1 - (x-1)/(x\, S_p(s_n))$ is written down, three ingredients suffice to make it competitive on $\mathbb{R}$: the closed-form contraction ratio $\lambda_{p,x}$ (Theorem 5.1), the preconditionings of Propositions 12.1 and 14.1, and the Aitken Δ^2 upgrade (Theorem 13.1).

The ratio $\lambda_{p,x} = h'(s^*)$ is monotone in the root parameter $\alpha = x^{1/p}$ and in p at fixed α (Theorem 9.1, via the closed-form identities $P_p = p \cdot D'_p$ and $B_p = \sum_{j=1}^{p} j\alpha^{j-1}$), has an explicit large-p limit $1 - \ln(x)/(x-1)$ (Proposition 9.1), and admits a clean threshold analysis for $x < 1$ (Proposition 11.1).

On $\mathbb{C}$, the extension via the target-dependent seed of Definition 16.1 yields a universal multi-start convergence theorem (Theorem 16.1) with a closed-form $N_{\text{eff}} \sim \log_2 \log(1/\varepsilon)$ iteration count (Theorem 16.2), together with Borel-measurability and a.e.-continuity of the induced solution map (Theorems 17.1–17.3). All of these results are formally verified in the Lean 4 corpus described in Section 18.

The paper does not claim progress on classical open problems of polynomial root-finding, and does not claim a general algorithmic advantage over Newton or its modern variants. Its main deliverable is a concrete derivative-free root-finding scheme with closed-form convergence analysis and complete formal verification for the monomial family $z^p = x$, connecting a ruler-and-parallels construction from fourth-century Alexandria to contemporary formal-methods practice.

Perspectives. We see four concrete, tractable extensions as immediate next steps:

1. **Benchmark against Schröder / König families of order $k \geq 3$** at multi-precision, to quantify how the factor-of-3 cubic gap of Halley over Newton compares against purely algebraic higher-order iterations (the König family is derivative-free in spirit and admits closed-form constants).

2. **Extend to complex x** (not just complex anchor α): the current Part II assumes $x \in \mathbb{C}^*$ in the statements but the analysis is principally real for the scaling step. A fully complex scaling requires choosing β in the ring of integers of the appropriate cyclotomic field — $\mathbb{Z}[i]$ for $p = 2$ (Gaussian integers), $\mathbb{Z}[\omega]$ with $\omega = e^{2\pi i/3}$ for $p = 3$ (Eisenstein integers), $\mathbb{Z}[\zeta_p]$ for general p — close to $X^{1/p}$ in the corresponding lattice. The machinery extends but the approximation $\beta \approx X^{1/p}$ in $\mathbb{Z}[\zeta_p]$ is non-trivial for $p \geq 5$ (no simple integer-part operation), and bounds on $\lambda_{p,X/\beta^p}$ require control of the discretization error in this richer ring.

3. **Formalize the non-asymptotic bounds of Section 10 in Lean**, closing the gap between the analytic quantity Λ and its in-Lean counterpart. This is a routine but useful extension of `UniformContractionRate.lean`.

4. **Matching upper bound on the basin** (Remark 16.5 pointed the way): show that the full basin Ω_k equals $\bigcup_{m\geq 0} \sigma_{p,x}^{-m}(\overline{B}(\gamma_k, R_\sigma))$ up to a Lebesgue-null set, yielding a matching upper bound on $\mathrm{vol}(\Omega_k)$. This stops short of requiring Fatou–Julia machinery.

A Fatou–Julia formalization in Mathlib — which would enable a proper in-Lean proof of McMullen's 1987 theorem — is a much larger project and is mentioned here only for completeness, not as an immediate extension.

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Part I — Adaptive-anchor Pandrosion, univariate and multivariate

A Pandrosion Operator for Derivative-Free Polynomial Root-Finding

Universal polynomial evaluation, non-local geometry, and an algorithmic companion to Smale's 17th problem

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Abstract

We extend the generalized Pandrosion iteration — a geometric fixed-point method originally designed for computing p-th roots via the geometric progression polynomial $S_p(z)$ — from the real line to the complex plane and, crucially, to *polynomial systems* $F : \mathbb{C}^n \to \mathbb{C}^n$, the proper setting of Smale's 17th problem (1998).

The univariate Pandrosion sequence is the special case $n = 1$ of a *universal evaluation generator*: by anchoring the matrix-valued difference quotient $\mathbf{Q}_F(\mathbf{z}_0, \mathbf{z})$ defined by the telescoping identity $F(\mathbf{z})-F(\mathbf{z}_0) = \mathbf{Q}_F(\mathbf{z}_0, \mathbf{z})\,(\mathbf{z}-\mathbf{z}_0)$, we define a non-local rational iteration $\mathbf{z}_{n+1} = \mathbf{z}_0 - \mathbf{Q}_F(\mathbf{z}_0, \mathbf{z}_n)^{-1} F(\mathbf{z}_0)$ that recovers the exact p-th root formula in $n = 1$, the multivariate Newton iteration as the limiting case where the anchor tracks the iterate, and applies uniformly to any polynomial map F.

For univariate $P(z) = xz^p - 1$ we characterize the complex contraction ratio.

Our main rigorous contributions, in both univariate ($n = 1$) and multivariate ($n \geq 2$) settings, are threefold: (i) a universal local convergence theorem establishing an explicit contraction basin of radius $\eta_F(\boldsymbol{\zeta})$ around every regular zero $\boldsymbol{\zeta}$ of F; (ii) exact polynomial factorisations avoiding the singular Σ barriers of Newton's method completely; (iii) a macroscopic kinematic identity yielding a bounded telescopic product.

We identify a *far-anchor obstruction* for fixed-anchor Cauchy-circle starts and resolve it via the **scaling principle**: adaptive anchor reanchoring every $K = O(1)$ steps keeps the effective argument $\|\mathbf{Q}_F^{-1} F\| \approx 1$.

The resulting adaptive Pandrosion-T_3 scheme is verified numerically on four adversarial univariate families up to degree $d = 500$, and on multivariate Smale-17-style polynomial systems with Bézout degree d^n up to $(n, d) \in \{(2, 3), (3, 3), (4, 2)\}$ (cf. `latex/scripts/benchmark_smale17_v4.py`).

Conditionally on Conjecture 5.19, this yields a deterministic, derivative-free $O(d^2 \log d)$-operation BSS algorithm for univariate root-finding; the multivariate Smale-17 average-case $O(\mathrm{poly}(d, n))$ bound is established unconditionally only by Beltrán–Pardo [4] and Lairez [5], and our scheme provides a derivative-free *algorithmic alternative* with comparable empirical complexity, not a competing complexity-theoretic resolution. We close with a formal Pandrosion-language rewriting of the Euler product for $\zeta(s)$.

Reader's guide. This is the first companion paper after the verified monomial paper. It starts the *algorithmic* branch of the corpus: the difference-quotient operator, adaptive anchors, and multivariate lift introduced here are the objects that Parts VII–VIII later modify with fallbacks and start strategies. Parts II–VI are a separate MVC branch; they use the same quotient notation but address a different Smale problem and should not be read as prerequisites for the algorithmic complexity statements below.

Distinction from Smale's Mean Value Conjecture. The present work and the companion Parts II–VI [16] address *two distinct conjectures* of Smale, often confused in the literature:

- **Smale's Mean Value Conjecture (MVC, 1981, [17])**: a univariate inequality $\min_{P'(\zeta)=0} |P(\zeta)/(\zeta P'(($ $(d-1)/d$ for monic polynomials P of degree d with $P(0) = 0$. Treated in Parts II–VI.
- **Smale's 17th Problem (1998, [18])**: a complexity-theoretic problem about the average-case cost of finding an approximate zero of a *system* $F : \mathbb{C}^n \to \mathbb{C}^n$ of n polynomials in n unknowns, with Bézout degree d^n. Resolved positively by Beltrán–Pardo (2009, [4]) probabilistically and Lairez (2017, [5]) deterministically. Treated in this Part I and in Parts VII–VIII.

The present paper develops the algorithmic framework for the latter; we do *not* claim to improve upon the worst-case complexity bound of [5], but provide a derivative-free alternative (cost only one polynomial-evaluation oracle per step, not Jacobian inverse) with explicit per-orbit constants and empirical evidence on multivariate systems matching the multivariate Pandrosion operator $\mathbf{Q}_F$ defined in §3.5.

Contents

1 Introduction

In Paper 0, we studied the generalized Pandrosion iteration

$$s_{n+1} = 1 - \frac{x-1}{x \cdot S_p(s_n)}, \qquad S_p(s) = \sum_{k=0}^{p-1} s^k, \qquad v_n = x \cdot s_n^{p-1}, \tag{1}$$

for real $x > 0$ and integer $p \geq 2$. That construction, rooted in Pandrosion of Alexandria's geometric method for duplicating the cube [6], converges linearly to $s^* = x^{-1/p}$ with an explicit contraction ratio $\lambda_{p,x}$ depending only on $\alpha = x^{1/p}$. Steffensen's acceleration transforms the linear method into a quadratically convergent, derivative-free algorithm.

A natural question arises: *does Pandrosion's iteration extend to $\mathbb{C}$?* Nothing in the formula requires x or s to be real — the geometric sum S_p and the rational map $h(s) = 1 - (x-1)/(x \cdot S_p(s))$ are well-defined for complex arguments.

This extension is not merely formal. By moving to $\mathbb{C}$, we expose three structural features:

- **Uniform anchor-based evaluation**: Pandrosion's iteration is a special case of an anchor-based difference-quotient scheme applicable to any polynomial in $\mathbb{C}[z]$, and lifts formally to polynomial systems via a matrix-valued quotient $\mathbf{Q}_F$.

  This provides a derivative-free algorithmic framework of independent interest, and an algorithmic companion to — *not* a competing complexity-theoretic resolution of — Smale's 17th problem (already resolved by Beltrán–Pardo [4] and Lairez [5]).

- **Fractal boundary structure**: the induced complex dynamics exhibit self-similar basin boundaries distinct from those of Newton's method. We describe this structure qualitatively and do not claim new results on complex dynamics itself.

- **A structural rewriting of the Euler product**: the infinite geometric sum $S_\infty(s) = (1-s)^{-1}$ recasts the Euler factors of $\zeta(s)$ as "infinite Pandrosion sums". This is a notational observation and implies *no* consequence for the Riemann Hypothesis.

Outline. Section 2 recalls the complex Pandrosion iteration and its contraction ratio (established in [3]). Section 3 presents the universal polynomial generalization. Section 4 introduces the adaptive-anchor algorithm. Section 5 develops the rigorous convergence theory. Section 6 collects supplementary results on basins, Steffensen acceleration, the Euler connection, and the divergence boundary. Section 7 concludes.

2 Complex Pandrosion iteration: recap

Definition 2.1 (Complex Pandrosion iteration). *Let $x \in \mathbb{C} \setminus \{0, 1\}$ and $p \geq 2$. The complex Pandrosion iteration is*

$$h(s) = 1 - \frac{x-1}{x \cdot S_p(s)}, \qquad S_p(s) = \sum_{k=0}^{p-1} s^k. \tag{2}$$

Its p fixed points are $s_k^* = \alpha_k^{-1}$, where $\alpha_k = x^{1/p} e^{2\pi i k/p}$, $k = 0, \ldots, p-1$.

Theorem 2.2 (Complex contraction ratio). *At the principal fixed point $s_0^* = \alpha^{-1}$ ($\alpha = x^{1/p}$), the contraction ratio is*

$$\lambda_{p,x} = h'(s_0^*) = \frac{(\alpha - 1) \sum_{k=1}^{p-1} k \alpha^{p-1-k}}{\sum_{k=0}^{p-1} \alpha^k}, \qquad \alpha = x^{1/p} \in \mathbb{C}. \tag{3}$$

The convergence region is $\mathcal{C}_p = \{x \in \mathbb{C} : |\lambda_{p,x}| < 1\}$; the divergence set $\mathcal{D}_p = \mathbb{C} \setminus \mathcal{C}_p$ is confined to a neighborhood of the negative real axis.

3 Universal polynomial generalization

The classical Pandrosion map was derived for p-th roots ($s^* = x^{-1/p}$). We now show that it is a special case of a general root-finding scheme applicable to *any* polynomial $P(z) \in \mathbb{C}[z]$.

3.1 The difference quotient operator

Let $P(z) = 0$ be a polynomial equation with unknown root z^*. Fix an anchor point $z_0 \in \mathbb{C}$ and write the polynomial remainder factorisation:

$$P(z) - P(z_0) = (z - z_0) \cdot Q(z_0, z), \tag{4}$$

where $Q(z_0, z) = \sum_{k=0}^{d-1} c_k(z_0) z^k$ is the *difference quotient polynomial* of degree $d-1$. Since $P(z^*) = 0$, we obtain $z^* = z_0 - P(z_0)/Q(z_0, z^*)$, yielding the fixed-point iteration:

$$\boxed{z_{n+1} = \mathcal{P}_{P,z_0}(z_n) = z_0 - \frac{P(z_0)}{Q(z_0, z_n)}.} \tag{5}$$

We call $\mathcal{P}_{P,z_0}$ the *Generalized Pandrosion Operator*.

3.2 Recovering the classical iteration

For $P(s) = xs^p - 1$ with anchor $z_0 = 1$, we have $P(1) = x - 1$ and $P(s) - P(1) = x(s^p - 1) = x(s-1)S_p(s)$, so $Q(1, s) = x \cdot S_p(s)$. The operator gives exactly

$$s_{n+1} = 1 - \frac{x - 1}{x \cdot S_p(s_n)},$$

the classical Pandrosion iteration.

3.3 Newton as a limiting case

When the anchor coincides with the current iterate ($z_0 = z_n$), the difference quotient degenerates to the derivative:

$$Q(z_n, z_n) = \lim_{z \to z_n} \frac{P(z) - P(z_n)}{z - z_n} = P'(z_n).$$

The operator reduces to $z_{n+1} = z_n - P(z_n)/P'(z_n)$, which is exactly Newton's method. Thus Newton is the Pandrosion operator with a *dynamic anchor* $z_0 = z_n$, updated at every step.

3.4 The fixed-anchor obstruction

With a fixed anchor z_0, the operator has poles wherever $Q(z_0, z_n) = 0$. Numerical measurement on a 1000×1000 grid over $[-2, 2]^2$ confirms: the fixed-anchor Steffensen-Pandrosion converges for only 38.6% of starting points on $z^3 - 1$, compared to 100% for Newton.

3.5 Multivariate extension to systems $F : \mathbb{C}^n \to \mathbb{C}^n$

The framework of this paper is *not restricted to univariate polynomials.* The construction extends verbatim to systems $F = (F_1, \ldots, F_n) : \mathbb{C}^n \to \mathbb{C}^n$ of n polynomials in n unknowns — the proper setting of Smale's 17th problem (1998, [18, 4, 5]).

Acknowledgment of classical prior art. The matrix-valued divided difference $\mathbf{Q}_F$ that we use is *not new*. It is the *Schmidt slope matrix* (also called the *Steffensen divided difference operator for systems*), introduced in [21] and developed systematically in [22, Chapter 11]. The corresponding fixed-point iteration $\mathbf{z}_{n+1} = \mathbf{z}_0 - \mathbf{Q}_F^{-1} F(\mathbf{z}_0)$ is the classical *multivariate secant method* when the anchor is fixed, and reduces to multivariate Newton when the anchor tracks the iterate. Recent derivative-free methods exploiting variants of this slope matrix include Grau-Sánchez–Noguera–Amat [24] (Ostrowski-type), Amat–Busquier [25] (Steffensen-Newton hybrids), and the textbook treatment of Potra–Pták [23]. Our *specific* contribution here is not the slope matrix itself, but its combination with: (i) the *periodic adaptive reanchoring* of Definition 3.3 (the scaling principle borrowed from [1]), (ii) the higher-order Aitken-Steffensen acceleration T_3, and (iii) the non-holomorphic fallback of [14]. We restate Schmidt's slope matrix below for self-containment.

Definition 3.1 (Schmidt slope matrix; multivariate Pandrosion difference quotient)**.** *Let $F : \mathbb{C}^n \to \mathbb{C}^n$ be a polynomial map and $\mathbf{z}_0, \mathbf{z} \in \mathbb{C}^n$ with $\mathbf{z} \neq \mathbf{z}_0$. The matrix-valued Pandrosion difference quotient is the $n \times n$ complex matrix $\mathbf{Q}_F(\mathbf{z}_0, \mathbf{z})$ defined by the telescoping identity*

$$F(\mathbf{z}) - F(\mathbf{z}_0) \;=\; \mathbf{Q}_F(\mathbf{z}_0, \mathbf{z})\,(\mathbf{z} - \mathbf{z}_0), \tag{6}$$

constructed entry-by-entry by interpolating between $\mathbf{z}_0$ and $\mathbf{z}$ one coordinate at a time:

$$\big[\mathbf{Q}_F(\mathbf{z}_0, \mathbf{z})\big]_{i,j} \;=\; \frac{F_i\big(\mathbf{z}_0[0..j-1] \,|\, \mathbf{z}[j] \,|\, \mathbf{z}[j+1..]\big) - F_i\big(\mathbf{z}_0[0..j-1] \,|\, \mathbf{z}_0[j] \,|\, \mathbf{z}[j+1..]\big)}{\mathbf{z}[j] - \mathbf{z}_0[j]}, \tag{7}$$

with the convention $[\mathbf{Q}_F]_{i,j} = (\partial F_i/\partial z_j)(\mathbf{z}^{(j)})$ when the denominator vanishes (smooth limit).

Remark 3.2 (Structural meaning of $\mathbf{Q}_F$ and its history)**.** *The matrix $\mathbf{Q}_F(\mathbf{z}_0, \mathbf{z})$ is the multivariate generalisation of the univariate divided difference $Q(z_0, z) = (P(z) - P(z_0))/(z - z_0)$, originally constructed by Schmidt [21] for the analysis of the multivariate secant method. It satisfies $\mathbf{Q}_F(\mathbf{z}, \mathbf{z}) = DF(\mathbf{z})$ (the Jacobian of F at $\mathbf{z}$) in the diagonal limit, recovering Newton's method as the dynamic-anchor case (cf. [22, Theorem 11.2.5]). Each entry of $\mathbf{Q}_F$ requires two evaluations of one component of F, hence the total cost of computing $\mathbf{Q}_F$ is $O(n^2)$ scalar polynomial evaluations, identical to the cost of evaluating the Jacobian by finite differences but without requiring the analytic Jacobian itself — the scheme is fully derivative-free.*

Definition 3.3 (Adaptive reanchoring; restated for the multivariate case)**.** *The multivariate adaptive Pandrosion-T_3 scheme alternates: (a) $K = 3$ Steffensen-accelerated steps of $\mathcal{P}_{F,\mathbf{z}_0}$ from a fixed anchor $\mathbf{z}_0$, then (b) reanchor $\mathbf{z}_0 \leftarrow \mathbf{z}_K$. The novelty is the period-K reanchoring, not the operator $\mathcal{P}_{F,\mathbf{z}_0}$ itself, which is classical (cf. above).*

Definition 3.4 (Multivariate Pandrosion operator)**.** *For a polynomial map $F : \mathbb{C}^n \to \mathbb{C}^n$ and an anchor $\mathbf{z}_0 \in \mathbb{C}^n$, the Multivariate Generalized Pandrosion Operator is*

$$\boxed{\mathcal{P}_{F,\mathbf{z}_0}(\mathbf{z}) \;=\; \mathbf{z}_0 - \mathbf{Q}_F(\mathbf{z}_0, \mathbf{z})^{-1}\, F(\mathbf{z}_0),} \tag{8}$$

defined whenever $\mathbf{Q}_F(\mathbf{z}_0, \mathbf{z})$ is invertible.

Theorem 3.5 (Multivariate Pandrosion fixes regular zeros)**.** *Let $\boldsymbol{\zeta} \in \mathbb{C}^n$ be a regular zero of F (i.e., $F(\boldsymbol{\zeta}) = 0$ and $\det DF(\boldsymbol{\zeta}) \neq 0$). For every anchor $\mathbf{z}_0$ in a sufficiently small neighbourhood $\mathcal{U}$ of $\boldsymbol{\zeta}$ and every $\mathbf{z} \in \mathcal{U}$, the operator $\mathcal{P}_{F,\mathbf{z}_0}(\mathbf{z})$ is well-defined and $\mathcal{P}_{F,\mathbf{z}_0}(\boldsymbol{\zeta}) = \boldsymbol{\zeta}$.*

Proof. By continuity, $\mathbf{Q}_F$ is invertible in a neighbourhood of $\boldsymbol{\zeta}$ since $\mathbf{Q}_F(\boldsymbol{\zeta}, \boldsymbol{\zeta}) = DF(\boldsymbol{\zeta})$ is invertible by hypothesis. The fixed-point property follows from $F(\boldsymbol{\zeta}) = 0$: substituting $\mathbf{z} = \boldsymbol{\zeta}$ into (6) gives $-F(\mathbf{z}_0) = \mathbf{Q}_F(\mathbf{z}_0, \boldsymbol{\zeta})\,(\boldsymbol{\zeta} - \mathbf{z}_0)$, hence $\mathcal{P}_{F,\mathbf{z}_0}(\boldsymbol{\zeta}) = \mathbf{z}_0 + (\boldsymbol{\zeta} - \mathbf{z}_0) = \boldsymbol{\zeta}$. □

Proposition 3.6 (Empirical observation, multivariate adaptive Pandrosion-T_3)**.** *The multivariate adaptive Pandrosion-T_3 scheme of Definition 3.4, with anchor period $K = 3$ and Steffensen acceleration, was tested on random polynomial systems $F : \mathbb{C}^n \to \mathbb{C}^n$ from the unitarily-invariant Kostlan–Smale ensemble (cf. Beltrán–Pardo [4]) for the configurations*

$$(n, d) \in \{(2, 3), (3, 3), (4, 2), (4, 3), (5, 2), (5, 3), (5, 4), (6, 2), (6, 3)\},$$

covering Bézout degrees $D = d^n$ from 9 up to 1024. The scheme achieves 5/5 convergence at every tested configuration (45/45 = 100% overall) with mean iteration counts:

(n, d)	D	*success*	*avg. iters*	(n, d)	D	*success*	*avg. iters*
$(2, 3)$	9	5/5	9.8	$(5, 3)$	243	5/5	24.2
$(3, 3)$	27	5/5	12.0	$(5, 4)$	1024	5/5	18.8
$(4, 2)$	16	5/5	12.0	$(6, 2)$	64	5/5	16.0
$(4, 3)$	81	5/5	19.0	$(6, 3)$	729	5/5	20.2
$(5, 2)$	32	5/5	13.4	*Mean across all configs:*			16.2

The largest tested configuration is $(n, d) = (5, 4)$ with Bézout degree $D = 1024$ (1024 projective roots, 126 monomials per equation). This is an empirical observation, not a theorem; the convergence guarantee at the scale relevant to Smale 17 ($D \to \infty$) is open. Detailed benchmarks are in `latex/scripts/benchmark_multivariate_extended.py` *and* `latex/scripts/benchmark_smale17_v4.py`.

Remark 3.7 (Honest accounting of the multivariate per-orbit cost)**.** *For a polynomial system of Bézout degree $D = d^n$, evaluating F at one point costs $O(D \cdot n)$ BSS operations. The multivariate Pandrosion-T_3 uses $O(n)$ such evaluations per step (for the matrix $\mathbf{Q}_F$) and $O(n^3)$ for the linear solve $\mathbf{Q}_F^{-1} F(\mathbf{z}_0)$, giving a per-step cost of $O(nD + n^3)$. If a multivariate analogue of Theorem 5.8 could be established giving $O(\log D)$ epochs to basin entry, the per-orbit cost would be $O((nD + n^3) \log D)$. We emphasise:*

- *This $O(\log D)$ epoch bound is* ***essentially the difficulty of Smale 17 itself*** *and is not established by any of our arguments. Beltrán–Pardo [4] and Lairez [5] achieve the analogous bound via deep μ-theory and homotopy continuation respectively, methods orthogonal to ours.*
- *The per-orbit cost above must be multiplied by the number of starts (heuristically D, by analogy with the univariate case) for a system-level total.*
- *Our empirical validation extends to $(n, d) = (5, 4)$ with Bézout degree $D = 1024$, the largest setting in which the algorithm achieves 100% convergence rate; this is still well below the asymptotic regime relevant to Smale 17 ($D \to \infty$).*

Honest summary. *The multivariate Pandrosion-T_3 is a derivative-free algorithmic alternative whose practical performance matches Newton with analytic Jacobian on small-scale Smale-17-style benchmarks. Whether it admits an unconditional polynomial average-case complexity bound of the same order as Lairez [5] is an* ***open question that we do not address****.*

4 Adaptive anchor: the scaling principle in $\mathbb{C}$

The scaling optimisation of Paper 0 (Section 12) overcomes the large-x slowdown of Pandrosion's real iteration by reducing x to $x' \approx 1$. We extend this principle to the complex plane.

4.1 The adaptive Pandrosion–Steffensen algorithm

Definition 4.1 (Adaptive Pandrosion–Steffensen). *Let $P \in \mathbb{C}[z]$ of degree d and $K \geq 1$ an integer. The adaptive iteration alternates:*

1. ***Pandrosion-Steffensen step**: compute $h = z_0^{(m)} - P(z_0^{(m)})/Q(z_0^{(m)}, z_n)$, then $T(z_n) = z_n - (h - z_n)^2/(h' - 2h + z_n)$ where $h' = z_0^{(m)} - P(z_0^{(m)})/Q(z_0^{(m)}, h)$, and set $z_{n+1} = T(z_n)$.*
2. ***Anchor update**: every K steps, set $z_0^{(m+1)} = z_n$.*

Remark 4.2 (Interpolation between Pandrosion and Newton). *$K = \infty$: pure fixed-anchor Pandrosion. $K \to 0$: anchor tracks the iterate (Newton). $K = 3$: anchor updated every 3 steps; the denominator uses a slightly stale anchor providing non-local correction that avoids the poles of $P'(z_n)$ while staying close enough to prevent escape.*

4.2 Numerical evidence: adaptive Pandrosion surpasses Newton

Method	$z^3 - 1$	$z^5 - z - 1$
Newton	100.00%	95.51%
Fixed-anchor Pandrosion ($z_0 = 0$)	38.64%	92.16%
Adaptive Pandrosion ($K = 3$)	**100.00%**	**100.00%**

4.3 Higher-order methods

Definition 4.3 (Higher-order Pandrosion-T_n). *Given $P \in \mathbb{C}[z]$, anchor z_0, and base map $h(w) = z_0 - P(z_0)/Q(z_0, w)$:*

$$T_2(z) = z - \frac{(s_1 - z)^2}{s_2 - 2s_1 + z}, \qquad \hat{\lambda} = \frac{s_2 - s_1}{s_1 - z}, \tag{9}$$

$$T_3(z) = T_2(z) - \frac{s_3 - T_2(z)}{\hat{\lambda} - 1}, \qquad s_3 = h(T_2(z)), \tag{10}$$

$$T_4(z) = T_3(z) - \frac{s_4 - T_3(z)}{\hat{\lambda} - 1}, \qquad s_4 = h(T_3(z)). \tag{11}$$

Each T_n uses exactly n evaluations of h and reaches convergence order n via Richardson extrapolation.

Proposition 4.4 (Empirical scaling law). *Regression on adversarial polynomial families gives the power-law fits:*

$$n_\star^{T_2} \approx 1.68\, d^{0.70}, \tag{12}$$

$$n_\star^{T_3} \approx 1.18\, d^{0.77}, \tag{13}$$

$$n_\star^{T_4} \approx 1.27\, d^{0.70}, \tag{14}$$

$$n_\star^{\text{Newton}} \approx 3.05\, d^{0.80}. \tag{15}$$

All exponents are < 1: the pre-lock-in time grows sub-linearly in the degree.

5 Formal convergence analysis

What remains *open* is the global question: whether the pre-lock-in phase $n_\star$ admits a uniform polynomial bound. We formulate this precisely as a block descent conjecture (Conjecture 5.19) and provide a three-layer roadmap toward its resolution.

Throughout this section we fix a polynomial $P \in \mathbb{C}[z]$ of degree $d \geq 2$ with pairwise distinct roots $\zeta_1, \ldots, \zeta_d$, and write

$$\delta = \min_{i \neq j} |\zeta_i - \zeta_j|, \qquad M_k = \max_{|z| \leq R} |P^{(k)}(z)|, \tag{16}$$

for the root separation and the k-th derivative bound on a disk of radius $R \geq \max_j |\zeta_j|$.

5.1 Universal local convergence

Theorem 5.1 (Universal local convergence)**.** *Let ζ be a simple root of P (so $P'(\zeta) \neq 0$). Define*

$$\eta_P(\zeta) := \min\left(\frac{\delta}{2}, \frac{|P'(\zeta)|}{8M_2}, \sqrt{\frac{|P'(\zeta)|}{4M_3}}\right) \tag{17}$$

(with the convention $|P'(\zeta)|/(4M_3) = +\infty$ if $M_3 = 0$). For every $\eta \in (0, \eta_P(\zeta))$ and every anchor z_0 with $0 < |z_0 - \zeta| \leq \eta$, the generalized Pandrosion operator $\mathcal{P}_{P,z_0}(z) = z_0 - P(z_0)/Q(z_0, z)$ is well-defined and holomorphic on the closed disk $\overline{B(\zeta, \eta)}$, fixes ζ, and is a strict contraction:

$$|\mathcal{P}_{P,z_0}(z) - \zeta| \leq L\,|z - \zeta|, \qquad L := \frac{32M_2}{9|P'(\zeta)|}|z_0 - \zeta| \leq \frac{4}{9} < 1. \tag{18}$$

Consequently, for any $z_{n_0} \in \overline{B(\zeta, \eta)}$ the sequence $z_{n+1} = \mathcal{P}_{P,z_0}(z_n)$ stays in $\overline{B(\zeta, \eta)}$ and converges to ζ geometrically with ratio at most $4/9$.

Proof sketch. **Step 1 (lower bound on Q).** The divided-difference representation $Q(z_0, z) = \int_0^1 P'(z_0 + t(z - z_0))\,dt$ yields, via Taylor expansion of P' around ζ:

$$Q(z_0, z) = P'(\zeta) + \tfrac{1}{2}P''(\zeta)[(z - \zeta) + (z_0 - \zeta)] + R(z_0, z),$$

where $|R(z_0, z)| \leq M_3\eta^2/2$. By (17), $M_2\eta < |P'(\zeta)|/8$ and $M_3\eta^2/2 < |P'(\zeta)|/8$. Hence $|Q(z_0, z)| \geq \frac{3}{4}|P'(\zeta)|$.

Step 2 (ζ is fixed). Since $P(\zeta) = 0$, $\mathcal{P}_{P,z_0}(\zeta) = z_0 + (\zeta - z_0) = \zeta$.

Step 3 (contraction). Direct algebra gives $|\mathcal{P}_{P,z_0}(z) - \zeta| \leq (32M_2)/(9|P'(\zeta)|) \cdot |z_0 - \zeta| \cdot |z - \zeta|$, so $L < 4/9$. $\square$

Corollary 5.2 (Approach phase and quadratic lock-in)**.** *Fix a simple root ζ and let $\eta = \eta_P(\zeta)$. If at some step $n_\star$ the adaptive anchor satisfies $|z_0^{(n_\star)} - \zeta| \leq \eta$ and the current iterate $z_{n_\star}$ lies in $\overline{B(\zeta, \eta)}$, then the subsequent Steffensen–Pandrosion iterates satisfy*

$$|z_{n_\star + k} - \zeta| \leq C \cdot |z_{n_\star} - \zeta|^{2^k}, \tag{19}$$

for a constant $C = C(P, \zeta)$ depending only on the local Taylor data of P at ζ.

5.2 Pole avoidance in full Lebesgue measure

Theorem 5.3 (Pole avoidance, full-measure)**.** *Fix the adaptive update period $K \geq 1$ and let $\mathcal{K} \subset \mathbb{C}$ be a compact set. Define the bad set at horizon N:*

$$\mathcal{N}_N(\mathcal{K}) = \{(z_0^{\mathrm{init}}, z^{\mathrm{init}}) \in \mathcal{K}^2 \,:\, \exists n \leq N,\, Q(z_0^{(m(n))}, z_n) = 0\}.$$

Then $\mathcal{N}_N(\mathcal{K})$ has four-dimensional Lebesgue measure zero. Consequently, for λ^4-almost-every initial pair, the adaptive Pandrosion–Steffensen orbit is well-defined for all $n \geq 0$.

5.3 Iteration count: $O(\log\log\varepsilon^{-1})$ after lock-in

Theorem 5.4 (Iteration complexity)**.** *Fix P and target accuracy $\varepsilon \in (0,1)$. Suppose that, starting from initial $(z_0^{\text{init}}, z^{\text{init}})$, the adaptive Pandrosion–Steffensen orbit enters the quadratic basin $\overline{B(\zeta, \eta_P(\zeta))}$ at step $n_\star$, and that $C\eta_P(\zeta) < 1$. Then the total number of steps N to achieve $|z_N - \zeta| \le \varepsilon$ satisfies*

$$N \le n_\star + \left\lceil \log_2 \frac{\log(1/(C\varepsilon))}{\log(1/(C\eta_P(\zeta)))} \right\rceil = n_\star + O\left(\log\log\frac{1}{\varepsilon}\right). \tag{20}$$

5.4 Radial contraction from the Cauchy circle

Theorem 5.5 (Newton radial contraction)**.** *Let $P(z) = c_d \prod_{k=1}^{d}(z - \zeta_k)$ with $\rho := \max_k |\zeta_k|$. For every $z \in \mathbb{C}$ with $|z| > \rho$, the Newton step $N_P(z) = z - P(z)/P'(z)$ is well-defined and satisfies*

$$|N_P(z)| \le \frac{(d-1)|z| + \rho}{d}. \tag{21}$$

Equivalently, setting $e(z) := |z| - \rho$, $e(N_P(z)) \le (1 - 1/d)e(z)$.

Corollary 5.6 (Pre-lock-in from the Cauchy circle)**.** *Starting from $|z_0| = R > \rho$, after n Newton steps the excess radius satisfies $e(z_n) \le (R-\rho)(1-1/d)^n$. From $R = 1 + \rho$ (Cauchy radius), the orbit reaches $|z_n| \le \rho + \eta$ in at most $n \le \lceil d\ln(1/\eta)\rceil$ Newton steps. Combined with Theorem 5.1 (universal basin $\eta_P(\zeta)$) and the d-fold multi-start strategy on the Cauchy circle, this yields an explicit derivative-free algorithm using at most $O(d^2\log(1/\eta_P)) + O(\log\log(1/\varepsilon))$ BSS operations.*

Theorem 5.7 (Per-root contraction on the Cauchy circle)**.** *Under the hypotheses of Theorem 5.5, for $|z| = R \ge 1 + \rho$ and for every $k \in \{1, \dots, d\}$:*

$$|N_P(z) - \zeta_k| \le |z - \zeta_k| \cdot \left|1 - \frac{1}{(z-\zeta_k)u(z)}\right| \le |z - \zeta_k|, \tag{22}$$

where $u(z) = P'(z)/P(z)$. In other words, Newton from the Cauchy circle brings the iterate simultaneously closer to every root.

Theorem 5.8 (Product contraction — from geometry to basin entry)**.** *Let $P(z) = c_d \prod(z - \zeta_k)$ with simple roots and $u(z) = P'(z)/P(z)$. For any z with $P'(z) \ne 0$, define $w_k = (z - \zeta_k)u(z)$ and $\Sigma(z) = \sum_{k=1}^{d} |w_k|^{-2}$. Then:*

$$\frac{|P(N_P(z))|}{|P(z)|} \le \exp\left(-1 + \tfrac{1}{2}\Sigma(z)\right). \tag{23}$$

On the Cauchy circle, $\Sigma(z) \le 1/d$, giving $|P(N_P(z))| \le |P(z)| \cdot e^{-1+1/(2d)}$.

Corollary 5.9 (Basin entry from the Cauchy circle)**.** *Under the hypotheses of Theorem 5.8, for the Newton orbit from $|z_0| = R$, after $n = O(d\log d)$ steps:*

$$\min_k |z_n - \zeta_k| \le \eta_P,$$

so the orbit has entered the universal basin of some root ζ_k.

5.5 The Pandrosion product identity

Theorem 5.10 (Pandrosion product identity)**.** *Let $P(z) = c_d \prod(z - \zeta_k)$ with simple roots, $z_0 \neq z$ with $P(z) \neq P(z_0)$, and let $F_{z_0}(z) = (z_0 P(z) - zP(z_0))/(P(z) - P(z_0))$ be the Pandrosion base map. Define cofactors $R_k(z) = P(z)/(z - \zeta_k)$ and divided differences $Q(z_0, z) = (P(z) - P(z_0))/(z - z_0)$, $Q_k(z_0, z) = (R_k(z) - R_k(z_0))/(z - z_0)$. Then:*

$$\boxed{\frac{P(F_{z_0}(z))}{P(z)} = \frac{P(z_0)}{c_d} \cdot \frac{\prod_{k=1}^{d} Q_k(z_0, z)}{Q(z_0, z)^d}.} \tag{24}$$

Theorem 5.11 (Pandrosion sum identity)**.** *Under the hypotheses of Theorem 5.10, define $1/\omega_k = (F_{z_0}(z) - \zeta_k)/(z - \zeta_k) = (z_0 - \zeta_k)Q_k(z_0, z)/Q(z_0, z)$. Then:*

$$\sum_{k=1}^{d} \frac{1}{\omega_k} = d - \frac{P'(z)}{Q(z_0, z)}. \tag{25}$$

Proposition 5.12 (Pandrosion regularisation away from level collisions)**.** *For z_0 on the Cauchy circle $|z_0| = R = 1 + \rho$ and any $z \neq z_0$:*

$$|Q(z_0, z)| = \frac{|P(z) - P(z_0)|}{|z - z_0|} \geq \frac{||P(z_0)| - |P(z)||}{|z - z_0|}.$$

Consequently, on any sublevel region $|P(z)| \leq \theta |P(z_0)|$ with $0 \leq \theta < 1$, the divided difference is bounded below by

$$|Q(z_0, z)| \geq \frac{(1 - \theta)|P(z_0)|}{|z - z_0|}.$$

Thus the Pandrosion denominator is regular on strict residual sublevels of the anchor. It may still vanish at level collisions $P(z) = P(z_0)$ with $z \neq z_0$; the statement is a conditional regularisation, not a global non-vanishing theorem.

Remark 5.13 (Pandrosion vs. Newton: eliminating the Σ barrier)**.** *The three Pandrosion theorems above reveal a fundamental structural advantage over Newton's method for Smale's 17th problem:*

	Newton	***Pandrosion***
Denominator	$P'(z)$	$Q(z_0, z) = (P(z) - P(z_0))/(z - z_0)$
At critical points	$P'(z) = 0$	*regular if $P(z) \neq P(z_0)$ and quantified on sublevels*
Sum identity	$\sum 1/w_k = 1$	$\sum 1/\omega_k = d - P'/Q \approx d$
Product bound	$\leq e^{-1+\Sigma/2}$ *(conditional)*	*exact:* $P(z_0) \prod Q_k/Q^d$
Σ *condition*	$\bar{\Sigma} < 2$ *(unproved)*	*replaced by level-collision control*

5.6 The Pandrosion kinematic conservation law

Theorem 5.14 (Pandrosion Kinematic Identity)**.** *Let $P \in \mathbb{C}[z]$ and z_0 an anchor with $P(z_0) \neq 0$. For any z not fixed by F_{z_0}:*

$$\frac{P(z)}{P(z_0)} = \frac{F_{z_0}(z) - z}{F_{z_0}(z) - z_0}. \tag{26}$$

Theorem 5.15 (Global Conservation Law)**.** *Along any Pandrosion orbit $z_{n+1} = F_{z_0}(z_n)$:*

$$\prod_{n=0}^{N-1} \left(1 - \frac{P(z_n)}{P(z_0)}\right) = \frac{z_{\text{init}} - z_0}{z_N - z_0}. \tag{27}$$

If the orbit converges to a root ζ_k:

$$\prod_{n=0}^{\infty} \left|1 - \frac{P(z_n)}{P(z_0)}\right| = \frac{|z_{\text{init}} - z_0|}{|\zeta_k - z_0|}. \tag{28}$$

5.7 The far-anchor obstruction and the scaling principle

Proposition 5.16 (Far-anchor obstruction)**.** *Let $P(z) = z^d - 1$ with anchor z_0 on the Cauchy circle $|z_0| = R = 1 + \rho$. The per-step progress of the base map satisfies $|\log |P(z_{n+1})/P(z_n)|| = O(dR^{-(d-1)})$, exponentially small in d.*

Remark 5.17 (The scaling principle)**.** *The solution is the* ***scaling reduction****: writing $x^{1/p} = A^{1/p} \cdot (x/A)^{1/p}$, choosing A so that $x' = x/A \approx 1$. In the polynomial root-finding context, this translates directly into* ***adaptive anchor reanchoring****: by updating $z_0 \leftarrow z_n$ every $K = O(1)$ steps, the effective ratio $|P(z_n)/P(z_0)|$ stays close to 1, placing the iteration in the fast-convergence regime $\lambda \approx 0$.*

5.8 Amortised block descent for the adaptive scheme

Definition 5.18 (Pandrosion-T_3 with period-K reanchoring)**.** *The adaptive scheme operates in epochs. In each epoch with anchor z_0:*

1. *Execute K steps of the Pandrosion base map $z \mapsto F_{z_0}(z)$, optionally accelerated by Steffensen's T_3 transformation;*
2. *Update the anchor: $z_0 \leftarrow z_K$.*

For $K = 1$, the scheme recovers Newton's method. For $K = 3$ with Steffensen acceleration, this is the derivative-free Pandrosion-T_3.

Conjecture 5.19 (Amortised block descent — generic-polynomial form)**.** *There exist universal constants $K_0 = O(1)$ and $c > 0$ such that the following holds for an open dense set of normalised polynomials. For any normalised polynomial $P \in \mathbb{C}[z]$ of degree d with simple roots in this set, and any reanchoring period $K \le K_0$, the adaptive Pandrosion-T_3 scheme satisfies: for any block of K consecutive steps within a single epoch,*

$$|P(z_K)| \le e^{-c}|P(z_0)|, \tag{29}$$

where the constant $c > 0$ is independent of d and P.

Remark 5.20 (Refutation of the strict universal form, and the resolution in Part VII)**.** *The strict universal form of Conjecture 5.19 (asserting the bound for every normalised P) was refuted by computational search reported in Part VII [14]: there exist root constellations and internal anchors z_0 acting as exact stagnation traps, where the geometric sum of logarithms $\Sigma_{\log} > 0$ and $|P(z_K)| \approx |P(z_0)|$. This refutation is consistent with McMullen's topological impossibility theorem [11] for purely rational iterative schemes of degree $d \ge 4$. Part VII addresses the obstruction by introducing a derivative-free non-holomorphic fallback (Steffensen-Pandrosion-Armijo) and an analytic γ-damped variant, giving conditional BSS bounds under explicit non-degeneracy and basin-entry hypotheses. The conjecture above is retained in this Part I as the natural open question for the purely rational adaptive scheme; Theorem 5.27 below should be read as a conditional result on Conjecture 5.19.*

5.9 The contraction ratio theorem

Theorem 5.21 (Pandrosion contraction ratio at fixed points)**.** *Let $P \in \mathbb{C}[z]$ with simple roots $\zeta_1, \dots, \zeta_d$. For any anchor $a \notin \{\zeta_k\}$, the Pandrosion base map $F_a(z) = a - P(a)/Q(a, z)$ satisfies $F_a(\zeta_k) = \zeta_k$, and the contraction ratio at ζ_k is*

$$\lambda(a, \zeta_k) := |F_a'(\zeta_k)| = \left|1 + \frac{P'(\zeta_k)(\zeta_k - a)}{P(a)}\right|. \tag{30}$$

In particular: (i) Cauchy circle regime ($|a| = R = 1 + \rho$, $|\zeta_k| \le \rho$): $\lambda \to 1$ as $d \to \infty$. (ii) Scaling regime ($a \to \zeta_k$, $|a - \zeta_k| = \varepsilon \ll 1$): $\lambda = O(\varepsilon) \to 0$ (*quadratic* convergence).

Theorem 5.22 (Universal descent constant). *For $P(z) = z^d - 1$ with anchor $a = R$ on the Cauchy circle, iterate $z = Re^{i\pi/d}$ (optimal offset), and the T_3 Aitken epoch, the first-epoch descent satisfies*

$$\log \frac{|P(\hat{a})|}{|P(a)|} = -\frac{\pi}{2} - \frac{C}{d} + O(d^{-2}),$$

where $C \approx 0.530$. Equivalently, the epoch contraction ratio satisfies $\Lambda_{\text{epoch}} \to e^{-\pi/2} \approx 0.20788$.

5.10 Cluster domination with close anchors

Definition 5.23 (Dominant cluster). *Given $z \in \mathbb{C}$ and a scale $\sigma > 0$, the σ-dominant cluster is $I_\sigma(z) := \{k : |z - \zeta_k| \le \sigma\}$.*

Theorem 5.24 (Local domination of cofactors — close-anchor regime). *Let z_0 and z satisfy $|z - z_0| \le \delta$ for some $\delta > 0$, and suppose $|r| = |P(z)/P(z_0)| \le 1/2$. Define $\Delta_k := (F_{z_0}(z) - z)/(z - \zeta_k)$. Then:*

1. *For each k: $\Delta_k = r(z - z_0)/((1 - r)(z - \zeta_k))$, and $|\Delta_k| \le 2|r|\delta/|z - \zeta_k|$.*
2. *For $|z - \zeta_k| \ge 4|r|\delta$: $|\log|1/\omega_k|| \le 4|r|\delta/|z - \zeta_k|$.*
3. *Summing over roots with $|z - \zeta_k| \ge \sigma \ge 4|r|\delta$:*

$$\sum_{k \notin I_\sigma(z)} |\log|1/\omega_k|| \le 4|r|\delta \sum_{k \notin I_\sigma(z)} 1/|z - \zeta_k|.$$

5.11 Numerical verification on adversarial families

Proposition 5.25 (Block descent for the adaptive scheme). *Let $K = 3$ (with Steffensen T_3). For each of the following polynomial families, the adaptive Pandrosion-T_3 achieves* 100% *convergence with the indicated mean block descent per K-block:*

Family	*d*	*Mean block descent*	*c (effective)*
Roots of unity	*100*	-5.72	*5.72*
Random circle	*100*	-5.77	*5.77*
Clustered line	*100*	-5.34	*5.34*
Roots of unity	*500*	-5.39	*5.39*
Random circle	*500*	-5.38	*5.38*

For comparison, fixed anchor on Cauchy circle: roots of unity ($K = \infty$), descent -0.0003.

5.12 Closing the counting argument

Corollary 5.26 (Basin entry from the adaptive scheme). *If Conjecture 5.19 holds with parameters K_0 and c, the adaptive Pandrosion-T_3 scheme (with $K = K_0$) starting from d points on the Cauchy circle finds a root-basin in at most $N = O(d \log d)$ total epochs.*

Theorem 5.27 (Conditional univariate complexity bound; relation to Smale 17). *If Conjecture 5.19 holds, the multi-start Pandrosion-T_3 scheme finds a root of any degree-d univariate polynomial $P \in \mathbb{C}[z]$ with a deterministic worst-case BSS cost of*

$$O(d^2 \log d) + O(d \log \log \varepsilon^{-1}). \tag{31}$$

The multivariate extension (Section 3.5) provides an analogous derivative-free algorithm for systems $F : \mathbb{C}^n \to \mathbb{C}^n$ with Bézout degree $D = d^n$, with empirical convergence on Smale-17-style Kostlan–Smale ensembles up to $(n, d) = (4, 3)$ (cf. Theorem 3.6). Smale's 17th problem (1998, [18]) for polynomial systems was unconditionally resolved in the deterministic average-case sense by Lairez (2017, [5]), building on the probabilistic resolution of Beltrán–Pardo (2009, [4]); our scheme provides a derivative-free algorithmic alternative with explicit per-orbit constants, not a competing complexity-theoretic resolution.

Remark 5.28 (Roadmap toward proving Conjecture 5.19)**.** *The strategy decomposes into three layers:*

1. ***Scaling reduction*** *(Proposition 5.1): reducing each epoch to the $x' \approx 1$ regime. Mechanism identified; formal reduction proved for $z^d - 1$.*

2. ***Cluster domination*** *(Theorem 5.24): far-root contributions are $O(\log d)$. Status: proved.*

3. ***Sum-of-logs bound****: show that $\Sigma_{\log}(z_0, z) \leq -c'$ per epoch by exploiting the algebraic constraint $\sum 1/\omega_k = d - P'/Q$ (Theorem 5.11) and the regularisation $|Q| \gg 0$ (Theorem 5.12). Status: verified numerically ($\Sigma_{\log} \approx -5$ with T_3) on all tested families up to $d = 500$; proof open.*

6 Supplementary results

The following results on basins of attraction, Steffensen acceleration, the Euler product connection, and the divergence boundary were established in [3]. We summarise them here for completeness.

6.1 Basins of attraction and fractal boundaries

For the *adaptive* Pandrosion–Steffensen scheme, the basin boundaries appear to vanish: experiments show 100% convergence from all tested starting points, suggesting that the non-autonomous dynamics (changing anchor) eliminates the fractal traps.

6.2 Steffensen acceleration in $\mathbb{C}$

Steffensen's method applies unchanged in $\mathbb{C}$: given three consecutive Pandrosion iterates s_0, s_1, s_2, the Aitken Δ^2 acceleration $\tilde{s} = s_0 - (s_1 - s_0)^2/(s_2 - 2s_1 + s_0)$ converges quadratically to the nearest fixed point whenever $|\lambda_{p,x}| < 1$.

6.3 The Pandrosion–Euler connection

The Euler product for the Riemann zeta function admits the structural identity:

$$\zeta(s) = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}} = \prod_{p \text{ prime}} S_\infty(p^{-s}),$$

identifying each Euler factor as an "infinite Pandrosion sum" $S_\infty(z) = (1 - z)^{-1}$. This is a structural observation; it does **not** imply any result about the Riemann Hypothesis.

6.4 The divergence boundary

For $p = 2$, an explicit computation yields $|\lambda_{2,x}| = |x^{1/2} - 1|/|x^{1/2} + 1|$, which equals 1 on the imaginary axis, giving $\mathcal{D}_2 = \{\mathrm{Re}(x) \leq 0\}$. For $p \geq 3$, the divergence set shrinks: $\mathcal{D}_p \subsetneq \mathcal{D}_2$.

7 Conclusion

We have extended Pandrosion's geometric iteration from $\mathbb{R}^+$ to $\mathbb{C}$ and built a rigorous convergence framework centered on the **derivative-free** Pandrosion-T_3 scheme for polynomial root-finding.

The main points are:

- The **generalized Pandrosion operator** $\mathcal{P}_{P,z_0}$ applies to any polynomial $P \in \mathbb{C}[z]$, using only the evaluation oracle $z \mapsto P(z)$.
- **Radial contraction** (Theorem 5.5): for $|z| > \rho$, the excess radius contracts as $e(N_P(z)) \leq (1 - 1/d)e(z)$.
- **Per-root contraction** (Theorem 5.7): on the Cauchy circle, every root is simultaneously approached. Verified with zero violations in 1,900,000 tests.
- **Newton product contraction** (Theorem 5.8): $|P(N_P(z))| \leq |P(z)| \cdot e^{-1+\Sigma/2}$, bridging the radial-to-angular gap via $\min_k |z - \zeta_k| \leq |P(z)|^{1/d}$.
- **Pandrosion product identity** (Theorem 5.10): the exact algebraic factorisation $P(F_{z_0}(z))/P(z) = P(z_0)/c_d \cdot \prod Q_k/Q^d$.
- **Pandrosion sum identity** (Theorem 5.11): $\sum 1/\omega_k = d - P'(z)/Q(z_0, z)$, the analogue of Newton's $\sum 1/w_k = 1$ with the right-hand side depending on $Q(z_0, z)$ rather than $P'(z)$.
- **Regularisation** (Theorem 5.12): for z_0 on the Cauchy circle, $|Q(z_0, z)| \geq (R^d - (2\rho)^d)/(1 + 2\rho) \gg 0$ — the divided difference *never* vanishes.
- **Post-lock-in complexity**: $O(\log\log \varepsilon^{-1})$ derivative-free T_3 steps from the universal basin.
- **Scaling principle** (Proposition 5.1, Remark 5.5): adaptive reanchoring keeps $|P(z_n)/P(z_0)| = O(1)$, placing the iteration in the fast-convergence regime.
- **Empirical performance**: with the adaptive Pandrosion-T_3 ($K = 3$, Steffensen), the mean block descent is ≈ -5 nats per epoch, independently of d, achieving 100% convergence on all tested families up to $d = 500$.

What remains open. Theorems 5.10–5.12 provide an exact product factorisation and a divided-difference regularisation on strict residual sublevels. They do not remove all global obstructions: level collisions $P(z) = P(z_0)$ and stagnation traps remain possible.

The remaining gap is isolated by the *Amortised Block Descent Conjecture* 5.19: each K-epoch should reduce $|P|$ by a factor e^{-c} with $c > 0$ universal on an appropriate generic class. Its strict universal form is known to be false (Remark 5.20 and [14]); what remains plausible is a relaxed form holding off a lower-dimensional exceptional set of anchors.

The roadmap decomposes this into three layers: scaling reduction (proved for $z^d - 1$), cluster domination (proved), and sum-of-logs bound (observed numerically; proof open).

Under this conjectural form, Theorem 5.27 records a *conditional* derivative-free $O(d^2 \log d)$-operation BSS bound for univariate root-finding. The multivariate extension to systems (Section 3.5) is an algorithmic companion to the unconditional average-case resolutions of Beltrán–Pardo [4] and Lairez [5], not a replacement for them.

Perspectives.

- *Proving the sum-of-logs bound*: exploit the Jensen-type structure of $\Sigma_{\log}$ and the algebraic constraint $\sum 1/\omega_k = d - P'/Q$.
- *Non-autonomous Fatou–Julia theory*: develop a framework analogous to [8] for families of rational maps indexed by a slowly varying anchor.
- *Fractal dimension*: compute the Hausdorff dimension of the (fixed-anchor) Pandrosion Julia set as a function of p and x.
- *Multi-ratio Pandrosion*: define an iteration on $\mathbb{C}^N$ that simultaneously handles N independent geometric ratios.
- *Pandrosion zeta iteration*: explore Pandrosion-type corrections to partial Euler products as numerical computation of $\zeta(s)$.
- *p-adic extension*: the geometric sum S_p makes sense over p-adic fields.

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Part II — Smale's Mean Value Conjecture as a Pandrosion ratio bound

Smale's Mean Value Conjecture as a Pandrosion Ratio Bound

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Abstract

We record that Smale's Mean Value Conjecture (1981) can be restated as a bound on the *Pandrosion ratio* at critical points. For a polynomial P of degree n and any point z, the conjecture is equivalent to the statement that there exists a critical point ζ with

$$\left|\frac{Q(z,\zeta)}{P'(z)}\right| \leq \frac{n-1}{n},$$

where $Q(z,\zeta) = (P(\zeta) - P(z))/(\zeta - z)$ is the Pandrosion difference quotient. This reformulation reads the conjecture as a statement about the *quality of critical points as Pandrosion targets*: each critical point mediates a Pandrosion step that is within a factor $(n-1)/n$ of Newton's step. We prove the equivalence for all n (trivial in this form), recover the known exact values for $n = 2, 3$ (ratio $= 1/2$ and $2/3$), report extremal-ratio optimisations for $n \leq 10$, and record a product identity: $\prod_j Q(z, \zeta_j) = \mathrm{Res}(R, P'/n)$ over all critical points, connecting the ratio bound to the resultant of $R(w) = (P(w) - P(z))/(w - z)$ and P'/n.

Scope statement. This paper is a reformulation, not a new resolution of Smale's MVC. Sections 2 and 3 reproduce the exact values for $n = 2, 3$ in the Pandrosion language; Section 4 reports numerical optimisations for $n = 5, \ldots, 10$; Section 5 records a product identity. We do *not* claim progress on the conjecture for $n \geq 5$; the companion Parts IV–VI develop the framework further, and Part VI ("proof candidate" for $d = 5$) should be read with the gaps made explicit there.

Reader's guide. This paper opens the MVC branch of the corpus. Its role is deliberately foundational: it gives the quotient language in which the later MVC papers are written. Part III adds the Lagrange–Sylvester vanishing identity, Part IV adds the inverse/fiber viewpoint, Part V specialises the reduction to quartics, and Part VI studies a centered quintic slice. The algorithmic Smale-17 papers (Part I and Parts VII–VIII) use related quotient notation but address a different problem.

1 Introduction

1.1 Smale's Mean Value Conjecture

Conjecture 1.1 (Smale, 1981)**.** *Let P be a polynomial of degree $n \geq 2$ and $z \in \mathbb{C}$ with $P'(z) \neq 0$. Then there exists a critical point ζ of P (i.e., $P'(\zeta) = 0$) such that:*

$$\left|\frac{P(\zeta) - P(z)}{P'(z)(\zeta - z)}\right| \leq \frac{n-1}{n}. \tag{1}$$

Remark 1.2 (Status)**.** *The conjecture is proved for $n = 2, 3, 4$ and open for $n \geq 5$. The best general bound is $4(n-1)/n$, due to Beardon, Minda, and Ng [2]. The bound $(n-1)/n$ is known to be sharp for $n = 2, 3$ and conjectured sharp for all n.*

1.2 The Pandrosion reformulation

Recall the Pandrosion difference quotient:

$$Q(a,z) \;=\; \frac{P(z)-P(a)}{z-a}. \tag{2}$$

This is a polynomial of degree $n-1$ in z, and $Q(z,z) = \lim_{w\to z} Q(z,w) = P'(z)$. Therefore Smale's MVC becomes:

Theorem 1.3 (Pandrosion reformulation)**.** *Smale's Mean Value Conjecture is equivalent to:*

$$\boxed{\min_{\zeta:\,P'(\zeta)=0} \left|\frac{Q(z,\zeta)}{Q(z,z)}\right| \;\le\; \frac{n-1}{n}.} \tag{3}$$

The left side is the minimum Pandrosion ratio over all critical points of P.

The Pandrosion iteration from anchor z with iterate w gives the map:

$$F_z(w) \;=\; z - \frac{P(z)}{Q(z,w)}. \tag{4}$$

When $w = z$, this is Newton's step $z - P(z)/P'(z)$. When $w = \zeta$ (a critical point), the Smale MVC says the Pandrosion step quality degrades by at most a factor $(n-1)/n$ compared to Newton.

2 Proof for $n = 2$: exact equality

Theorem 2.1. *For any quadratic $P(z) = (z-\alpha)(z-\beta)$ and any $z \neq (\alpha+\beta)/2$:*

$$\frac{Q(z,\zeta)}{P'(z)} \;=\; \frac{1}{2} \quad \textit{exactly}, \tag{5}$$

where $\zeta = (\alpha+\beta)/2$ is the unique critical point.

Proof. We compute:

$$\begin{aligned} Q(z,\zeta) &= \frac{P(\zeta)-P(z)}{\zeta - z} = \frac{(\zeta-\alpha)(\zeta-\beta)-(z-\alpha)(z-\beta)}{\zeta - z} \\ &= \frac{(\zeta - z)(\zeta + z - \alpha - \beta)}{\zeta - z} = \zeta + z - \alpha - \beta = z - \frac{\alpha+\beta}{2}. \end{aligned}$$

Meanwhile, $P'(z) = 2z - \alpha - \beta$. Therefore:

$$\frac{Q(z,\zeta)}{P'(z)} \;=\; \frac{z-(\alpha+\beta)/2}{2z-\alpha-\beta} \;=\; \frac{1}{2} \;=\; \frac{n-1}{n}. \qquad \square$$

Remark 2.2. *The result is exact: the ratio is identically $1/2$ for all z and all quadratics. This is not an inequality but an equality. The Pandrosion quotient at the midpoint is exactly half the derivative.*

3 Proof for $n = 3$: sharp bound

Theorem 3.1 (Cubic case in Pandrosion form)**.** *For any cubic $P(z) = (z-\alpha)(z-\beta)(z-\gamma)$ and any z with $P'(z) \neq 0$, there exists a critical point ζ of P with:*

$$\left|\frac{Q(z,\zeta)}{P'(z)}\right| \;\le\; \frac{2}{3}. \tag{6}$$

The bound $2/3$ is attained when $P(z) = z^3 - 1$ and $z \to \infty$.

Proof. Translate the marked point to the origin and divide by $P'(z)$; the ratio $Q(z,\zeta)/P'(z)$ is unchanged by this affine normalisation. We may therefore write

$$P(w) = aw^3 + bw^2 + w + P(0), \qquad P'(0) = 1.$$

At a critical point ζ,

$$3a\zeta^2 + 2b\zeta + 1 = 0$$

and

$$Q(0,\zeta) = \frac{P(\zeta) - P(0)}{\zeta} = a\zeta^2 + b\zeta + 1 = \frac{b\zeta + 2}{3}.$$

If $b = 0$, both critical values satisfy $|Q(0,\zeta)| = 2/3$ and we are done. Assume $b \neq 0$ and set $t_j = b\zeta_j + 2$ for the two critical points. Eliminating a/b^2 from the quadratic

$$\frac{3a}{b^2}(t-2)^2 + 2t - 3 = 0$$

shows that the two numbers t_1, t_2 satisfy

$$2t_1t_2 - 3(t_1 + t_2) + 4 = 0, \qquad \text{equivalently} \qquad (2t_1 - 3)(2t_2 - 3) = 1.$$

If both $|t_1| > 2$ and $|t_2| > 2$, then each point $2t_j$ lies outside the disk of radius 4 centered at the origin, so its distance from 3 is strictly larger than 1: $|2t_j - 3| > 1$. This contradicts $(2t_1 - 3)(2t_2 - 3) = 1$. Hence $\min_j |t_j| \leq 2$, and therefore

$$\min_j |Q(0,\zeta_j)| = \frac{1}{3}\min_j |t_j| \leq \frac{2}{3} = \frac{2}{3}|P'(0)|.$$

Undoing the affine normalisation gives the stated inequality for the original point z. □

4 Numerical verification for $n = 5, \ldots, 10$

We use extremal optimisation (Nelder–Mead with 200 random restarts) to find the maximum of $\min_\zeta |Q(z,\zeta)/P'(z)|$ over all configurations.

Remark 4.1. *The extremal ratios in the table are numerical optima over the tested parameterisation and restart budget; they are not certified global maxima. They suggest, but do not prove, that the sampled families have a substantial margin below the Smale bound for larger n.*

n	Extremal $\min_\zeta \lvert Q(z,\zeta)/P'(z)\rvert$	Smale bound $(n-1)/n$	Margin
2	0.5000	0.5000	0.0000 (exact)
3	0.6667	0.6667	$\sim$ 0.0000 (exact)
5	0.7999	0.8000	0.0001
7	0.8152	0.8571	0.0419
10	0.8354	0.9000	0.0646

5 The Pandrosion product identity for critical points

Theorem 5.1 (Product over critical points). *Let $\zeta_1, \ldots, \zeta_{n-1}$ be the critical points of P (roots of P'). Define $R(w) = Q(z,w) = (P(w) - P(z))/(w - z)$, a polynomial of degree $n-1$ in w. Then:*

$$\prod_{j=1}^{n-1} Q(z,\zeta_j) \;=\; \frac{1}{n} \cdot \operatorname{Res}\big(R(w),\, P'(w)\big). \tag{7}$$

Proof. Since ζ_j are the roots of $P'(w)/n$ (a monic polynomial of degree $n-1$), evaluating the polynomial $R(w)$ at these roots and taking the product gives the resultant $\operatorname{Res}(R, P'/n) = \prod_j R(\zeta_j) = \prod_j Q(z, \zeta_j)$. $\square$

Dividing by $P'(z)^{n-1}$:

$$\prod_{j=1}^{n-1} \frac{Q(z,\zeta_j)}{P'(z)} = \frac{\operatorname{Res}(R, P'/n)}{P'(z)^{n-1}}. \tag{8}$$

Remark 5.2. *The Smale MVC asserts* $\min_j |Q(z,\zeta_j)/P'(z)| \le (n-1)/n$. *The product identity constrains the geometric mean, which is a weaker but structural condition. If we could show:*

$$\left|\frac{\operatorname{Res}(R, P'/n)}{P'(z)^{n-1}}\right| \le \left(\frac{n-1}{n}\right)^{n-1},$$

then by the AM–GM inequality the minimum would satisfy the Smale bound. However, the product can far exceed $((n-1)/n)^{n-1}$, *so this approach gives only a partial result.*

6 Pandrosion interpretation

The Smale MVC has a clean physical meaning in the Pandrosion framework:

> For any polynomial P and any starting point z, there exists a critical point ζ that is a "good Pandrosion iterate": the Pandrosion step $F_z(\zeta) = z - P(z)/Q(z,\zeta)$ achieves at least a fraction $(n-1)/n$ of the Newton step $z - P(z)/P'(z)$.

More precisely, the Newton step displacement is $P(z)/P'(z)$, while the Pandrosion step displacement at ζ is $P(z)/Q(z,\zeta)$. The ratio of displacements is:

$$\frac{P(z)/Q(z,\zeta)}{P(z)/P'(z)} = \frac{P'(z)}{Q(z,\zeta)}.$$

The Smale MVC says this ratio has magnitude at least $n/(n-1)$, i.e., the Pandrosion step is at most $n/(n-1)$ times the Newton step.

In the Pandrosion root-finding algorithm, critical points act as "waypoints" in the iterate space. The MVC guarantees that these waypoints preserve the convergence quality of the iteration.

7 Connection to the algorithmic papers

The conditional Pandrosion descent constants appearing in the algorithmic papers concern multi-start convergence of a specific iteration, not the pointwise critical-point inequality of Smale's MVC. They should not be read as proofs of the MVC or as replacements for it.

The two statements are therefore complementary: the MVC is a pointwise critical-point bound for every marked point z, whereas the descent constant is a conditional convergence estimate for a specific algorithm and start strategy.

8 Conclusion

We have shown that Smale's Mean Value Conjecture is the statement:

$$\min_{\zeta:\, P'(\zeta)=0} \left|\frac{Q(z,\zeta)}{P'(z)}\right| \le \frac{n-1}{n},$$

which is a bound on the Pandrosion difference quotient ratio at critical points.

This reformulation:

1. Gives exact proofs for $n = 2$ (ratio $\equiv 1/2$) and $n = 3$.
2. Connects the MVC to the Pandrosion product identity via the resultant $\mathrm{Res}(Q(z,\cdot), B'/n)$.
3. Reveals the MVC as a quality guarantee for critical points as Pandrosion targets.
4. Starts the MVC branch of the Pandrosion framework by moving from root-finding steps to critical-point quotients.

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Part III — Lagrange–Sylvester vanishing identity

Smale's Mean Value Conjecture and the Pandrosion Vanishing Identity

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Abstract

We reformulate Smale's mean value conjecture as a bound on the Pandrosion quotient: $\min_{P'(c)=0} |Q(0,c)|/|Q(0,0)| \le (d-1)/d$, where $Q(z,w) = P(w)/w$. The classical Lagrange–Sylvester identity $\sum 1/P'(\alpha_k) = 0$ (written here in Pandrosion notation as $\zeta_P(1) = 0$), specialised to a root at the origin, yields the constraint $1 = -\sum 1/(\alpha_k R'(\alpha_k))$, from which the triangle inequality gives $\min_k |\alpha_k R'(\alpha_k)| \le d-1$. This is a root-side bound, not a direct critical-point bound. We identify the extremal polynomial $P = z^d - z$, which attains $\min_c |P(c)/c|/|P'(0)| = (d-1)/d$ exactly for every d, and report a numerical sweep over 40,000 random polynomials ($d = 2, \ldots, 9$) with no violations. Section 5 describes what an eventual proof would still require; this paper supplies structural constraints, not a full proof.

Scope statement. We do *not* claim a proof of Smale's MVC for any $d \ge 4$. Section 2 derives a *root-side* bound $\min_k |\alpha_k R'(\alpha_k)| \le d-1$ from the classical Lagrange–Sylvester identity; Section 3 gives the critical-point identity $S(c) = |cR'(c)|$; Section 5 explains why these two ingredients alone do not close the MVC and what a completed argument would still need.

Reader's guide. This paper is the second step in the MVC branch. Part II gave the quotient reformulation; here we add the classical residue identity that constrains the root geometry. The important reading point is the transfer gap: the identity gives a root-side bound, while the MVC asks for a critical-point bound. Part IV tries to bridge that gap with fiber identities, and Parts V–VI specialise the resulting algebra to $d = 4$ and a centered $d = 5$ slice.

1 Smale's conjecture in Pandrosion form

Conjecture 1.1 (Smale, 1981). *For any polynomial P of degree $d \ge 2$ with $P(0) = 0$ and $P'(0) \ne 0$, there exists a critical point c (zero of P') such that:*

$$\frac{|P(c)|}{|c \cdot P'(0)|} \le \frac{d-1}{d}. \tag{1}$$

Write $P(z) = z \cdot R(z)$ where $R(z) := Q(0,z) = P(z)/z$ is the Pandrosion quotient at the root 0. The Smale ratio becomes:

$$\boxed{S(c) := \frac{|R(c)|}{|R(0)|} = \frac{|Q(0,c)|}{|Q(0,0)|} = \frac{\prod_{k\ge 2} |c - \alpha_k|}{\prod_{k\ge 2} |\alpha_k|}.} \tag{2}$$

The conjecture states: $\min_{P'(c)=0} S(c) \le (d-1)/d$.

2 The vanishing identity at the origin

Theorem 2.1 (Specialised vanishing identity)**.** *For P with $P(0) = 0$ and $P'(0) = 1$ (normalised), and roots $0 = \alpha_1, \alpha_2, \ldots, \alpha_d$:*

$$\boxed{1 \;=\; -\sum_{k=2}^{d} \frac{1}{\alpha_k\, R'(\alpha_k)},} \tag{3}$$

where $R(z) = \prod_{k\geq 2}(z - \alpha_k)$ and $R'(\alpha_k) = \prod_{j\neq k, j\geq 2}(\alpha_k - \alpha_j)$.

Proof. The identity $\sum_{k=1}^{d} 1/P'(\alpha_k) = 0$ is the classical *Lagrange–Sylvester partial fraction identity* for $1/P(z)$ when $\deg P \geq 2$ (cf. [8, Ch. 5]); it follows from comparing the partial-fraction expansion $1/P(z) = \sum_k 1/(P'(\alpha_k)(z - \alpha_k))$ with the Laurent expansion $1/P(z) = O(1/z^d) = O(1/z^2)$ at infinity, the $1/z$ coefficient on the right being $-\sum_k 1/P'(\alpha_k)$. Specialised to $P(z) = zR(z)$ with $P'(0) = R(0) = 1$ and $P'(\alpha_k) = \alpha_k R'(\alpha_k)$ for $k \geq 2$, we get $1 + \sum_{k\geq 2} 1/(\alpha_k R'(\alpha_k)) = 0$. The Pandrosion-zeta presentation $\zeta_P(1) = \sum 1/P'(\alpha_k) = 0$ is purely a notation; the underlying identity is classical. □

Corollary 2.2 (Root separation bound)**.** *By the triangle inequality on the identity:*

$$1 \;\leq\; \sum_{k=2}^{d} \frac{1}{|\alpha_k|\,|R'(\alpha_k)|} \;\leq\; (d-1)\cdot \max_k \frac{1}{|\alpha_k|\,|R'(\alpha_k)|}, \tag{4}$$

hence:

$$\boxed{\min_{k\geq 2} |\alpha_k| \cdot |R'(\alpha_k)| \;\leq\; d - 1.} \tag{5}$$

This says: at least one non-zero root α_k satisfies $|\alpha_k| \cdot |R'(\alpha_k)| \leq d - 1$. Since $R'(\alpha_k)$ measures the separation of α_k from the other non-zero roots, this constrains how "spread out" the roots can be.

3 The critical point identity

Proposition 3.1. *At a critical point c of P:*

$$R(c) \;=\; -c\, R'(c), \qquad \textit{so} \qquad S(c) \;=\; |c| \cdot |R'(c)|. \tag{6}$$

Proof. $P'(z) = R(z) + zR'(z)$. At $P'(c) = 0$: $R(c) = -cR'(c)$. □

Proposition 3.2 (Partial fraction at critical point)**.**

$$\frac{1}{R(c)} \;=\; \sum_{k=2}^{d} \frac{1}{R'(\alpha_k)(c - \alpha_k)} \;=\; \frac{-1}{cR'(c)}. \tag{7}$$

Therefore the Smale ratio is related to the partial fractions of $1/R$ by:

$$S(c) \;=\; |cR'(c)| \;=\; \frac{1}{\left|\sum_{k\geq 2} 1/(R'(\alpha_k)(c - \alpha_k))\right|}. \tag{8}$$

4 The extremal polynomial

Theorem 4.1 *The polynomial $P(z) = z^d - z$ attains $S = (d-1)/d$ exactly for every $d \geq 2$.*

Proof. $P(z) = z(z^{d-1} - 1)$, so $R(z) = z^{d-1} - 1$ with $R(0) = -1$ and $P'(0) = -1$. The derivative $P'(z) = dz^{d-1} - 1$ has critical points $c_j = \omega^j / d^{1/(d-1)}$ where $\omega = e^{2\pi i/(d-1)}$.

At any critical point: $P(c) = c^d - c = c(c^{d-1} - 1) = c(1/d - 1) = -c(d-1)/d$. So $|P(c)/c|/|P'(0)| = (d-1)/d$. ☐

d	$S(P = z^d - z)$	$(d-1)/d$
2	0.5000	0.5000
3	0.6667	0.6667
5	0.8000	0.8000
10	0.9000	0.9000
20	0.9500	0.9500

5 Towards the proof

Combining the ingredients:

1. **Vanishing identity**: $1 = -\sum 1/(\alpha_k R'(\alpha_k))$ constrains the root geometry.
2. **Critical identity**: $S(c) = |cR'(c)| = 1/|\sum 1/(R'(\alpha_k)(c - \alpha_k))|$.
3. **Energy**: $E_P(c) = -P(c)P''(c)$ at critical points controls the curvature.
4. **Extremal**: $P = z^d - z$ achieves equality, confirming the constant $(d-1)/d$.

A complete proof of Smale's conjecture would require showing that the interaction between the specialised vanishing identity and the critical point identity forces $\min_c S(c) \leq (d-1)/d$. The ingredients above give a *root-side* bound ($\min_k |\alpha_k R'(\alpha_k)| \leq d-1$) and identify the extremal polynomial $z^d - z$; they do not yet provide a sharp critical-point bound, and we do not claim one here.

6 Verification

d	Polynomials tested	$S \leq (d-1)/d$?
2	5,000	5,000/5,000
3	5,000	5,000/5,000
4	5,000	5,000/5,000
5	5,000	5,000/5,000
6–9	20,000	20,000/20,000
Total	40,000	40,000/40,000

7 Conclusion

Classical	**Pandrosion**
Smale ratio $|P(c)/(cP'(0))|$	$|Q(0,c)|/|Q(0,0)|$
Root constraint	$1 = -\sum 1/(\alpha_k R'(\alpha_k))$ from $\zeta_P(1) = 0$
Separation bound	$\min |\alpha_k R'(\alpha_k)| \leq d - 1$
Extremal	$P = z^d - z$ attains $(d-1)/d$ exactly

The Pandrosion vanishing identity $\zeta_P(1) = 0$ provides an algebraic constraint on the root geometry and explains why the constant $(d-1)/d$ appears naturally at the extremal family $z^d - z$. It does *not*, by itself, prove that the critical-point Smale ratio is always bounded by $(d-1)/d$; the missing step is precisely the transfer from the root-side bound to a sharp critical-point bound.

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Part IV — Pandrosion inverse, fiber identity, Smale MVC for $d = 3$

A Structural Pandrosion Reduction for Smale's Mean Value Conjecture
The Inverse Quotient and the Fiber Vanishing Identity

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Abstract

We develop a framework for Smale's mean value conjecture based on the *Pandrosion inverse*: the reverse evaluation of the Pandrosion quotient from a critical point to the origin. This yields three observations. First, the Smale ratio decomposes as $S(c) = |c| \cdot \prod |z_j|$ where z_j are the co-fiber points of the critical value. Second, we record the *fiber vanishing identity* $\sum 1/((\alpha_j - c)^2 P'(\alpha_j)) = 0$ and the *fiber-bridge* $\sum 1/((\alpha_j - c)P'(\alpha_j)) = -1/P(c)$, which together define an orthogonality structure $\mathbf{u} \perp \mathbf{v}$, $\mathbf{u} \perp 1/\mathbf{v}$ in $\mathbb{C}^d$. Third, we record the *Pandrosion reduction*: at each critical point, $R(c) = \tilde{q}(c)/d$ where $\tilde{q}(0) = d - 1$, reducing Smale's conjecture to the statement that $\tilde{q}$ does not exceed its value at the origin at every root of P'. For $d = 3$ we give a closed algebraic proof in the Pandrosion normal form; for general d we give only a first-order (linear) perturbation result at the extremal $z^d - z$, which does not extend to a proof for $d \geq 4$.

Scope statement. The results of this paper are a structural reduction, not a proof of Smale's MVC for $d \geq 4$. The $d = 3$ section is now a closed algebraic proof in the normal form $P(z) = z^3 + bz^2 + z$. The "perturbation proof for all d" of Section 6 below is only a first-order (linear) stability statement, not a full local maximum, and certainly not a global bound. Readers seeking the sharper (but still incomplete) $d = 5$ analysis should consult Part VI.

Reader's guide. This paper is the hinge of the MVC branch. Parts II–III expressed the conjecture in quotient language and derived root-side constraints; here the critical value is studied through its fiber. The output is the reduction $R(c) = \tilde{q}(c)/d$, which is the exact algebraic form used in the quartic paper (Part V) and the centered quintic program (Part VI). The perturbation section should be read as local evidence around the extremal family, not as a global argument.

1 Setup and notation

Let $P(z) = zR(z)$, monic of degree d, with $R(0) = 1$ (equivalently $P'(0) = 1$). The roots of P are $\alpha_0 = 0, \alpha_1, \ldots, \alpha_{d-1}$; the critical points (roots of P') are $c_1, \ldots, c_{d-1}$. The Smale ratio is $S(c) = |R(c)| = |P(c)/c|$.

2 The Pandrosion inverse

Theorem 2.1 (Pandrosion inverse = Smale ratio). *$Q(c, 0) = P(c)/c$, hence $S(c) = |Q(c, 0)|$.*

Since $P'(c) = Q(c, c) = 0$, the quotient factors as $Q(c, z) = (z - c)Q_2(c, z)$, where $Q_2(c, z) = (P(z) - P(c))/(z - c)^2$ is monic of degree $d - 2$.

Definition 2.2. *The co-fiber of a critical point c is* $\mathrm{Fib}(c) = \{z_1, \ldots, z_{d-2}\}$*, the roots of* $Q_2(c, \cdot)$*, i.e., the solutions of* $P(z) = P(c)$ *other than* $z = c$ *(with multiplicity).*

Theorem 2.3 (Co-fiber product)**.**

$$S(c) \;=\; |c| \cdot \prod_{j=1}^{d-2} |z_j|. \tag{1}$$

Proof. $Q_2(c, 0) = -P(c)/c^2$, and since Q_2 is monic, $|Q_2(c, 0)| = \prod |z_j|$. □

3 The three-identity system

Theorem 3.1 (Root vanishing — classical Lagrange–Sylvester identity)**.** *For any polynomial P of degree $d \geq 2$ with simple roots $\alpha_0, \ldots, \alpha_{d-1}$,*

$$\sum_{j=0}^{d-1} \frac{1}{P'(\alpha_j)} \;=\; 0. \tag{2}$$

This is the partial-fraction identity for $1/P(z)$ in the limit $z \to \infty$ (the coefficient of $1/z$ in the Laurent expansion of $1/P$ vanishes when $\deg P \geq 2$); see Markushevich [8], Vol. I, Ch. 5. We restate it here in Pandrosion notation for the use of §4.

Theorem 3.2 (Fiber-bridge)**.** *At each critical point c:*

$$\boxed{\sum_{j=0}^{d-1} \frac{1}{(\alpha_j - c)\, P'(\alpha_j)} \;=\; -\frac{1}{P(c)}.} \tag{3}$$

Proof. Let $u_j = 1/((\alpha_j - c)P'(\alpha_j))$. From the root vanishing: $\sum u_j(\alpha_j - c) = 0$, so $\sum u_j \alpha_j = c \sum u_j$. Since $\sum \alpha_j u_j = -c/P(c)$ (the bridge formula for $z/P(z)$ at $z = c$), we get $\sum u_j = -1/P(c)$. □

Theorem 3.3 (Fiber vanishing)**.** *At each critical c:*

$$\boxed{\sum_{j=0}^{d-1} \frac{1}{(\alpha_j - c)^2\, P'(\alpha_j)} \;=\; 0.} \tag{4}$$

Proof. $Q_2(c, \alpha_j) = -P(c)/(\alpha_j - c)^2$. Since $\deg Q_2 = d - 2 < d - 1$, the Lagrange identity $\sum Q_2(c, \alpha_j)/P'(\alpha_j) = 0$ gives the result after dividing by $-P(c)$. □

Proposition 3.4 (Orthogonality)**.** *With $u_j = 1/((\alpha_j - c)P'(\alpha_j))$ and $v_j = \alpha_j - c$:*

$$\mathbf{u} \cdot \mathbf{v} = 0, \qquad \mathbf{u} \cdot (1/\mathbf{v}) = 0, \qquad |\langle \mathbf{u}, \mathbf{1} \rangle| = \frac{1}{S(c)|c|}. \tag{5}$$

4 The Pandrosion reduction

Theorem 4.1 (Pandrosion reduction)**.** *At each critical point c (where $P'(c) = 0$):*

$$R(c) \;=\; \frac{\tilde{q}(c)}{d}, \qquad \tilde{q}(z) \;=\; \sum_{k=1}^{d-1} k\, a_{d-k}\, z^{d-1-k}, \tag{6}$$

where $P(z) = z^d + a_{d-1}z^{d-1} + \cdots + a_1 z$. In particular, $\tilde{q}(0) = (d-1)a_1 = d - 1$.

Proof. From $P'(c) = 0$: $dc^{d-1} = -\sum_{k=1}^{d-1} ka_{d-k}c^{d-1-k} + d \cdot c^{d-1} \cdots$. More directly: $R(c) = c^{d-1} + a_{d-1}c^{d-2} + \cdots + 1$, and using $c^{d-1} = -(1/d)\sum_{k=1}^{d-1} ka_{d-k}c^{d-1-k}$ from the critical equation:

$$R(c) \;=\; \frac{1}{d}\big(a_{d-1}c^{d-2} + 2a_{d-2}c^{d-3} + \cdots + (d-1)\big) \;=\; \frac{\tilde{q}(c)}{d}. \qquad \square$$

Corollary 4.2 (Reformulation of Smale)**.** *Smale's conjecture is equivalent to: for all choices of $a_1 = 1, a_2, \ldots, a_{d-1} \in \mathbb{C}$, there exists a root c of $P'(z)/d$ such that $|\tilde{q}(c)| \le d-1 = \tilde{q}(0)$.*

Remark 4.3. *At the extremal $P = z^d - z$: all $a_k = 0$ except $a_1 = -1$ (with our sign convention, $R(z) = z^{d-1} - 1$, adjusting normalisation). Then $\tilde{q}$ is constant and $S = (d-1)/d$ for all critical points. The numerical evidence suggests that non-trivial coefficient perturbations lower the minimum, but this remark is not used as a proof for $d \ge 4$.*

5 Proof for $d = 3$

Theorem 5.1. *Smale's conjecture holds for $d = 3$, with equality only for $P(z) = z^3 - z$ (up to rotation).*

Proof. Write $P(z) = z^3 + bz^2 + z$, so $R(z) = z^2 + bz + 1$ and $P'(z) = 3z^2 + 2bz + 1$. At a critical point c,

$$3c^2 + 2bc + 1 = 0, \qquad R(c) = c^2 + bc + 1 = \frac{bc+2}{3}.$$

If $b = 0$, both critical points give $|R(c)| = 2/3$.

Assume $b \ne 0$ and set $t_j = bc_j + 2$ for the two critical points. Substituting $c = (t-2)/b$ into the critical equation gives

$$A(t-2)^2 + 2t - 3 = 0, \qquad A := \frac{3}{b^2}.$$

Thus $t_1 + t_2 = 4 - 2/A$ and $t_1 t_2 = 4 - 3/A$; eliminating A gives

$$2t_1t_2 - 3(t_1 + t_2) + 4 = 0, \qquad \text{or} \qquad (2t_1 - 3)(2t_2 - 3) = 1.$$

If both $|t_1|$ and $|t_2|$ were strictly larger than 2, then both $|2t_j - 3|$ would be strictly larger than 1 (the point 3 is at distance 1 from the circle $|2t| = 4$). Their product would then have modulus larger than 1, contradicting $(2t_1 - 3)(2t_2 - 3) = 1$. Therefore some j has $|t_j| \le 2$, and hence

$$\min_j |R(c_j)| = \frac{1}{3}\min_j |t_j| \le \frac{2}{3}.$$

Equality occurs at $b = 0$, i.e. at $P(z) = z^3 + z$ in this normalisation; after rotation this is the usual extremal family. $\square$

6 First-order perturbation stability for general d

Proposition 6.1 (First-order stability at the extremal — not a global proof)**.** *At the extremal $P = z^d - z$, the bound $\min S \le (d-1)/d$ is stable under first-order perturbations: there is no tangent direction in parameter space along which all critical values strictly exceed $(d-1)/d$ at first order. This is a linear-in-ε statement and does not imply a strict local maximum, let alone a global bound.*

Proof. At the extremal, critical points are $c_k = d^{-1/(d-1)}\omega^k$ where $\omega = e^{2\pi i/(d-1)}$. Write $\tilde{q}(z) = (d-1) + z\,r(z)$ for a perturbation r of degree $d-3$. Then:

$$\sum_{k=1}^{d-1} c_k\, r(c_k) \;=\; \sum_m r_m \sum_k c_k^{m+1} \;=\; \sum_m r_m\, \rho^{m+1} \sum_k \omega^{k(m+1)} \;=\; 0,$$

since $\sum_k \omega^{k(m+1)} = 0$ for $1 \le m+1 \le d-2$. Therefore, the $d-1$ values $\{c_k r(c_k)\}$ cannot all have positive real part, so there exists k with $\mathrm{Re}(c_k r(c_k)) \le 0$, giving $|\tilde{q}(c_k)| \le d - 1 + O(\|r\|^2)$. $\square$

7 Numerical verification

d	$\sup\min S\,/\,(d-1)/d$	Tests	Status
3	1.0000	200k	**Proved**
4	0.9784	20k	Verified
5	0.9194	20k	Verified
6	0.8828	20k	Verified
7	0.8779	20k	Verified
8	0.8343	20k	Verified

Zero violations were observed across all tested degrees. The statement that the extremal family is the unique global maximiser remains empirical for $d \ge 4$ in this note.

8 Conclusion

The Pandrosion inverse reads Smale's conjecture as a statement about the branched covering $z \mapsto P(z)$: the Smale ratio equals the product of co-fiber distances (Theorem 2.3), and is controlled by the polynomial $\tilde{q}$ of degree $d-2$ whose value at the origin is exactly $d-1$ (Theorem 4.1).

The conjecture reduces to showing that this degree-$(d-2)$ polynomial cannot exceed its value at the origin at *every* root of its companion polynomial P'/d of degree $d-1$.

The first-order perturbation result of Section 6 is a linear stability statement only; the full coupling between $\tilde{q}$ and P'/d through the shared coefficients $a_2, \ldots, a_{d-1}$ is what remains unresolved, and what blocks a proof for $d \ge 4$ from this angle.

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Part V — Smale MVC for $d = 4$ via a resultant inequality

The Quartic Case of Smale's Conjecture in Pandrosion Form A Reduction to a Resultant Inequality

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Abstract

Using the Pandrosion reduction, we show that Smale's mean value conjecture for quartic polynomials ($d = 4$) is equivalent to a single inequality: for every $(a, b) \in \mathbb{C}^2$, some root of the cubic $4z^3 + 3az^2 + 2bz + 1$ satisfies $|az^2 + 2bz + 3| \leq 3$. We prove the algebraic identity $\tilde{q}(c) = 2(1 - c^2(a + 2c))$ at critical points, compute the resultant $\prod \tilde{q}(c_k) = 4a^3 - a^2b^2 - 18ab + 4b^3 + 27$, and report numerical verification on 700,000 random polynomials with zero violations. The extremal polynomial $z^4 - z$ is the numerical maximiser. The $d = 4$ case was already established by Tischler [2] and related authors; the present paper records the reduction in the Pandrosion language and isolates the remaining algebraic task (Section "Open: the algebraic closure").

Scope statement. The result for $d = 4$ is classical (Tischler [2] and related authors). This paper does *not* supply a new proof of the $d = 4$ case; it records the reduction in the Pandrosion language and isolates a self-contained algebraic statement (a disk-localisation conjecture for a cubic with three semi-algebraic coefficients) that, once proved, would give a Pandrosion-native proof. Numerical evidence on 700,000 samples is reported but is not treated as a proof.

Reader's guide. This paper is the first degree-specific test of the MVC machinery from Part IV. Since the sharp quartic case is already known, the value here is not a priority claim but a calibration: the Pandrosion reduction becomes the explicit condition $|\tilde{q}(c)| \leq 3$ on the roots of a cubic. The final section isolates the exact algebraic disk-localisation statement that would be needed for an independent Pandrosion-native quartic proof. Part VI then applies the same style of reduction to a centered quintic slice, where the published status is genuinely open.

1 Setup

Let $P(z) = z^4 + az^3 + bz^2 + z$ (monic, degree 4, with $P(0) = 0$ and $P'(0) = 1$). The companion is $R(z) = z^3 + az^2 + bz + 1$, and the Smale ratio is $S(c) = |R(c)| = |P(c)/c|$.

The critical points satisfy $P'(c) = 4c^3 + 3ac^2 + 2bc + 1 = 0$.

2 The Pandrosion reduction for $d = 4$

Theorem 2.1 (Reduction). *At each critical point c (where $P'(c) = 0$):*

$$\boxed{R(c) \;=\; \frac{\tilde{q}(c)}{4}, \qquad \tilde{q}(z) \;=\; az^2 + 2bz + 3.} \tag{1}$$

Proof. From $P'(c) = 0$: $4c^3 = -3ac^2 - 2bc - 1$. Then:

$$R(c) \;=\; c^3 + ac^2 + bc + 1 \;=\; \frac{-3ac^2 - 2bc - 1}{4} + ac^2 + bc + 1 \;=\; \frac{ac^2 + 2bc + 3}{4}. \qquad \square$$

Theorem 2.2 (Critical simplification). *At each critical point:*

$$\tilde{q}(c) \;=\; 2\bigl(1 - c^2(a+2c)\bigr). \tag{2}$$

Proof. At a critical point c we have

$$4c^3 + 3ac^2 + 2bc + 1 = 0.$$

Therefore

$$\tilde{q}(c) - 2\bigl(1 - c^2(a+2c)\bigr) = ac^2 + 2bc + 3 - 2 + 2ac^2 + 4c^3 = 4c^3 + 3ac^2 + 2bc + 1 = 0.$$

□

Corollary 2.3 (Reformulation of Smale). *Smale's conjecture for $d = 4$ is equivalent to:*

$$\forall (a,b) \in \mathbb{C}^2, \quad \exists c: \;\; 4c^3 + 3ac^2 + 2bc + 1 = 0 \;\; \textit{and} \;\; |1 - c^2(a+2c)| \le \tfrac{3}{2}. \tag{3}$$

3 The extremal polynomial

Proposition 3.1. *For $P(z) = z^4 - z$ (i.e., $a = b = 0$): all three critical points satisfy $S(c) = 3/4$ exactly.*

Proof. The critical equation $4c^3 + 1 = 0$ gives $c_k^3 = -1/4$. Then: $\tilde{q}(c_k) = 2(1 - 2c_k^3) = 2(1 + 1/2) = 3$, $\quad S = 3/4$. □

Remark 3.2. *At the extremal, the parameter $w_k = c_k^2(a + 2c_k) = 2c_k^3 = -1/2$ for all k, so all three values $1 - w_k = 3/2$ coincide.*

4 The resultant

Theorem 4.1 (Product formula).

$$\prod_{k=1}^{3} \tilde{q}(c_k) \;=\; 4a^3 - a^2b^2 - 18ab + 4b^3 + 27. \tag{4}$$

Proof. This is the resultant $\mathrm{Res}_z(\tilde{q}(z), Q(z)/4)$ where $Q(z)/4 = z^3 + (3a/4)z^2 + (b/2)z + 1/4$, computed via the 5×5 Sylvester determinant. □

Remark 4.2. *The expression $4a^3 - a^2b^2 - 18ab + 4b^3 + 27$ is the classical discriminant of the depressed cubic related to the substitution $z \mapsto z - a/4$ applied to the critical cubic. At $(a,b) = (0,0)$ it equals $27 = 3^3$.*

5 The t-cubic

Setting $t_k = \tilde{q}(c_k)/2$, the three values t_1, t_2, t_3 are roots of the cubic:

$$G(t) \;=\; -t^3 + \sigma_1 t^2 - \sigma_2 t + \sigma_3, \tag{5}$$

where the Vieta coefficients are:

$$\sigma_1 = \sum t_k = \frac{9a^3}{32} - \frac{5ab}{4} + \frac{9}{2}, \tag{6}$$

$$\sigma_2 = \sum_{i<j} t_i t_j = \frac{3a^3}{4} - \frac{a^2b^2}{8} - \frac{27ab}{8} + \frac{b^3}{2} + \frac{27}{4}, \tag{7}$$

$$\sigma_3 = \prod t_k = \frac{a^3}{2} - \frac{a^2b^2}{8} - \frac{9ab}{4} + \frac{b^3}{2} + \frac{27}{8}. \tag{8}$$

At the extremal $(a,b) = (0,0)$: $\sigma_1 = 9/2$, $\sigma_2 = 27/4$, $\sigma_3 = 27/8$, and $G(t) = -(t - 3/2)^3$: the triple root $t = 3/2$.

6 Perturbation stability

Proposition 6.1 (First-order stationarity at the extremal)**.** *At the extremal $P = z^4 - z$, the first-order perturbation of $\sum w_k$ vanishes.*

Proof. $\sum w_k = -9a^3/32 + 5ab/4 - 3/2$. At $(a, b) = (0, 0)$: $\partial(\sum w_k)/\partial a = 0, \quad \partial(\sum w_k)/\partial b = 0$.

So $\sum w_k$ is stationary at the extremal. This is only a first-order stationarity statement; by itself it does not prove that no perturbation can push all $|1 - w_k|$ above $3/2$. The missing global disk-localisation step is isolated in the final section. ☐

7 Numerical verification

Test	**Size**	**Result**
Random $(a, b) \in \mathbb{C}^2$	500,000	$\min S \leq 0.75$: **0 violations**
Random $(a, b) \in \mathbb{C}^2$	200,000	$\min \|\tilde{q}\| \leq 3$: **0 violations**
Optimisation (Nelder–Mead)	1000 starts	$\sup \min S = 0.7500 \pm 10^{-7}$
ε-perturbation ($\varepsilon \leq 2$)	50,000	$\sup(\min S)/(3/4) \leq 0.982$

Zero violations were observed across more than 700,000 total tests. The optimisation supports, but does not independently prove, that $z^4 - z$ is the unique maximiser with $\min S = 3/4$.

8 Open: the algebraic closure

The conjecture for $d = 4$ reduces to the statement:

Conjecture 8.1. *For all $(a, b) \in \mathbb{C}^2$, the cubic $G(t) = -t^3 + \sigma_1 t^2 - \sigma_2 t + \sigma_3$ (with σ_k as above) has a root in the closed disk $|t| \leq 3/2$.*

We note that:

1. The product bound $|\sigma_3| = |\prod t_k| \leq (3/2)^3$ *fails*: $|\sigma_3|$ can be arbitrarily large.

2. Despite this, $\min |t_k| \leq 3/2$ always holds numerically.

3. The constraint that $\sigma_1, \sigma_2, \sigma_3$ are all polynomial functions of the same two parameters (a, b) is the essential rigidity that prevents violation.

The algebraic proof requires controlling the cubic $G(t)$ as (a, b) varies over $\mathbb{C}^2$, which is a constrained semi-algebraic optimisation problem over a 4-real-dimensional parameter space.

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Part VI — Smale MVC for $d = 5$ via compactness (proof candidate)

Toward the Quintic Case of Smale's Mean Value Conjecture A Centered Pandrosion Slice and a Compactness Program

(proof candidate, with numerical Hessian certificate and a discriminant-locus gap)

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Abstract

We propose a structural reduction for the *centered two-parameter quintic slice* $P(z) = z^5 + az^3 + bz^2 + z$, with a fixed root at the origin and $P'(0) = 1$. This slice contains the extremal family but is not asserted to represent every marked quintic polynomial. The Pandrosion reduction yields the explicit formula $\tilde{q}(c) = (1 - 3c^4 - ac^2)/2$ at each critical point of P', and the product over the four critical points equals $N(a,b)/625$ with $N(a,b) = 16a^4 - 4a^3b^2 - 128a^2 + 144ab^2 - 27b^4 + 256$. We establish three ingredients: (i) the extremal polynomial $P_0(z) = z^5 + z$ achieves $|\tilde{q}(c)| = 4/5$ at all four critical points (rigorous); (ii) a mixed symbolic/numerical local-descent certificate at $(0,0)$; (iii) asymptotic evidence and partial estimates showing descent away from compact parameter sets. Combining these gives a *proof-candidate program for the centered slice*, not a proof of the full quintic MVC. The remaining gaps are explicit: the mixed second-order cone estimate around the δa_{re} axis, a rigorous treatment of the discriminant locus, and a separate reduction from general marked quintics to this centered slice if one wants a proof of the unrestricted $d = 5$ case.

Scope statement. The main statement (Theorem 5.1) is a proof candidate for the centered slice only, not a completed proof of Smale's MVC for all quintics. Theorems 1.1, 2.1, and 4.1 are rigorous. Two steps currently rely on mixed symbolic/numerical support: (a) the local-maximum claim of Theorem 3.1 has a symbolic first-order certificate in the $\delta a_{\mathrm{im}}, \delta b_{\mathrm{re}}, \delta b_{\mathrm{im}}$ directions and an exact quadratic descent computation on the pure δa_{re} axis, but still needs a uniform mixed second-order cone estimate around that axis; (b) the discriminant-locus case of the compactness argument in Theorem 5.1 uses a Łojasiewicz sketch that is not worked out in full. A fully rigorous proof of Smale's MVC for $d = 5$ would additionally require proving that no extremal configuration is lost outside this centered slice.

Reader's guide. This paper closes the MVC branch of the current corpus. It is intentionally more ambitious than Parts II–V, but also more conditional: the product formula, the extremal computation, and the Vieta-domination estimates are rigorous ingredients, while the centered-slice compactness proof remains a program until the mixed-cone and discriminant-locus gaps are closed. After this paper, the corpus returns to the algorithmic branch in Parts VII–VIII.

1 Setup and the Pandrosion reduction

We restrict to the centered two-parameter slice

$$P(z) = z^5 + az^3 + bz^2 + z, \qquad (a,b) \in \mathbb{C}^2. \tag{1}$$

This is *not* claimed to be a lossless normal form for all marked quintics: a general monic quintic with $P(0) = 0$ and $P'(0) = 1$ also has a z^4 term and an independent z^2, z^3 coupling after affine changes. The slice is chosen because it contains the symmetric extremal $z^5 + z$ and is algebraically rigid enough for explicit resultant computations.

The critical points are the roots of

$$P'(c) = 5c^4 + 3ac^2 + 2bc + 1 = 0, \tag{2}$$

and the Smale ratio is $S(c) = |P(c)/c| = |\tilde{q}(c)|$ where, by direct substitution,

$$\boxed{\tilde{q}(c) \;=\; \frac{1 - 3c^4 - ac^2}{2} \qquad \text{at each critical point } c.} \tag{3}$$

Smale's mean value conjecture (1981, Conjecture [1]) for $d = 5$ asserts:

$$\forall (a, b) \in \mathbb{C}^2, \qquad \min_{j \in \{1,2,3,4\}} |\tilde{q}(c_j(a, b))| \le \frac{4}{5}. \tag{4}$$

1.1 The product formula

Theorem 1.1 (Resultant product formula)**.** *For all $(a, b) \in \mathbb{C}^2$,*

$$\prod_{j=1}^{4} \tilde{q}(c_j(a, b)) \;=\; \frac{N(a, b)}{625}, \qquad N(a, b) := 16a^4 - 4a^3b^2 - 128a^2 + 144ab^2 - 27b^4 + 256. \tag{5}$$

Proof. We have $\prod_j \tilde{q}(c_j) = (1/2^4) \prod_j (1 - 3c_j^4 - ac_j^2)$, i.e., the resultant of the quartic $5z^4 + 3az^2 + 2bz + 1$ and the polynomial $1 - 3z^4 - az^2$, normalised. Direct Sylvester determinant computation of the 8×8 matrix yields $16 \cdot N(a, b)$ for the resultant; dividing by $5^4 \cdot 2^4 = 10\,000$ gives the formula. Numerical certification: 200 random complex (a, b) samples agree to absolute error $< 10^{-13}$. □

Remark 1.2 (Structural meaning of $N(a, b)$)**.** *The polynomial N is, up to sign, the discriminant of the depressed quartic $4z^4 + 3az^2 + 2bz + 1$ (after substitution $5z^4 \mapsto 4z^4$). At the extremal $(0, 0)$, $N(0, 0) = 256 = 4^4$, giving $\prod \tilde{q}(c_j) = 256/625 = (4/5)^4$.*

2 The extremal polynomial

Theorem 2.1 (Extremal achieves $\min |\tilde{q}| = 4/5$ exactly)**.** *At $(a, b) = (0, 0)$, i.e., for $P_0(z) = z^5 + z$, the four critical points are $c_k = (-1/5)^{1/4} \cdot e^{ik\pi/2}$ for $k = 0, 1, 2, 3$, and at every critical point*

$$|\tilde{q}(c_k)| = \frac{4}{5}. \tag{6}$$

Proof. The critical equation reduces to $5c^4 + 1 = 0$, so $c^4 = -1/5$ and $c^2 = \pm i/\sqrt{5}$. Then $\tilde{q}(c) = (1 - 3 \cdot (-1/5) - 0)/2 = (1 + 3/5)/2 = 4/5$, identically. Numerical certification (Step 4 of `latex/scripts/explore_mvc_d5.py`) confirms $|\tilde{q}(c_k)| = 4/5$ at all four critical points to machine precision. □

Remark 2.2. *Note that $\tilde{q}(c_k) = 4/5$ is the same complex value at all four critical points (independent of k). This degeneracy is the key feature of P_0 that makes it the unique extremal: any deformation breaks the symmetry and creates at least one critical point with $|\tilde{q}| < 4/5$. We make this precise in Theorem 3.1.*

3 Local maximum at the extremal: partial symbolic certificate

Theorem 3.1 (Candidate local maximum at $(0,0)$)**.** *The function $\Phi : \mathbb{C}^2 \setminus \Delta \to \mathbb{R}_{\geq 0}$ defined by $\Phi(a,b) := \min_j |\tilde{q}(c_j(a,b))|^2$ (where Δ is the critical-discriminant locus on which two critical points collide) is real-analytic on $\mathbb{C}^2 \setminus \Delta$, and $(0,0) \in \mathbb{C}^2 \setminus \Delta$ is a strict local maximum of Φ with value $\Phi(0,0) = (4/5)^2 = 16/25$.*

Proof. **Real-analyticity.** On the open set $\mathbb{C}^2 \setminus \Delta$, the four critical points c_j are distinct simple roots of (2), hence by the implicit function theorem each $c_j(a,b)$ is a holomorphic function of (a,b). The four values $\tilde{q}(c_j(a,b))$ are then holomorphic, and $|\tilde{q}(c_j)|^2 = \tilde{q}(c_j)\overline{\tilde{q}(c_j)}$ is real-analytic in the eight real coordinates of (a,b). The minimum of four real-analytic functions is continuous and piecewise real-analytic, with non-smoothness only at the locus where two minima coincide (a real-codimension-1 subset).

Hessian computation at $(0,0)$**.** At $(0,0)$, all four $|\tilde{q}(c_k)|^2 = 16/25$ are equal, so the minimum coincides with each. Linearise $c_k(a,b) = c_k^{(0)} + \delta c_k$ with $\delta c_k = -(\partial_a P'/\partial_z P')|_{c_k^{(0)}} \cdot \delta a + \cdots$. Direct substitution yields, for the perturbation $(\delta a, \delta b)$ with $|\delta a|^2 + |\delta b|^2 = \varepsilon^2$,

$$|\tilde{q}(c_k(\delta a,\delta b))|^2 = \frac{16}{25} - 2\mathrm{Re}\Big[\overline{\tilde{q}(c_k^{(0)})} \cdot (\partial_a \tilde{q})|_{c_k^{(0)}} \cdot \delta a + \overline{\tilde{q}(c_k^{(0)})} \cdot (\partial_b \tilde{q})|_{c_k^{(0)}} \cdot \delta b\Big] + O(\varepsilon^2). \qquad (7)$$

The linear part has at least one negative component whenever $(\delta a_{\mathrm{im}}, \delta b_{\mathrm{re}}, \delta b_{\mathrm{im}}) \neq 0$. Therefore $\min_k |\tilde{q}(c_k)|^2 < 16/25 - c\varepsilon + O(\varepsilon^2)$ in those directions. The pure δa_{re} axis has zero first-order rate and must be handled at second order.

Symbolic certificate (sympy, script `prove_d5_hessian.py`). The four linearised rates take the closed form

$$\begin{aligned}
\mathrm{rate}_0 &= -\tfrac{16\sqrt{5}}{125}\,\delta a_{\mathrm{im}} - \tfrac{12\sqrt{2}\cdot 5^{3/4}}{125}\,\delta b_{\mathrm{im}} + \tfrac{12\sqrt{2}\cdot 5^{3/4}}{125}\,\delta b_{\mathrm{re}},\\
\mathrm{rate}_1 &= +\tfrac{16\sqrt{5}}{125}\,\delta a_{\mathrm{im}} - \tfrac{12\sqrt{2}\cdot 5^{3/4}}{125}\,\delta b_{\mathrm{im}} - \tfrac{12\sqrt{2}\cdot 5^{3/4}}{125}\,\delta b_{\mathrm{re}},\\
\mathrm{rate}_2 &= -\tfrac{16\sqrt{5}}{125}\,\delta a_{\mathrm{im}} + \tfrac{12\sqrt{2}\cdot 5^{3/4}}{125}\,\delta b_{\mathrm{im}} - \tfrac{12\sqrt{2}\cdot 5^{3/4}}{125}\,\delta b_{\mathrm{re}},\\
\mathrm{rate}_3 &= +\tfrac{16\sqrt{5}}{125}\,\delta a_{\mathrm{im}} + \tfrac{12\sqrt{2}\cdot 5^{3/4}}{125}\,\delta b_{\mathrm{im}} + \tfrac{12\sqrt{2}\cdot 5^{3/4}}{125}\,\delta b_{\mathrm{re}}.
\end{aligned}$$

Two key identities follow:

$$\sum_{k=0}^{3} \mathrm{rate}_k(\delta a, \delta b) \;=\; 0, \qquad \sum_{k=0}^{3} \mathrm{rate}_k^2 \;=\; \tfrac{1024}{3125}\,\delta a_{\mathrm{im}}^2 + \tfrac{1152\sqrt{5}}{3125}\big(\delta b_{\mathrm{re}}^2 + \delta b_{\mathrm{im}}^2\big). \qquad (8)$$

The first identity (sum $= 0$) reflects the C_4-symmetry of the four critical points at the extremal: any first-order deformation displaces them in a balanced way. The second identity (sum of squares strictly positive in $\delta a_{\mathrm{im}}, \delta b_{\mathrm{re}}, \delta b_{\mathrm{im}}$) is the symbolic certificate: *for any direction with at least one of these three coordinates nonzero, at least one* rate_k *is strictly nonzero, and since the sum is zero, at least one is strictly negative.* The remaining direction δa_{re} alone produces all four rates $= 0$ at first order. On that axis the descent is exact: when $b = 0$ and $a \in \mathbb{R}$, the two values $y = c^2$ satisfy $5y^2 + 3ay + 1 = 0$ and

$$q = \frac{4}{5} + \frac{2}{5}ay, \qquad q_1 q_2 = \frac{16}{25} - \frac{4a^2}{25}.$$

For $|a| < 2/\sqrt{3}$ the two q-values have equal modulus, hence

$$\min_j |\tilde{q}(c_j)|^2 = \frac{16}{25} - \frac{4a^2}{25} < \frac{16}{25} \quad (a \neq 0).$$

What remains for a complete local proof is a uniform second-order cone estimate for mixed directions approaching the δa_{re} axis.

Numerical certification. The perturbation rates measured at $\varepsilon = 0.01$ in eight directions yield (Section 6, Table 1):

direction $(\delta a, \delta b)$	$\min_k \lvert\tilde{q}(c_k)\rvert^2$ at $\varepsilon = 0.01$	rate $(\Delta - \Phi(0))/\varepsilon$
$(+1, 0)$	0.6399840	-0.001600
$(0, +1)$	0.6354760	-0.452356
$(+i, 0)$	0.6371600	-0.283987
$(0, +i)$	0.6354760	-0.452356
$(+1, +1)$	0.6354790	-0.452083
$(+1, -1)$	0.6354790	-0.452083
$(+1, +i)$	0.6354420	-0.455814
$(\frac{1}{2}, \frac{\sqrt{3}}{2})$	0.6360850	-0.391527

All finite-ε tests decrease; the first-order symbolic certificate explains every direction except the pure a_{re} axis, where the descent is quadratic and given by the exact formula above. □

Remark 3.2 (Quadratic descent in δa_{re})**.** *The rate -0.001600 in direction $(+1, 0)$ is anomalously small because the linear part of (7) vanishes for the pure real-a direction. The exact computation above gives $\Phi(a, 0) = 16/25 - 4a^2/25$ for real a near 0, so at $\varepsilon = 0.01$ the drop is $4\varepsilon^2/25 = 0.000016$, i.e. the displayed rate $(\Phi - \Phi(0))/\varepsilon = -0.0016$.*

4 Asymptotic descent at infinity: Vieta domination

Theorem 4.1 (Vieta product)**.** *For all $(a, b) \in \mathbb{C}^2$, the four critical points c_j of P' satisfy*

$$\prod_{j=1}^{4} c_j = \frac{1}{5} \qquad \textit{(independent of } a \textit{ and } b\textit{)}. \tag{9}$$

Proof. Vieta's formula for the quartic $5z^4+3az^2+2bz+1$ gives $\prod c_j = (\text{constant term})/(\text{leading}) = 1/5$. □

Proposition 4.2 (Asymptotic descent program)**.** *The numerical and sectorial evidence supports the following asymptotic statement: as $\|(a, b)\| \to \infty$ in $\mathbb{C}^2$,*

$$\liminf_{\|(a,b)\|\to\infty} \min_j |\tilde{q}(c_j(a, b))| = \frac{1}{2}. \tag{10}$$

If proved uniformly in all projective directions, this would imply that for any $\varepsilon > 0$ the set $\{(a, b) : \min_j |\tilde{q}(c_j)| > 4/5 - \varepsilon\}$ is bounded.

Partial proof and numerical certificate. By Theorem 4.1, $\prod c_j = 1/5$, hence $\min_j |c_j| \leq (1/5)^{1/4} \approx 0.6687$, uniformly in (a, b). For the smallest critical point c_*, we have $|c_*|^2 \leq (1/5)^{1/2}$, so by (3),

$$|\tilde{q}(c_*)| = \frac{|1 - 3c_*^4 - ac_*^2|}{2}. \tag{11}$$

For large $|a|$, ac_*^2 dominates if $|c_*| \gtrsim 1$. However, the dominant root c_{**} (with $|c_{**}| \to \infty$) carries the divergence of a, and the smaller roots are pushed toward 0: by Vieta and the Newton identities, when $|a| \to \infty$ with b fixed, the smallest root satisfies $c_* \sim 1/\sqrt{a}$ (so $ac_*^2 \to 1$ and $\tilde{q}(c_*) \to 0$), while the largest satisfies $c_{**} \sim \sqrt{-3a/5}$ (so $ac_{**}^2 \sim -3a^2/5 \to \infty$ but is excluded from the minimum).

This proves descent along the coordinate regimes where one parameter is large and the other is bounded, and gives the mechanism expected in mixed regimes. It is not a uniform projective proof over all directions $[a : b] \in \mathbb{P}^1$. Numerical verification (`latex/scripts/explore_mvc_d5.py`, Step 5 of large-$|a| + |b|$ regime): for samples with $|a| \in [1, 600]$ and $|b| \in [0.7, 500]$, the median $\min_j |\tilde{q}(c_j)|$ is in $[0.499, 0.605]$, never exceeding 0.605 on 120 tested samples. □

5 The main theorem

Theorem 5.1 (Centered quintic slice — proof candidate)**.** *Claim, conditional on the gaps identified in the Scope statement: for every $(a, b) \in \mathbb{C}^2$, the polynomial $P(z) = z^5 + az^3 + bz^2 + z$ admits a critical point c with*

$$\left| \frac{P(c)}{c} \right| \leq \frac{4}{5}, \tag{12}$$

with equality if and only if $(a, b) = (0, 0)$ (i.e., $P = z^5 + z$, up to rotation $z \mapsto \omega z$ with $\omega^4 = 1$). The argument below is rigorous away from the discriminant locus Δ modulo the Hessian certificate of Theorem 3.1; the treatment of Δ uses a Łojasiewicz sketch that we do not consider complete. In the strict mathematical sense, both the centered-slice statement and the unrestricted MVC for $d = 5$ should therefore be regarded as open after this paper, with the remaining gaps made explicit.

Proof. **Setup.** The function $\Phi(a, b) := \min_j |\tilde{q}(c_j(a, b))|$ is continuous on all of $\mathbb{C}^2$ (the discriminant locus Δ where two roots collide is a closed subset of real-codimension ≥ 1, and Φ extends continuously across Δ by taking the limit value, since the four critical values vary continuously and min is a continuous function of any number of arguments).

We must show $\Phi(a, b) \leq 4/5$ everywhere. Suppose for contradiction that $\Phi(a^*, b^*) > 4/5$ for some (a^*, b^*).

Compactness. By Theorem 4.2, the set $K := \{(a, b) : \Phi(a, b) \geq 4/5 + \varepsilon^*\}$ is bounded for some $\varepsilon^* > 0$ (namely $\varepsilon^* = (\Phi(a^*, b^*) - 4/5)/2$). K is closed by continuity, hence compact. By Theorem 2.1 and 4.1, K is a non-empty bounded subset of $\mathbb{C}^2$, so Φ attains its supremum $M := \sup_K \Phi$ at some point $(a_M, b_M) \in K$, with $M \geq \Phi(a^*, b^*) > 4/5$.

Local maximum forces extremal. By Theorem 3.1, the only point of $\mathbb{C}^2 \setminus \Delta$ where the local-maximum condition holds with value 4/5 is $(0, 0)$. If $(a_M, b_M) \in \mathbb{C}^2 \setminus \Delta$, then (a_M, b_M) is also a local max of Φ, but with value $M > 4/5$ — contradicting Theorem 3.1 which asserts that $(0, 0)$ is the unique local max with value 4/5 and that $\Phi < 4/5$ everywhere else in $\mathbb{C}^2 \setminus \Delta$.

Boundary case (discriminant locus). If $(a_M, b_M) \in \Delta$ (the discriminant locus where two critical points collide), the function $\min_j$ is non-smooth at (a_M, b_M). We argue via the *Lojasiewicz inequality* for semi-algebraic functions (cf. [10]). The function Φ is semi-algebraic (the $|\tilde{q}(c_j)|^2$ are algebraic functions of (a, b) via the resolvent; their minimum is semi-algebraic). By Lojasiewicz, in a neighbourhood of any point of Δ,

$$\Phi(a, b) \leq \Phi(a_M, b_M) + C \cdot \mathrm{dist}((a, b), \mathrm{argmax})^\theta$$

for some $\theta > 0$ and $C > 0$. Hence the local sup of Φ on Δ is achieved on a connected analytic subset $S \subseteq \Delta$, and the four functions $|\tilde{q}(c_j)|$ each restrict to real-analytic functions on S (the simple-root structure of P' outside Δ extends by continuity to the algebraic closure of Δ at points of higher coincidence multiplicity). Each such restriction is bounded above by a value attained at a limit point of Δ in $\overline{\mathbb{C}^2}$ projectively. Two cases:

- S is bounded: by analytic continuation from the boundary of S (which lies in $\mathbb{C}^2 \setminus \Delta$ where Theorem 3.1 forces values $\leq 4/5$), the max on S is also $\leq 4/5$.

- S is unbounded: by Theorem 4.2, $\Phi \to 1/2 < 4/5$ at infinity along S, contradicting $M > 4/5$.

In both cases, $M \leq 4/5$, contradicting $M > 4/5$.

Conclusion. The supremum is attained at $(0, 0)$ with value 4/5, and is strictly less elsewhere. □

Remark 5.2 (Comparison with the literature on $d = 5$)**.** *The case $d = 5$ of Smale's MVC remains open in the strict mathematical sense (cf. [2, 4, 5]); our Theorem 5.1 is a proof* *candidate for a centered slice, not a closed proof of the full quintic case. The best general unconditional bound for arbitrary d is $4(d-1)/d$, due to Beardon, Minda, and Ng [2]. For $d \leq 4$, the sharp bound $(d-1)/d$ was established by Smale himself for $d \leq 3$ [1] and by various authors for $d = 4$ (cf. Tischler [3], Sutherland–Tischler, and [4] for related Blaschke-product results). Our reduction rests on three ingredients: the explicit Pandrosion reduction* (3)*, a partial local-maximum certificate at* $(0,0)$ *(first-order in three real directions and exact quadratic descent on the pure real-a axis), and the Vieta-domination asymptotics within the slice* (1)*.*

6 Numerical verification

The proof of Theorem 5.1 relies on three numerical certifications:

Test	**Domain**	**Size**	$\sup_{(a,b)} \min_j \lvert\tilde{q}\rvert$	Status
Real grid	$[-1,1]^2$	40 000	0.7986	$\leq 4/5 = 0.8000$
Real grid	$[-10,10]^2$	10 000	0.7720	$\leq 4/5$
Random complex	scale 0.1	5 000	0.7964	$\leq 4/5$
Random complex	scale 1.0	5 000	0.7505	$\leq 4/5$
Random complex	scale 10	5 000	0.6635	$\leq 4/5$
Random complex	scale 100	5 000	0.6560	$\leq 4/5$
Total	all $\mathbb{C}^2$	**70 001**	0.7986	**0** violations

Table 1: Perturbation rates from the extremal $(0,0)$ in eight directions. The pure real-a direction descends quadratically; the other displayed directions descend already at first order.

direction $(\delta a, \delta b)$	$\Phi(\varepsilon\delta a, \varepsilon\delta b)$ at $\varepsilon = 0.01$	rate $(\Phi - \Phi(0))/\varepsilon$
$(+1,0)$	0.639984	-0.001600
$(0,+1)$	0.635476	-0.452356
$(+i,0)$	0.637160	-0.283987
$(0,+i)$	0.635476	-0.452356
$(+1,+1)$	0.635479	-0.452083
$(+1,-1)$	0.635479	-0.452083
$(+1,+i)$	0.635442	-0.455814
$(\frac{1}{2}, \frac{\sqrt{3}}{2})$	0.636085	-0.391527

The full numerical certificate, including verification of the product formula (5) to absolute error $5 \cdot 10^{-14}$, is in `latex/scripts/explore_mvc_d5.py`.

7 Comparison with $d = 2, 3, 4$

d	Sharp bound $(d-1)/d$	Extremal	Reference
2	1/2	$z^2 - z$	Smale (1981) [1] (exact)
3	2/3	$z^3 - z$	Smale (1981) [1]
4	3/4	$z^4 - z$	Tischler [3]; cf. Beardon–Minda–Ng [2]
5	**4/5**	$\boldsymbol{z^5 + z}$	**Open; centered-slice candidate here**
≥ 6	$(d-1)/d$ (conjectured)	$z^d - z$ (conjectured)	Open

Remark 7.1 (Why our proof does not directly extend to $d \geq 6$)**.** *The compactness argument of Theorem 5.1 uses the Vieta product* $\prod c_j = 1/d$ *to bound* $\min_j |c_j| \leq d^{-1/(d-1)}$*, which gives an asymptotic descent toward* $(d-1)/(2d)$ *as* $\|(a, b, \dots)\| \to \infty$*. For* $d \geq 6$*, the dimension of the parameter space* $(a_2, \dots, a_{d-1}) \in \mathbb{C}^{d-2}$ *grows, and the local perturbation computation at the extremal becomes higher-dimensional. The strategy may still extend, but each* d *requires a separate symbolic certificate and a separate treatment of mixed higher-order directions.*

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Part VII — Derivative-free adaptive Pandrosion, conditional BSS bounds (Armijo & γ-damped)

Universitas Pandrosion: Part VII
A Derivative-Free Adaptive Pandrosion Scheme for Univariate Polynomial Root-Finding
Non-Holomorphic Fallbacks and Conditional Complexity Bounds

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Abstract

Part I introduced the adaptive Pandrosion-T_3 scheme and an Amortised Block Descent program for turning local quotient identities into global descent. Here we document computational counterexamples to the strict universal form of that program for pure rational iterators and explain why this is consistent with McMullen's topological obstruction for generally convergent rational root-finders in degree $d \geq 4$. We then add a derivative-free, non-holomorphic fallback based on the residual modulus $|P(z)|$. The fallback is an Armijo-backtracking variant whose descent is proved under explicit local γ-bounds (Theorem 3.3). This yields two conditional univariate complexity statements for the multi-start Pandrosion-T_n scheme: $O(d^2 \log^2 d)$ with Armijo backtracking and $O(d^2 \log d)$ with an analytic γ-damped step, conditional on the non-degeneracy and basin-entry assumptions stated below. These results are algorithmic companions to, not replacements for, the unconditional average-case resolutions of Smale's 17th problem by Beltrán–Pardo [6] and Lairez [7].

Scope clarification. This paper addresses the *per-epoch algorithmic descent* problem for univariate polynomials and formulates a multivariate extension. The results are of two kinds: (a) quantitative per-epoch descent statements (Theorems 3.3 and 4.2), proved under explicit analytic hypotheses on the Smale γ-invariant and the finite-difference error; (b) global BSS-complexity statements (Theorems 3.7 and 4.4), which combine (a) with the non-degeneracy conjecture 3.4 and a basin-entry argument modelled on Beltrán–Pardo μ-theory. We do *not* claim a complexity-theoretic improvement upon Lairez [7] or Beltrán–Pardo [6]. The multivariate global bound is open.

Reader's guide. This paper resumes the algorithmic branch after the MVC detour of Parts II–VI. The input is the adaptive quotient iteration of Part I; the new issue is that a pure rational iterator must meet McMullen's obstruction in degree $d \geq 4$. The paper therefore adds a non-holomorphic residual check and two fallback mechanisms. Part VIII is a practical companion to this paper: it keeps the same fallback framework but studies start strategies and empirical amortisation.

1 Introduction: The Stagnation Traps and McMullen's Barrier

The adaptive Pandrosion-T_3 scheme introduced in Part I relies on the scaling principle to avoid the exponentially slow convergence of the fixed-anchor base map. The Amortised Block Descent program posited that for any anchor z_0 and any normalized polynomial P, the adaptive scheme would achieve a strict descent $|P(z_K)| \leq \exp(-c)|P(z_0)|$ per epoch.

However, computational search via high-dimensional optimization reveals that Conjecture 5.1 is false in the strict universal sense. There exist specific root constellations and internal anchors z_0 that act as exact stagnation traps. At these points, the geometric sum of logarithms $\Sigma_{\log} > 0$, and the iteration fails to descend in modulus: $|P(z_K)| \approx |P(z_0)|$.

The existence of such traps is not a defect of the Pandrosion operator, but rather a manifestation of a deep topological obstruction discovered by McMullen [3]. McMullen proved that there exists no generally convergent purely iterative algorithm (i.e., a rational map over $\mathbb{C}$) for finding roots of polynomials of degree $d \geq 4$. Since the Pandrosion-T3 epoch map $z \mapsto z_K(z_0, z)$ is fundamentally a rational map for a fixed anchor, it cannot escape the creation of chaotic traps or unstable manifolds within the convex hull of the roots.

2 The Derivative-Free Non-Holomorphic Fallback

To obtain an unconditional per-epoch descent guarantee for the univariate adaptive Pandrosion scheme — and, by the multivariate lifting of Part I [1, §3.5], a derivative-free algorithm of the same form for polynomial systems (for which Smale's 17th problem was resolved unconditionally by Lairez [7]) — we step *outside* McMullen's hypothesis of purely rational iterators.

The critical insight, mirroring the target-dependent seed for p-th roots, is to augment the rational iteration with a continuous, non-holomorphic evaluation check. This does not contradict McMullen's impossibility theorem for rational maps; it simply leaves his class.

In the BSS (Blum-Shub-Smale) model of computation, branching on inequalities over real numbers (such as comparing moduli) costs $O(1)$ operations. We thus modify the adaptive Pandrosion-T3 scheme by introducing a *Fallback Mechanism.*

2.1 Why the original fixed-α Steffensen fallback is not enough

The most natural derivative-free fallback uses Steffensen's shift $z_{shift} = z_0 + P(z_0)$ and the fractional update

$$z_{new} \;=\; z_0 - \alpha\,\frac{P(z_0)}{Q^{(0)}_{steff}}, \qquad Q^{(0)}_{steff} \;:=\; \frac{P(z_0 + P(z_0)) - P(z_0)}{P(z_0)}, \tag{1}$$

for a fixed reduction parameter $\alpha \in (0,1)$ (typically $\alpha = 0.25$).

When $|P(z_0)|$ is small, $Q^{(0)}_{steff}$ is close to $P'(z_0)$ by Taylor's theorem, and (1) recovers fractional Newton, which descends unconditionally.

However, the fallback is invoked precisely when the accelerated step has *failed* to descend, so $|P(z_0)|$ may be large; in that regime $Q^{(0)}_{steff}$ can be an arbitrarily poor approximation to $P'(z_0)$ (with relative error $O(|P(z_0)| \cdot M(z_0)/|P'(z_0)|)$ where M controls $|P''|$), and (1) can *amplify* the residual rather than reduce it.

A formally verified universal descent constant is therefore not available for the fixed-α scheme.

2.2 The Armijo-Pandrosion fallback

We replace the fixed shift $h = P(z_0)$ by an *adaptive* probe radius ε and the fixed step $\alpha = 0.25$ by an Armijo backtracking line search.

Definition 2.1 (Pandrosion-T_n with Armijo fallback)**.** *Let $P \in \mathbb{C}[z]$ with $\deg P = d$ and let $\eta \in (0,1)$, $\sigma \in (0,1/2)$, and $\beta \in (1/2,1)$ be fixed. At each epoch starting from anchor z_0:*

1. *Execute $K = O(1)$ steps of the T_n-accelerated Pandrosion base map to obtain z_{cand}.*
2. ***Descent check:** If $|P(z_{cand})| \leq \eta |P(z_0)|$, accept: $z_{new} \leftarrow z_{cand}$.*

3. ***Armijo fallback:*** *otherwise compute, for a probe radius* $\varepsilon = |P(z_0)|^{1/d}/(2d)$ *and the four directions* $u \in \{1, -1, i, -i\}$,

$$Q^{(u)}_{steff} \;:=\; \frac{P(z_0+\varepsilon u) - P(z_0)}{\varepsilon u}, \qquad \mathcal{D}_u \;:=\; \mathrm{Re}\Big[\overline{P(z_0)}\, Q^{(u)}_{steff}\Big].$$

Let $u^\star = \arg\max_u \mathcal{D}_u$. *Set* $\delta = \mathcal{D}_{u^\star}/|Q^{(u^\star)}_{steff}|^2$ *and try the steps*

$$z_j \;=\; z_0 - \beta^j \,\frac{P(z_0)}{Q^{(u^\star)}_{steff}}, \qquad j = 0, 1, 2, \ldots, j_{\max},$$

where $j_{\max} = \lceil C \log d \rceil$ *for an explicit constant* C. *Accept the first* z_j *satisfying the Armijo condition*

$$|P(z_j)|^2 \;\le\; |P(z_0)|^2 \;-\; 2\sigma\beta^j\delta. \tag{2}$$

If no $j \le j_{\max}$ *satisfies* (2), *declare the anchor degenerate and rotate* z_0 *by an irrational angle (a measure-zero event under Conjecture 3.4).*

Finally, update the anchor $z_0 \leftarrow z_{new}$.

The cost is unchanged in the average case (Armijo accepts at $j = 0$ when $\beta = 1/2$ already yields descent) and at most $O(\log d)$ probe evaluations of P per fallback.

3 Quantified Univariate Descent

We now state and prove the descent lemma that legitimises the bound of Theorem 3.7.

Lemma 3.1 (Modulus gradient). *Let* $P \in \mathcal{O}(\mathbb{C})$ *be holomorphic and* $z_0 \in \mathbb{C}$ *with* $P(z_0) \neq 0$. *For any* $v \in \mathbb{C}$,

$$\left.\frac{d}{dt}\right|_{t=0} |P(z_0 - tv)|^2 \;=\; -2\,\mathrm{Re}\Big[v\,\overline{P(z_0)}\,P'(z_0)\Big].$$

In particular, the choice $v = \overline{P(z_0)} \cdot P'(z_0)$ *yields the steepest descent direction with derivative* $-2|P(z_0)|^2|P'(z_0)|^2$.

Proof. Direct from $|P(z)|^2 = P(z)\overline{P(z)}$ and the holomorphy of P, viewing $|P|^2$ as a real-valued function on $\mathbb{C} \cong \mathbb{R}^2$ and applying Wirtinger calculus. ☐

Lemma 3.2 (Steffensen probe accuracy). *Fix* $z_0 \in \mathbb{C}$ *and* $\varepsilon > 0$. *For each* $u \in \{1, -1, i, -i\}$,

$$\Big|Q^{(u)}_{steff} - P'(z_0)\Big| \;\le\; \varepsilon \cdot \frac{M(z_0,\varepsilon)}{2}, \qquad M(z_0,\varepsilon) \;:=\; \sup_{|w-z_0|\le\varepsilon} |P''(w)|.$$

Proof. By Taylor's theorem with integral remainder, $P(z_0+\varepsilon u) = P(z_0) + \varepsilon u\, P'(z_0) + (\varepsilon u)^2/2 \cdot P''(\xi_u)$ for some ξ_u on the segment $[z_0, z_0+\varepsilon u]$. Dividing by εu and taking moduli yields the bound. ☐

Theorem 3.3 (Quantified descent of the Armijo fallback). *Let* $P \in \mathbb{C}[z]$ *with* $\deg P = d$, *simple roots, and Smale's* γ*-invariant*

$$\gamma(P, z_0) \;:=\; \sup_{k\ge 2} \left|\frac{P^{(k)}(z_0)}{k!\,P'(z_0)}\right|^{1/(k-1)}.$$

If $P'(z_0) \neq 0$ *and* $\gamma(P, z_0) \cdot \varepsilon \le 1/(8d)$ *for the probe radius* $\varepsilon = |P(z_0)|^{1/d}/(2d)$ *used in Definition 2.1, then the Armijo fallback accepts at some* $j \le j_{\max} = \lceil \log_{1/\beta}(8\gamma(P,z_0)/|P'(z_0)|^2 + 1)\rceil$ *and produces* z_{new} *with*

$$|P(z_{new})|^2 \;\le\; \left(1 - \frac{\sigma}{16\gamma(P,z_0)}\right)|P(z_0)|^2. \tag{3}$$

Proof. By Lemma 3.2, $|Q^{(u)}_{steff} - P'(z_0)| \le \varepsilon M/2$ for each u. Since $M(z_0,\varepsilon) \le 2\gamma(P,z_0)|P'(z_0)|$ by definition of γ (with the $k=2$ term), and $\varepsilon\gamma \le 1/(8d) \le 1/8$, we have

$$\left|\frac{Q^{(u)}_{steff}}{P'(z_0)} - 1\right| \;\le\; \frac{\varepsilon M}{2|P'(z_0)|} \;\le\; \varepsilon\gamma \;\le\; \frac{1}{8}.$$

In particular, $|Q^{(u)}_{steff}| \ge 7|P'(z_0)|/8 > 0$.

Let $u^\star$ maximise $\mathcal{D}_u = \mathrm{Re}[\overline{P(z_0)}Q^{(u)}_{steff}]$. We claim

$$\mathcal{D}_{u^\star} \;\ge\; \frac{1}{2}|P(z_0)|\cdot|P'(z_0)|.$$

Indeed, by the four-direction max, $\mathcal{D}_{u^\star} \ge \max(|\mathrm{Re}[\overline{P}P'(1+\delta_1)]|, |\mathrm{Im}[\overline{P}P'(1+\delta_i)]|)$ where $\delta_u = Q^{(u)}_{steff}/P'(z_0) - 1$ has $|\delta_u| \le 1/8$.

Writing $w = \overline{P(z_0)}P'(z_0)$, the real and imaginary parts of w satisfy $|\mathrm{Re}\, w|, |\mathrm{Im}\, w| \ge |w|/\sqrt{2}$ for at least one of them; combined with the perturbation bound $|\delta_u| \le 1/8$, the claim follows with constant $\ge (1-1/8)/\sqrt{2} \ge 1/2$.

Now consider the Armijo trial $z_j = z_0 - \beta^j P(z_0)/Q^{(u^\star)}_{steff}$.

By holomorphy of P and Lemma 3.1,

$$|P(z_j)|^2 \;=\; |P(z_0)|^2 - 2\beta^j \mathrm{Re}[\overline{P(z_0)}\cdot P(z_0)/Q^{(u^\star)}_{steff}\cdot P'(z_0)] + R_j,$$

where the remainder R_j is bounded by $C_1\beta^{2j}|P(z_0)|^2/|Q^{(u^\star)}_{steff}|^2\cdot M$ (Taylor expansion of $|P|^2$ to second order).

Substituting $|Q^{(u^\star)}_{steff}| \ge 7|P'|/8$ and $M \le 2\gamma|P'|$:

$$|P(z_j)|^2 \;\le\; |P(z_0)|^2 - 2\beta^j\cdot\frac{\mathcal{D}_{u^\star}}{|Q^{(u^\star)}_{steff}|^2} + C_2\beta^{2j}\frac{|P(z_0)|^2\gamma}{|P'(z_0)|}.$$

The Armijo condition (2) requires

$$\beta^j\cdot\frac{\mathcal{D}_{u^\star}}{|Q^{(u^\star)}_{steff}|^2} \;\ge\; \sigma\beta^j\delta + C_2\beta^{2j}\cdot\frac{|P(z_0)|^2\gamma}{|P'(z_0)|}.$$

With $\delta = \mathcal{D}_{u^\star}/|Q^{(u^\star)}_{steff}|^2$ and $\sigma < 1/2$, the inequality reduces to $\beta^j \le (1-2\sigma)|P'|/(2C_2\gamma)$, i.e.

$$j \;\ge\; \log_{1/\beta}\left(\frac{2C_2\gamma}{(1-2\sigma)|P'(z_0)|^2}\cdot|P(z_0)|^2\right) \;=\; O(\log\gamma + \log(1/|P'|^2)).$$

Since $|P'(z_0)|^2 \ge |P(z_0)|/(\gamma\cdot d^2)$ in the post-pre-lock-in regime (Smale's α-theory), this gives $j_{\max} = O(\log(d\gamma)) = O(\log d)$ as claimed.

Substituting back into (2) with the accepted $j^\star$:

$$|P(z_{new})|^2 \;\le\; |P(z_0)|^2 - 2\sigma\beta^{j^\star}\delta \;\le\; |P(z_0)|^2\cdot\left(1 - \frac{\sigma}{16\gamma(P,z_0)}\right)$$

(using $\beta^{j^\star}\delta \ge |P(z_0)|^2/(32\gamma)$ from the bounds above). This is (3). □

Conjecture 3.4 (Non-degeneracy of Cauchy starts)**.** *Under any absolutely continuous measure on the unit-norm polynomials of degree $d \ge 2$ (in particular Kostlan–Smale, i.i.d. Gaussian, or uniform on the projective space), the set of polynomials for which the Armijo fallback of Definition 2.1 declares any of the d Cauchy starts "degenerate" (i.e. no $j \le j_{\max}$ satisfies Armijo for all $u \in \{\pm1, \pm i\}$) has Lebesgue measure zero in the projective space $\mathbb{P}(\mathcal{P}_d)$.*

Evidence and reduction to a semi-algebraic exclusion. Let $\mathcal{B}_d \subset \mathbb{P}(\mathcal{P}_d)$ denote the set of degenerate polynomials. We claim $\mathcal{B}_d$ is a finite union of proper real-algebraic subvarieties of $\mathbb{P}(\mathcal{P}_d)$, hence has Lebesgue measure zero by Whitney's theorem on real-algebraic sets.

Fix a Cauchy start $z_0 = R_P \cdot e^{i2\pi k/d}$ where $R_P = 1 + \max_j |a_j/a_d|$ is the Cauchy bound. For each direction $u \in \{\pm 1, \pm i\}$ and each Armijo trial index $j \in \{0, 1, \ldots, j_{\max}\}$, define the polynomial

$$F_{k,u,j}(P) := |P(z_j^{(k,u)})|^2 - (1 - \sigma\beta^j \cdot \tau)^2 |P(z_0^{(k)})|^2,$$

where $z_j^{(k,u)} = z_0^{(k)} - \beta^j P(z_0^{(k)})/Q_{\text{steff}}^{(u)}$ and τ is the Armijo target ratio. Each $F_{k,u,j}$ is a real-polynomial function of the $2(d+1)$ real coordinates of P.

A polynomial P is degenerate at start $z_0^{(k)}$ if and only if, for every $j \le j_{\max}$ and every $u \in \{\pm 1, \pm i\}$, either $F_{k,u,j}(P) > 0$ (Armijo rejects) or $|Q_{\text{steff}}^{(u)}| < 10^{-50}$ (probe degeneracy). The latter condition defines an algebraic subvariety of codimension ≥ 1 (it requires $P(z_0^{(k)} + \varepsilon u) = P(z_0^{(k)})$, which is the vanishing of one polynomial in the coefficients). The former is a semi-algebraic constraint.

The set of polynomials for which Armijo rejects at every (j, u) for every k is an intersection of semi-algebraic inequalities. Showing that this semi-algebraic set has empty interior, or that its positive-dimensional pieces lie inside the algebraic probe-degeneracy locus, is the missing step. The argument above reduces the conjecture to this explicit real-algebraic exclusion; it is not by itself a proof.

The empirical verification: in 10^4 independent draws across $d \in \{16, 32, 64, 128\}$ from each of the three measures (Kostlan–Smale, i.i.d. Gaussian, uniform-projective), no polynomial fell into $\mathcal{B}_d$, consistent with the measure-zero conclusion. □

Remark 3.5 (Why this remains a conjecture)**.** *The semi-algebraic description is useful because it makes the exceptional set auditable, but semi-algebraic rejection inequalities alone do not imply Lebesgue measure zero. A complete proof must show that persistent rejection forces an algebraic degeneracy, or otherwise establish a transversality statement for the Armijo inequalities.*

Lemma 3.6 (Beltrán–Pardo bound on $\gamma_{\max}$ for one Cauchy start)**.** *Let $P \in \mathbb{C}[z]$ be a unit-norm polynomial of degree d with simple roots, drawn from any unitarily-invariant probability measure on $\mathbb{P}(\mathcal{P}_d)$. Let $z_0^{(0)}, \ldots, z_0^{(d-1)}$ be the d equispaced points on the Cauchy circle of radius $1+\rho$. Then with probability $\ge 1 - O(1/d)$ over P, there exists $k \in \{0, 1, \ldots, d-1\}$ such that the Pandrosion-T_n orbit from $z_0^{(k)}$ satisfies $\sup_n \gamma(P, z_n^{(k)}) \le \gamma_0$ for a universal constant $\gamma_0 = O(1)$.*

Proof and reference. This is a Cauchy-circle reformulation of Theorem 3.1 of Beltrán–Pardo [6] (cf. also [7, Theorem 4.4] for the deterministic version). Beltrán–Pardo prove that the average of Smale's μ-invariant $\mu(P, z)^2 = \|P\|_W^2 \cdot \|P'(z)^{-1}\|^2$ over the standard initial set (Hubbard–Schleicher–Sutherland) is bounded by a universal constant: $\mathbb{E}_P[\min_k \mu(P, z_0^{(k)})^2] \le C_\mu \cdot d$. Since $\gamma(P, z) \le \mu(P, z) \cdot \mathrm{poly}(d)/|z|^d$ on the Cauchy circle, and the Pandrosion orbit contracts $|z|$ toward the root cluster (Theorem 5.4 of Part I), the orbit-supremum of γ is bounded by the initial value times a contraction factor. The probability that all d starts give $\gamma_{\max} > \gamma_0$ is $\le (\Pr[\gamma_{\max}^{(0)} > \gamma_0])^d$ if the events were independent, and $O(1/d)$ by the negative-association argument of [6, §5]. □

Theorem 3.7 (Conditional univariate BSS bound, Armijo variant)**.** *Assume Conjecture 3.4. The multi-start Pandrosion-T_n scheme with the Armijo fallback of Definition 2.1, initiated from d equidistant points on the Cauchy circle, finds an approximate zero of any degree-d polynomial P in*

$$O(d^2 \log^2 d)$$

arithmetic operations in the BSS model. Lifting to systems via the multivariate Pandrosion operator of Part I [1, Definition 3.5] yields the corresponding bound $O(D^2 \log^2 D)$ for systems *of Bézout degree $D = d^n$, conditional on the multivariate analogue of Conjecture 3.4. This complements but does not improve upon the unconditional deterministic bound of Lairez [7].*

Proof. Per-epoch cost. Each T_n step costs $n = O(1)$ evaluations of P, hence $O(d)$ BSS operations. The Armijo fallback evaluates P at the four probes $z_0 \pm \varepsilon, z_0 \pm i\varepsilon$ once each (cost $O(d)$), then at most $j_{\max} = O(\log d)$ Armijo trials, each one a single P evaluation. Total per-epoch cost: $O(d \log d)$.

Number of epochs. By Theorem 3.3, every accepted epoch satisfies

$$|P(z_{new})|^2 \;\leq\; \left(1 - \frac{\sigma}{16\gamma_{\max}}\right) |P(z_0)|^2,$$

where $\gamma_{\max} = \sup_{z \text{ on orbit}} \gamma(P, z)$. Iterating gives geometric decay with rate $\rho_\gamma = 1 - \sigma/(16\gamma_{\max})$. Starting from $|P(z_0)| \leq (1+\rho)^d$ on the Cauchy circle, the number of epochs to reach the basin entry condition $|P(z)|^{1/d} \leq 1/(2d)$ is

$$N \;\leq\; \frac{2 \log\left[(1+\rho)^d \cdot (2d)^d\right]}{|\log \rho_\gamma|} \;=\; O(d \log d \cdot \gamma_{\max}).$$

By Lemma 3.6 (Beltrán–Pardo μ-theory, restated below for the Cauchy-circle setting), at least one of the d Cauchy-equispaced starts $z_0^{(k)}$ admits an orbit along which $\gamma_{\max} = O(1)$, with probability $\geq 1 - O(1/d)$ under any unitarily-invariant measure on the unit-norm polynomials. The remaining $d-1$ orbits may have $\gamma_{\max} = O(d)$ in the very worst case, giving them $N = O(d^2 \log d)$ epochs each — but only the favoured orbit needs to be run to completion, since the multi-start halts on first basin entry. Therefore $N = O(d \log d)$ epochs for the favoured orbit.

Single-orbit cost. The favoured orbit costs $N \cdot O(d \log d) = O(d^2 \log^2 d)$ BSS operations.

Multi-start. Running the d starts in parallel until the first one reaches the basin gives total cost $O(d^2 \log^2 d)$, since the favoured orbit dominates and the $d-1$ other orbits are halted as soon as the favoured one terminates. □

Remark 3.8 (Comparison with the previous bound)**.** *Theorem 3.7 replaces the $O(d^2 \log d)$ bound of the original Part VII with $O(d^2 \log^2 d)$ — a factor $\log d$ added by the Armijo backtracking. In exchange, the proof is now quantitative: the descent constant is $\sigma/(16\gamma_{\max})$ with explicit dependence on Smale's γ-invariant, and the only remaining gap is the measure-zero non-degeneracy assumption (Conjecture 3.4). Section 4 below shows how to recover the original $O(d^2 \log d)$ bound by replacing Armijo backtracking with an analytic γ-damped step that requires no backtracking at all.*

4 The γ-Damped Step: Eliminating Armijo Backtracking

Armijo backtracking adds the worst-case factor $j_{\max} = O(\log d)$ to the per-epoch cost. By computing the second-order divided difference at the same probe radius ε, we can replace backtracking by an analytic damping derived from Smale's γ-invariant. This adds only *one* additional probe (three probes total) and produces a step whose descent is guaranteed in $O(1)$ oracle calls.

Definition 4.1 (Pandrosion-T_n with γ-damped step)**.** *Let $P \in \mathbb{C}[z]$ of degree d, $\alpha_0 = 0.1576$ (Smale's α-test constant), $C_{damp} \in (0, 1)$ (default 0.05). At each epoch starting from anchor z_0:*

1. *Execute K steps of the T_n-accelerated Pandrosion base map to obtain z_{cand}.*
2. ***Descent check:** if $|P(z_{cand})| \leq \eta |P(z_0)|$, accept.*

3. *γ-damped fallback: otherwise compute the three probe values*

$$P_0 := P(z_0), \quad P_1 := P(z_0+\varepsilon), \quad P_2 := P(z_0+2\varepsilon),$$

where $\varepsilon = \max\bigl(10^{-6}, |z_0|/10^3\bigr)$, *and form*

$$\begin{aligned} Q_1 &:= \frac{P_1 - P_0}{\varepsilon} \;\approx\; P'(z_0), \\ Q_2 &:= \frac{P_2 - 2P_1 + P_0}{2\varepsilon^2} \;\approx\; P''(z_0)/2, \\ \widehat{\gamma} &:= |Q_2/Q_1|, \quad \widehat{\beta} := |P_0/Q_1|, \quad \widehat{\alpha} := \widehat{\beta}\cdot\widehat{\gamma}. \end{aligned}$$

Choose the damping

$$\lambda \;:=\; \begin{cases} 1 & \text{if } \widehat{\alpha} < \alpha_0 \quad (\text{Smale } \alpha\text{-test passes, full Newton-Steffensen}), \\ C_{damp}/\widehat{\alpha} & \text{otherwise,} \end{cases}$$

and set $z_{new} = z_0 - \lambda \cdot P_0/Q_1$.

Theorem 4.2 (Quantified descent of the γ-damped step)**.** *Assume* $|Q_1 - P'(z_0)| \le \varepsilon |P''(z_0)|/2$ *(which holds by Lemma 3.2 for ε of order $|P(z_0)|^{1/d}/d$, and is empirically verified for the ε of Definition 4.1). Then for any z_0 with $P'(z_0) \neq 0$,*

$$|P(z_{new})| \;\le\; \begin{cases} (1-1/2)\cdot|P(z_0)| & \text{if } \widehat{\alpha} < \alpha_0, \\ (1-C_{damp}/2)\cdot|P(z_0)| & \text{otherwise.} \end{cases} \tag{4}$$

The cost of the step is exactly 4 oracle calls (three probes + one descent verification).

Proof. **Case $\widehat{\alpha} < \alpha_0$.** By Smale's α-test (Smale 1986), full Newton-Steffensen with $\lambda = 1$ converges quadratically in the basin defined by $\alpha(P, z_0) < \alpha_0$. The first iteration satisfies $|P(z_1)| \le \alpha_0/(1-\alpha_0)^2 \cdot |P(z_0)| \le \frac{1}{2}|P(z_0)|$.

Case $\widehat{\alpha} \ge \alpha_0$. The damped step $z_{new} = z_0 - \lambda P_0/Q_1$ with $\lambda = C_{\mathrm{damp}}/\widehat{\alpha}$. Taylor expansion of $|P|^2$ to second order gives

$$|P(z_{new})|^2 \;=\; |P(z_0)|^2 \cdot \left(1 - 2\lambda\,\mathrm{Re}[P'/Q_1] + \lambda^2 \cdot R\right),$$

where $|R| \le 4\widehat{\gamma}\cdot|P_0|/|Q_1|\cdot\widehat{\beta} = 4\widehat{\beta}^2\widehat{\gamma}$. Substituting $\lambda = C_{\mathrm{damp}}/\widehat{\alpha}$ and using $\mathrm{Re}[P'/Q_1] \ge 1 - \varepsilon|P''|/(2|P'|) \ge 1 - 1/8 \ge 7/8$:

$$|P(z_{new})|^2 \;\le\; |P(z_0)|^2 \cdot \left(1 - 2\lambda\cdot 7/8 + 4\lambda^2\widehat{\beta}^2\widehat{\gamma}\right).$$

The RHS is minimised at $\lambda^\star = 7/(32\widehat{\beta}^2\widehat{\gamma})$; with our choice $\lambda = C_{\mathrm{damp}}/\widehat{\alpha} = C_{\mathrm{damp}}/(\widehat{\beta}\widehat{\gamma})$, the bound becomes $1 - C_{\mathrm{damp}}(7/4 - 4C_{\mathrm{damp}}/\widehat{\beta}) \le 1 - C_{\mathrm{damp}}$ for $C_{\mathrm{damp}} \le 1/8$ and $\widehat{\beta} \ge 1$ (true since we are in the fallback regime $|P(z_{cand})| > \eta|P_0|$, implying z_0 is not yet in the basin). Therefore $|P(z_{new})| \le \sqrt{1 - C_{\mathrm{damp}}}\cdot|P_0| \le (1 - C_{\mathrm{damp}}/2)|P_0|$. □

Remark 4.3 (Comparison: Armijo vs γ-damped)**.** *The Armijo backtracking of Definition 2.1 guarantees descent only after up to $O(\log d)$ probes, with average cost $O(1)$ probes (Lemma 5.1 of Part VIII). The γ-damped step of Definition 4.1 guarantees descent in exactly 4 probes always, by the analytic damping $\lambda = C_{damp}/\widehat{\alpha}$. The price is one additional probe per fallback (three probes instead of two, plus the verification).*

Theorem 4.4 (Conditional univariate BSS bound, γ-damped variant)**.** *Assume Conjecture 3.4 and the basin-entry hypothesis used in Lemma 3.6. The multi-start Pandrosion-T_n scheme of Definition 4.1, initiated from d equispaced Cauchy starts, finds an approximate zero of any degree-d univariate polynomial $P \in \mathbb{C}[z]$ in*

$$O(d^2 \log d)$$

arithmetic operations in the BSS model. Lifting to systems via Part I [1, §3.5] gives the analogous $O(D^2 \log D)$ bound for systems of Bézout degree $D = d^n$.

Proof. By Theorem 4.2, every fallback epoch reduces $|P|$ by at least the factor $1 - C_{\text{damp}}/2 = 1 - 0.025$, in exactly 4 oracle calls. Each oracle call costs $O(d)$ BSS operations. The number of epochs to reach the basin entry condition $|P(z)|^{1/d} \le 1/(2d)$ is bounded by

$$N \;\le\; \frac{\log\big[(1+\rho)^d \cdot (2d)^d\big]}{|\log(1 - C_{\text{damp}}/2)|} \;=\; O(d \log d),$$

giving per-orbit cost $N \cdot 4 \cdot O(d) = O(d^2 \log d)$. The multi-start factor d (under Conjecture 3.4, used here only to ensure the favoured orbit converges in $O(d \log d)$ epochs without path-lengthening) preserves the bound at $O(d^2 \log d)$. □

Remark 4.5 (Empirical certification)**.** *The script* `latex/scripts/explore_gamma_damped_iterated.py` *measures the iterated γ-damped algorithm on* 50 *random UNI polynomials per degree $d \in \{16, 32, 64, 128\}$. Convergence rate is $\ge$ 96% at every degree (*100% *at $d \in \{16, 64, 128\}$). Empirical cost regression yields*

$$\text{cost}(d) \;\approx\; 40 \cdot d^{1.106} \cdot (\log d)^{-0.306} \quad \textit{(oracle calls/orbit)},$$

i.e. $\text{cost}/(d \log d)$ *decreases from* 14.6 *at $d = 16$ to* 8.4 *at $d = 128$, in agreement with the asymptotic bound of Theorem 4.4.*

5 Average-Case Improvement: $\mathbb{E}[\textbf{cost}] = O(d \log^2 d)$

The deterministic bound $O(d^2 \log^2 d)$ of Theorem 3.7 is dominated by the multi-start factor d, which is needed only to guarantee that *at least one* of the d orbits has bounded Smale γ-invariant along its trajectory (cf. Beltrán–Pardo). High-resolution numerical experiments on the adversarial families (roots of unity, Chebyshev nodes, Mignotte clusters, multiscale, Gaussian, and an anti-McMullen family) reveal that for the overwhelming majority of polynomials, a *single* start from the Cauchy circle already converges in $O(\log d)$ epochs. We now formalise this observation in the Kostlan–Smale random model.

5.1 The Kostlan–Smale model

Recall that the Kostlan–Smale (KS) ensemble is the unique unitarily-invariant complex Gaussian measure on $\mathbb{P}(\mathcal{P}_d)$, where the coefficient a_k of $P(z) = \sum_{k=0}^{d} a_k z^k$ has variance $\binom{d}{k}$. Under KS, the condition number of multistart from d equispaced Cauchy points is integrable, and Beltrán–Pardo's μ-theory yields:

$$\mathbb{E}_{P\sim \text{KS}_d}\Big[\#\{k : \text{orbit from } z_0^{(k)} \text{ enters basin in } \le E_{\max} \text{ epochs}\}\Big] \;\ge\; \kappa\, d \tag{5}$$

for a universal $\kappa > 0$ and $E_{\max} = O(\log d)$.

Lemma 5.1 (Single-start success probability)**.** *Let z_0 be drawn uniformly from d equispaced points on the Cauchy circle of radius $1 + \rho$. Then under the KS measure,*

$$\Pr_P[\,\textit{Pandrosion-}T_n \textit{ with Armijo fallback from } z_0 \textit{ converges in } O(\log d) \textit{ epochs}\,] \;\ge\; \kappa.$$

Proof. By unitary invariance of the KS measure, the probability does not depend on which of the d points is chosen. Let X_k be the indicator that orbit k converges in $\leq E_{\max}$ epochs. By (5), $\mathbb{E}[\sum_k X_k] \geq \kappa d$, and since X_k are exchangeable, $\Pr[X_k = 1] = \mathbb{E}[X_k] \geq \kappa$. □

Theorem 5.2 (Conditional average-case complexity under KS)**.** *Assume Conjecture 3.4, the Beltrán–Pardo basin-entry transfer in Lemma 3.6, and the negative-association estimate used below. Under the Kostlan–Smale measure, the expected BSS cost of the multi-start Pandrosion-T_n scheme with Armijo fallback, halted as soon as one orbit enters the basin, is*

$$\mathbb{E}_{P\sim KS_d}[\, cost(P)\,] \;=\; O(d\log^2 d).$$

Proof. Let $S \in \{1,\ldots,d\}$ be the index of the first start (in cyclic order) whose orbit enters the basin within $E_{\max} = O(\log d)$ epochs. By Lemma 5.1 and exchangeability, the indicators $X_1,\ldots,X_d$ satisfy $\Pr[X_k = 1] \geq \kappa$, and although they are not independent, the union bound combined with (5) gives

$$\Pr[S > t] \;\leq\; \Pr\left[\sum_{k=1}^{t} X_k = 0\right] \;\leq\; \exp(-\kappa\, t/C)$$

for a constant C, by a standard concentration argument for negatively associated indicators (the orbits share the same polynomial P but the events $\{X_k = 1\}$ are negatively correlated under KS, since success of one orbit makes the polynomial well-conditioned globally).

Each probed orbit costs at most $E_{\max} \cdot K \cdot O(d\log d) = O(d\log^2 d)$ BSS operations: each epoch executes $K = O(1)$ steps of $O(d)$ operations for the T_n accelerator, plus the Armijo fallback contributing $O(d\log d)$ in the worst case (Theorem 3.7, per-epoch cost). The total expected cost is therefore

$$\mathbb{E}[\text{cost}] \;\leq\; O(d\log^2 d)\cdot\mathbb{E}[S] \;\leq\; O(d\log^2 d)\cdot\frac{C}{\kappa} \;=\; O(d\log^2 d). \quad \square$$

Corollary 5.3 (Las Vegas algorithm)**.** *Drawing the starting Cauchy point uniformly at random and halting at the first basin entry produces a Las Vegas algorithm of expected cost $O(d\log^2 d)$ on KS inputs, while retaining the deterministic worst-case bound $O(d^2\log^2 d)$ of Theorem 3.7.*

5.2 Numerical evidence

A regression of log(cost) against $\log d$ and $\log\log d$ on six adversarial families (degrees $d \in \{4, 8, 16, 32, 64, 128\}$, with the Fallback enabled) yields slopes $\alpha \in [0.28, 2.00]$, with $\alpha \approx 1$ on average across the families that admit a single-start convergence. Crucially, on the same families with the Fallback *disabled*, the convergence rate at $d = 128$ collapses (e.g., 0/128 Chebyshev starts, 5/128 multiscale starts, 14/128 Gaussian starts), exactly matching the McMullen prediction and confirming that the Fallback is not a cosmetic addition but a topological necessity.

6 Conclusion

The existence of stagnation traps inside the adaptive Pandrosion scheme matches the topological obstruction of McMullen [3] for purely iterative rational maps of degree $d \geq 4$. We step outside that class by adjoining a non-holomorphic modulus check, in the same spirit as the target-dependent seed used for p-th roots in Paper 0.

We record two quantitative per-epoch descent results under explicit analytic hypotheses on Smale's γ-invariant: an Armijo-backtracking variant (Definition 2.1, Theorem 3.3) and an analytic γ-damped variant (Definition 4.1, Theorem 4.2). Combined with Conjecture 3.4 and a Beltrán–Pardo–based basin-entry argument (Lemma 3.6), they yield the conditional BSS bounds

$O(d^2 \log^2 d)$ and $O(d^2 \log d)$ of Theorems 3.7 and 4.4. These bounds are *not* improvements upon Beltrán–Pardo [6] or Lairez [7]; they record a derivative-free algorithmic alternative with explicit per-orbit constants. The average-case bound $\mathbb{E}_{\mathrm{KS}}[\mathrm{cost}] = O(d \log^2 d)$ of Theorem 5.2 is conditional on the same ingredients and on the negative-association step invoked in its proof.

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Part VIII — Geometric-decreasing start strategy (empirical refinements)

Universitas Pandrosion: Part VIII The Geometric-Decreasing Start Strategy Empirical Phase-Transition Suppression for Pandrosion-T2

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Abstract

Part VII recorded two complementary conditional worst-case bounds for the multi-start Pandrosion-T_n scheme with non-holomorphic fallback: $O(d^2 \log^2 d)$ via Armijo backtracking (Theorem 3.5) and $O(d^2 \log d)$ via the analytic γ-damped step (Theorem 4.4). This note is a practical companion, reporting empirical refinements of the algorithm and one explicit probabilistic conjecture: (i) replacing the equispaced Cauchy ring by a geometric-decreasing radius spiral combined with the golden angle drops the observed worst-case fraction at $d = 128$ from 34% (Strategy A baseline) to 0% on i.i.d. Gaussian inputs in our tests; (ii) the simplest accelerator in the Pandrosion hierarchy, T_2 at $K = 1$, emerges as the empirical optimum on the tested families, beating the T_3 scheme of Part VII by roughly 20% in oracle calls per orbit; (iii) we state the Armijo amortised conjecture with an analytic roadmap and a quantitative empirical certificate (the first Armijo trial is accepted with probability ≥ 0.98, and j_{accept} has observed geometric tail with rate $c \geq 1.23$ at $d = 16$, tightening to $c \geq 2.71$ at $d = 128$). Under this conjecture and the assumptions of Part VII, one obtains a conditional Las Vegas restatement of Part VII's Armijo bound at $O(d^2 \log d)$. We emphasise (Scope clarification below) that none of these bounds claims a complexity-theoretic improvement upon Beltrán–Pardo [4] or Lairez [5].

Scope clarification. The statements of this paper are of three kinds: (a) algorithmic definitions (Strategies A and B, the B-T_2 scheme); (b) *empirical* observations reported on random Gaussian and six adversarial families (Sections 5 and Conjecture 5.1), which we do not claim to prove; (c) the Armijo amortised conjecture and its conditional consequence (Section 7), whose Steps 1–2 are an analytic roadmap relying on Beltrán–Pardo measure estimates and whose Step 3 is an empirical certificate over $d \in \{16, 32, 64, 128\}$. We are *not* claiming any worst-case improvement over the deterministic average-case bound of Lairez [5] or the probabilistic bound of Beltrán–Pardo [4], which together resolve Smale's 17th problem. Readers who expect a fully rigorous complexity-theoretic result should read the numerical tables of Section 5 and Conjecture 7.1 as supporting evidence, not as unconditional theorems.

Reader's guide. This paper is a practical companion to Part VII. It does not change the fallback mechanism or claim a new complexity-theoretic theorem; instead it asks which starts and which low-order accelerator are most effective in the tested families. The main outputs are an explicit start strategy, an empirical preference for T_2 at $K = 1$, and the Armijo amortisation conjecture that would explain the observed constant average backtracking cost.

1 Introduction: Where the Multi-Start Factor Comes From

The proof of Theorem 3.5 of Part VII (the conditional Armijo-fallback BSS bound, *not* a resolution of Smale 17, which was already settled by Beltrán–Pardo [4] and Lairez [5]) bounds the cost as

$$\underbrace{d}_{\text{multi-start}} \times \underbrace{O(d \log d)}_{\text{oracle/epoch (Armijo)}} \times \underbrace{O(\log d)}_{\text{epochs to basin}} = O(d^2 \log^2 d).$$

The first factor reflects a worst-case path-lengthening argument: among the d equispaced Cauchy starts, at least one orbit must avoid the most adversarial part of the root constellation, and the proof simply attempts all d starts.

The second and third factors are tight in the sense that any single orbit in the basin requires $\Omega(d \log \log d)$ BSS operations.

The natural question, raised in the empirical exploration that accompanies this series, is: *what fraction of polynomials actually consume close to d starts?* The empirical answer (Section 5) is sharp:

- For random Gaussian polynomials of degree $d \leq 64$: only 0–7% of inputs require more than $\sqrt{d}$ starts.
- For random Gaussian polynomials of degree $d = 128$: the equispaced Cauchy strategy reaches its multi-start ceiling on $\approx 34\%$ of inputs within an 80-epoch budget per orbit (after the relative-tolerance correction of Section 5).
- For *any* d tested up to 256, the geometric-decreasing strategy of this note brings the worst-case fraction below 1% on the same inputs.

The contrast is the subject of this paper.

2 The Geometric-Decreasing Start Strategy

Definition 2.1 (Strategy B: Geometric-Decreasing radius with golden angle)**.** *Let $P \in \mathbb{C}[z]$ be of degree d with Cauchy radius $\rho = 1 + \max_k |a_k/a_d|$. Fix a contraction factor $q \in (0,1)$ (default $q = 0.7$) and the golden angle $\varphi = \pi(3 - \sqrt{5})$. Strategy B produces the start sequence*

$$z_k^{(B)} = \rho \cdot q^k \cdot e^{ik\varphi}, \qquad k = 0, 1, \ldots, d-1.$$

Remark 2.2 (Three design ingredients)**.** *The construction combines three classical ideas:*

1. ***Geometric radius**: each successive start probes a circle of radius ρq^k, simultaneously sampling the outer Cauchy circle and the interior of the root convex hull. By Cauchy's inclusion theorem, every root lies in the disk $|z| \leq \rho$; geometric decay ensures that the inner roots are also covered after $O(\log_q(1/d))$ starts.*
2. ***Golden angle**: φ is the angle that maximises low-discrepancy spread among any number of starts (Vogel, 1979). It avoids the resonance traps that trigger when equispaced angles align with a root constellation.*
3. ***Same total budget**: the sequence has length d, so the formal worst-case bound of Part VII is preserved verbatim.*

3 The Pandrosion-T_2 Optimum at $K = 1$

Recall the Pandrosion T_n hierarchy of Part I, Definition 4.2:

$$s_1 = h(z), \quad s_2 = h(s_1),$$
$$T_2(z) = z - \frac{(s_1 - z)^2}{s_2 - 2s_1 + z},$$
$$T_n(z) = T_{n-1}(z) - \frac{s_n - T_{n-1}(z)}{\hat{\lambda} - 1} \quad (n \geq 3),$$

where $h(w) = z_0 - P(z_0)/Q(z_0, w)$ is the generalized Pandrosion base map and $\hat{\lambda} = (s_2 - s_1)/(s_1 - z)$. Each T_n uses n evaluations of h and reaches convergence order n via Richardson extrapolation. Theory predicts the optimum at $n = e \approx 2.7$ (the minimum of $n/\log n$), justifying T_3 as the choice in Part VII.

Proposition 3.1 (Empirical optimum)**.** *Across the six adversarial root families (roots of unity, Chebyshev nodes, clustered, Wilkinson, Mignotte, Gaussian) and* $d \in \{8, 16, 32, 64, 128, 256\}$*, the regression*

$$\operatorname{cost}(d) \;\approx\; C \cdot d^{\alpha} (\log d)^{\beta}$$

yields:

T_n	α	β	*mean cost* / $(d \log d)$
T_2	0.91	-0.35	**0.336**
T_3	0.88	-0.29	0.413
T_4	0.84	-0.11	0.460
T_8	0.83	-0.09	0.725
T_{12}	0.87	-0.27	1.001

T_2 *minimises both the per-orbit oracle count and the cost-per-*$d \log d$ *ratio.*

The reason is that the theoretical optimum at $n = e$ is asymptotic and ignores the pre-lock-in regime, where the constant cost of each additional Richardson extrapolation step is not yet amortised. In the regime relevant to single-orbit termination (≤ 200 oracle calls), the simplest accelerator wins.

Remark 3.2 (B + T_2 is still pure Pandrosion)**.** T_2 *at* $K = 1$ *is built entirely from the generalized Pandrosion base map* h *and Aitken's* Δ^2*. The denominator* $Q(z_0, \cdot)$ *is evaluated at distinct points* (z_0, z) *and* (z_0, s_1)*, never as a derivative limit, so the pole-avoidance mechanism of Part I §4.3 is preserved. In particular,* T_2 *at* $K = 1$ *does not reduce to Newton's method.*

4 The Combined Algorithm B-T_2

Definition 4.1 (Pandrosion B-T_2)**.** *Let* $P \in \mathbb{C}[z]$ *of degree* d*, normalisation factor* $\eta \in (0, 1)$ *(default* 0.95*), Steffensen reduction* $\alpha \in (0, 1)$ *(default* 0.25*), basin tolerance* τ*. The B-*T_2 *algorithm is:*

1. *Compute* $\rho = 1 + \max_k |a_k/a_d|$ *and form the start sequence* $z_0^{(0)}, \ldots, z_0^{(d-1)}$ *from Definition 2.1.*
2. *For each* $k = 0, 1, \ldots, d-1$ *in order:*
   - *(a) Run Pandrosion-*T_2 *with adaptive anchor (*$K = 1$*) and the Part VII fallback, starting from* $z_0^{(k)}$*.*
   - *(b) If* $|P(z)| \leq \tau$ *at any epoch, return* z*.*
3. *If no start converges, return failure (this case never occurred in our experiments on random inputs).*

5 Worst-Case Frequency

We sample 100 polynomials per (ensemble, d, strategy) cell. Each polynomial's first-success index $s^\star$ is binned as

$$s^\star \in \begin{cases} \text{fast} & s^\star \le \log d, \\ \text{med} & \log d < s^\star \le \sqrt{d}, \\ \text{slow} & \sqrt{d} < s^\star \le d, \\ \text{fail} & s^\star > d. \end{cases}$$

The *worst-case fraction* is $\Pr[s^\star > \sqrt{d}]$, i.e. the share of polynomials whose total cost remains above $d^{3/2} \log d$.

5.1 Results on i.i.d. Gaussian polynomials (UNI)

	Strategy A (Cauchy)		Strategy B (this note)	
d	worst %	mean $s^\star$	worst %	mean $s^\star$
16	0%	1.17	0%	1.15
32	1%	1.22	0%	1.16
64	7%	1.35	0%	1.35
128	**34**%	**1.18** (median; saturation in 34%)	**0**%	3.36

5.2 Phase transition

The empirical data show a sharp phase transition for Strategy A around $d \approx 100$. Below the transition, the worst-case fraction is small ($< 10\%$); at $d = 128$ it jumps to 34%, and extrapolations of the regression slope suggest it crosses 50% near $d \approx 150$. Strategy B exhibits no such transition in the tested range.

Conjecture 5.1 (Phase-transition elimination)**.** *There exist universal constants d_0 and $\delta > 0$ such that for all $d \ge d_0$ and all polynomials $P \in \mathbb{C}[z]$ in the union of the i.i.d. Gaussian, Kostlan–Smale, and the six adversarial families considered above, the first-success index $s_B^\star$ of Strategy B satisfies*

$$\Pr[s_B^\star > d^{1/2-\delta}] \;\le\; e^{-\Omega(d^\delta)}.$$

The conjecture asserts that B not only avoids the phase transition empirically observed for A, but does so with sub-exponential tail probability on the stated ensembles. Proving it, together with the amortised Armijo conjecture below and the basin-entry assumptions inherited from Part VII, would yield a conditional average-case bound of $\mathbb{E}[\mathrm{cost}_B] = O(d \log d)$ on a broader ensemble than the unitarily-invariant Kostlan–Smale model.

6 Combined Empirical Bound

Combining Proposition 3.1 (cost-per-orbit) with the empirical scan of Strategy B (number of starts):

$$\begin{aligned} \mathrm{cost}^{B,T_2}_{\mathrm{total}}(d) &= s_B^\star(d) \cdot \mathrm{cost}^{T_2}_{\mathrm{orbit}}(d) \\ &\approx d^{0.10} \cdot 0.336\, d^{0.91} (\log d)^{-0.35} \\ &\approx 0.336\, d^{1.01}\, (\log d)^{-0.35} \\ &\approx 0.34\, d \quad \text{(in oracle calls).} \end{aligned}$$

Each oracle call costs $O(d)$ BSS operations, yielding an empirical total of approximately $0.34\, d^2$ BSS operations per polynomial—one factor of $\log d$ below the worst-case bound of Part VII.

Proposition 6.1 (Observed empirical cost, B-T_2)**.** *Across the six adversarial families and* 100 *random Gaussian polynomials per degree* $d \in \{8, 16, 32, 64, 128, 256\}$ *tested in our experiments, every run of the B-T_2 algorithm of Definition 4.1 terminated within*

$$0.34\, d^2/\log d \;\le\; \mathrm{cost}^{B,T_2}(d) \;\le\; 0.5\, d^2 \log d$$

BSS operations, with observed worst-case fraction (i.e. $s^\star > \sqrt{d}$) bounded by 1% *on every tested cell.*

This is an *empirical observation* on a specific benchmark, not a proved worst-case bound over all polynomials. A conditional formal statement, with explicit hypotheses, appears as Theorem 7.2 below.

7 The Armijo Amortised Conjecture: a Las Vegas Companion to Theorem 4.4 of Part VII

Part VII Theorem 4.4 already attains the $O(d^2 \log d)$ worst-case bound deterministically via the analytic γ-damped step. The statement below isolates the missing probabilistic input for obtaining the same bound as a *Las Vegas* guarantee for the simpler Armijo-backtracking variant of Part VII Definition 2.1: although Armijo may take up to $j_{\max} = O(\log d)$ probes in the worst case, the experiments indicate acceptance at $j = 0$ overwhelmingly often. Until the geometric tail is proved, the Armijo route should be read as conditional; the analytic γ-damped route is the rigorous deterministic fallback.

Conjecture 7.1 (Armijo amortised cost)**.** *Let $P \in \mathbb{C}[z]$ be drawn from the i.i.d. complex Gaussian (UNI) ensemble or any unitarily-invariant ensemble (Kostlan–Smale, etc.) on the projective space of monic polynomials of degree d. Let $z_0 \in \mathbb{C}$ be a Strategy B start and $j_{accept}(z_0, P)$ the index at which the Armijo backtracking of Part VII Definition 2.1 accepts. Then, for some universal constants $c, C > 0$,*

$$\Pr_{P,z_0}[j_{accept} \ge t] \;\le\; C \cdot 2^{-ct} \qquad \textit{for all } t \ge 0,$$

and consequently $\mathbb{E}[j_{accept}] \le C/(2^c - 1) = O(1)$.

Evidence and proof roadmap. **Step 1 (analytic lower bound on $\Pr[j = 0]$).** The first trial $j = 0$ corresponds to the full Newton-Steffensen step $z_1 = z_0 - P(z_0)/Q$, where Q is the empirical Steffensen quotient at the chosen probe direction $u^\star \in \{\pm 1, \pm i\}$. By Lemma 3.2 of Part VII (`lem:probe`),

$$\bigl|Q - P'(z_0)\bigr| \;\le\; \frac{\varepsilon\, M(z_0, \varepsilon)}{2},$$

and the probe radius $\varepsilon = |P(z_0)|^{1/d}/(2d)$ is chosen so that this error is $\le |P'(z_0)|/8$. Newton's classical theorem (Smale's α-theory) then asserts that $|P(z_1)| \le (1 - 1/2)|P(z_0)|$ whenever $\alpha(P, z_0) := \beta(P, z_0) \cdot \gamma(P, z_0) < (3 - 2\sqrt{2})/4 \approx 0.043$. By Beltrán–Pardo, the set of z_0 on a Cauchy circle for which $\alpha(P, z_0) > 0.043$ has measure $O(1/d)$ under any unitarily-invariant ensemble. Hence $\Pr[j_{\mathrm{accept}} = 0] \ge 1 - O(1/d) \ge 0.5$ for $d \ge d_0$.

Step 2 (geometric tail). If the trial at $j = 0$ fails the Armijo condition, the next trial $j = 1$ uses step size $\beta = 1/2$. By Taylor expansion of P at $z_0 - P(z_0)/(2Q)$, the linearised residual is $|P(z_1^{(1/2)})| \approx |P(z_0)|/2 + O(|P(z_0)|/4)$, which satisfies the Armijo condition $|P(z_1^{(1/2)})| \le (1 - \sigma/2)|P(z_0)|$ with $\sigma = 0.1$ as long as the second-order term is bounded, which fails only on a set of measure $O(1/d^2)$. Inductively, $\Pr[j_{\mathrm{accept}} \ge t] \le \prod_{s=0}^{t-1} O(1/d^s) \le O(2^{-t})$ for $d \ge 2$, with constant $c \to \log_2 d$ as d grows.

Step 3 (empirical certification). The script `latex/scripts/prove_armijo_01.py` measures $\Pr[j_{\mathrm{accept}} \ge t]$ on 80 random UNI polynomials per degree $d \in \{16, 32, 64, 128\}$ using

Strategy B and Pandrosion-$T_2/K = 1$, summing over more than 8 000 fallback events. The geometric-tail regression $\log_2 \Pr[j \geq t] \approx a - c \cdot t$ yields:

d	$\Pr[j_{\text{accept}} = 0]$	decay rate c	$\mathbb{E}[j_{\text{accept}}]$	max observed j
16	98.2%	1.23	0.031	7
32	99.2%	1.45	0.016	6
64	99.1%	2.10	0.018	5
128	$\geq 99.6\%$	2.71	≤ 0.01	4

The decay rate c is ≥ 1.23 at all tested degrees and *tightens* as d grows, in agreement with the analytic roadmap of Step 2. This evidence is not a substitute for the missing measure estimate controlling the bad Armijo set at every backtracking depth. □

Theorem 7.2 (Conditional Las Vegas $O(d^2 \log d)$ bound, Armijo variant)**.** *Assume Conjecture 7.1 and Conjecture 5.1. Combining this amortised Armijo input with Theorem 3.5 of Part VII, the multi-start Pandrosion-$T_2/K = 1$ scheme with Strategy B starts and the Armijo fallback of Part VII Definition 2.1 admits the conditional bound*

$$\mathbb{E}_P\left[\operatorname{cost}^{B,T_2}_{total}(P)\right] \;=\; O(d^2 \log d)$$

as a Las Vegas guarantee under any of the ensembles of Conjecture 7.1. This matches the deterministic bound of Part VII Theorem 4.4 (the γ-damped variant) only after the Strategy B and Armijo amortisation conjectures are supplied.

Proof. By Conjecture 7.1, the expected number of P-evaluations per fallback is $4+\mathbb{E}[j_{\text{accept}}]+1 = O(1)$ (four direction probes, one Armijo trial in expectation, one descent check). The expected per-epoch cost is therefore $O(d) \cdot O(1) = O(d)$, replacing the worst-case $O(d \log d)$ used in the proof of Theorem 3.5 of Part VII. The number of epochs is unchanged ($O(d \log d)$), and the multi-start factor reduces from d to $\mathbb{E}[s^\star_B] = O(1)$ under Conjecture 5.1, giving expected total cost $O(1) \cdot O(d \log d) \cdot O(d) = O(d^2 \log d)$. □

8 Conclusion

Three practical refinements of the Pandrosion-T_3 algorithm of Part VII are reported:

- *Strategy B starts* (geometric-decreasing radius, golden angle) brought the observed worst-case fraction on i.i.d. Gaussian polynomials at $d = 128$ from 34% (Strategy A) to 0% in our tests.
- *T_2 at $K = 1$* was the empirical optimum on the tested families, cutting per-orbit oracle cost by $\approx 20\%$ (from $0.413\, d \log d$ for T_3 to $0.336\, d \log d$ for T_2).
- *Conjecture 7.1 (Armijo amortised cost)* combines an analytic roadmap with an empirical certificate suggesting $\mathbb{E}[j_{\text{accept}}] = O(1)$ on the UNI / KS ensembles we sampled, with observed decay rate $c \geq 1.23$. Combined with Theorem 3.5 of Part VII, it gives the conditional Las Vegas guarantee of Theorem 7.2: $\mathbb{E}[\operatorname{cost}^{B,T_2}_{\text{total}}] = O(d^2 \log d)$.

The reframing is deliberately sober: the Pandrosion-T_2 scheme with Strategy-B starts is presented as a practical, derivative-free root-finding heuristic with promising empirical behaviour and a conditional complexity bound.

It is not a new resolution of Smale's 17th problem — that problem is already resolved, unconditionally in the deterministic-average-case sense, by Lairez [5] and, probabilistically, by Beltrán–Pardo [4].

What we offer is a concrete, simple, derivative-free algorithmic alternative whose behaviour on adversarial and random inputs is worth documenting.

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